

# CSE 4615:: Wireless Networks

## Overview of Wireless Access Networks

### Lecture 2

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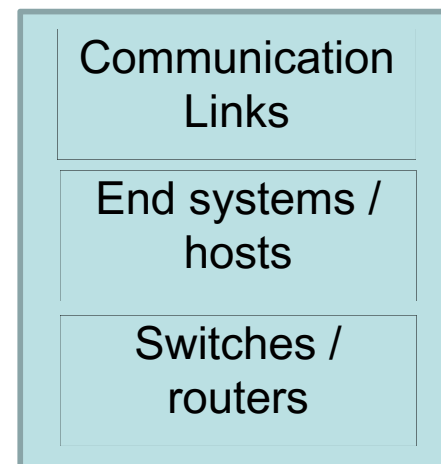
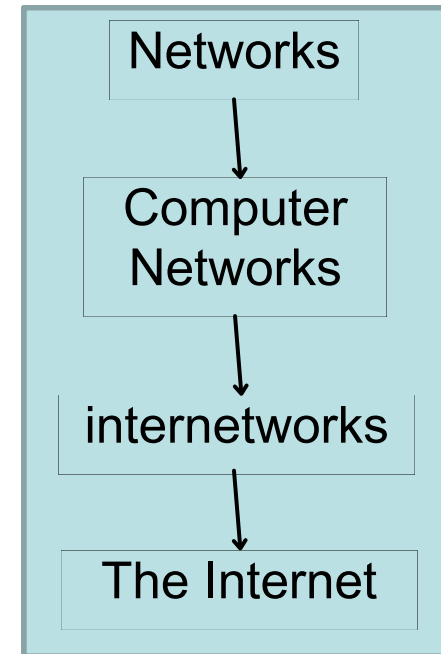
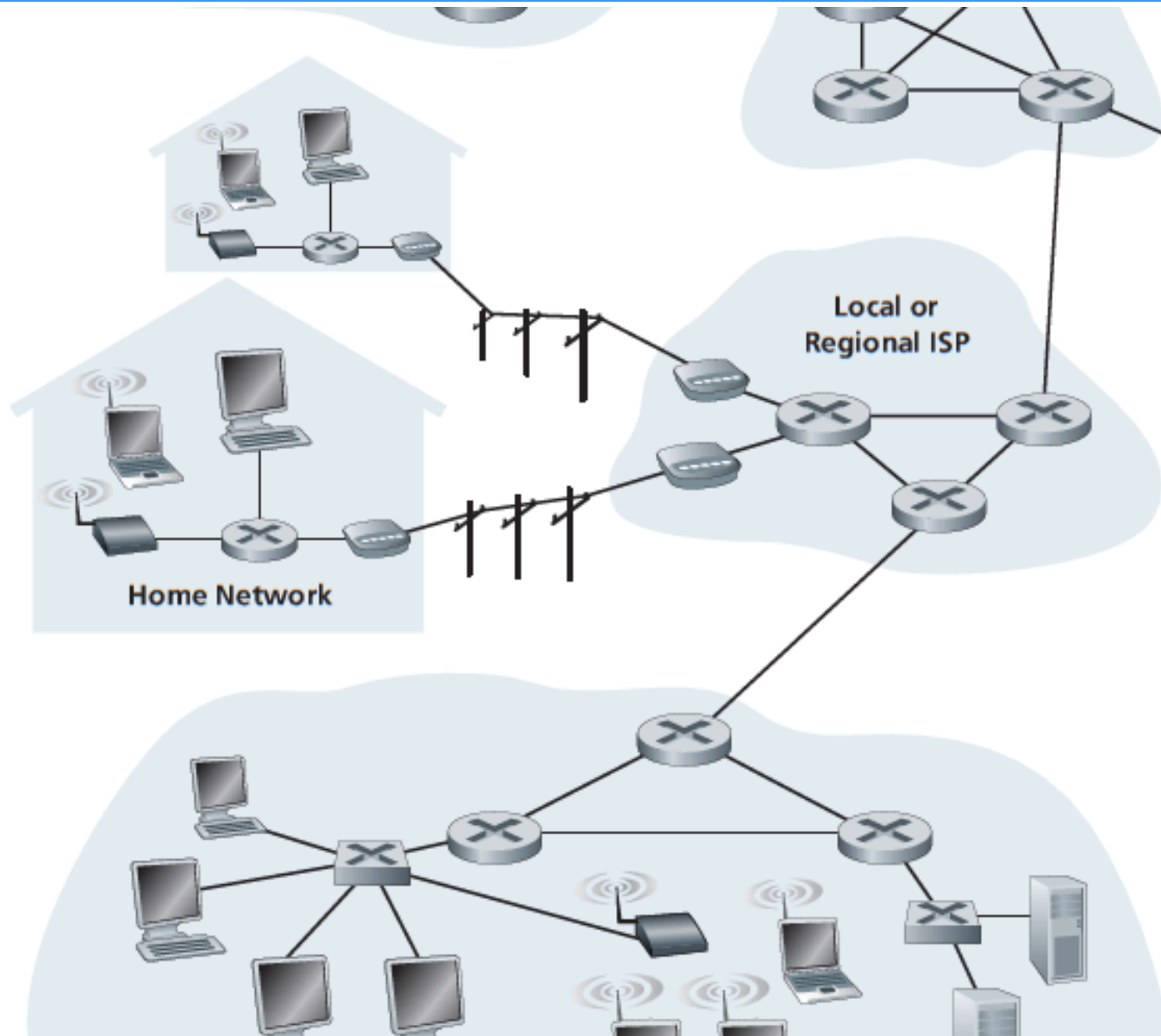
Volkan Rodoplu, UCSB

Andrea Goldsmith, Princeton University

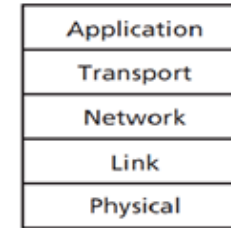
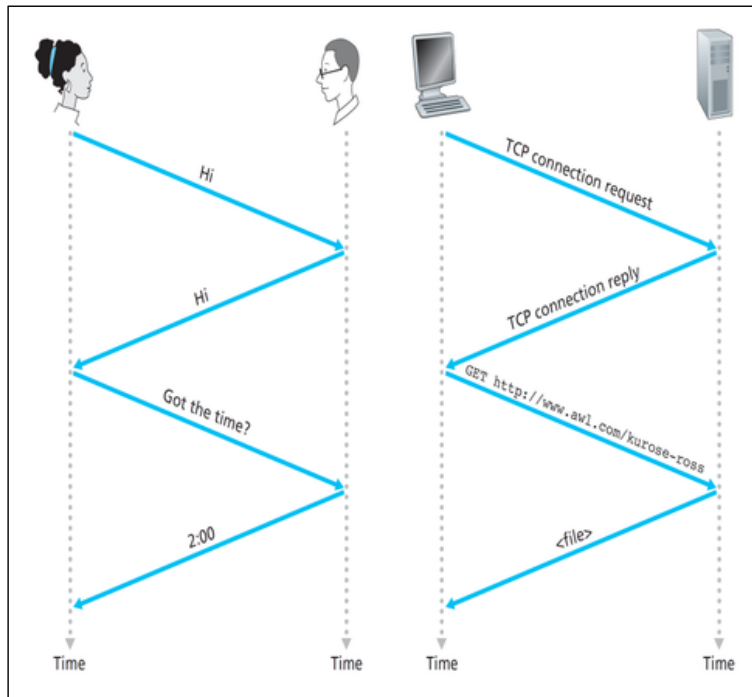
James F. Kurose and Keith W. Ross, University of Massachusetts  
Amherst

Christoph Sommer & Falko Dressler, Paderborn University

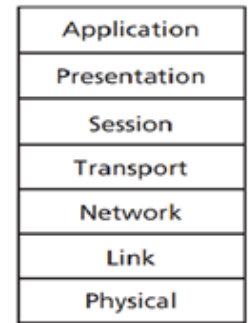
# Internet – the bigger picture



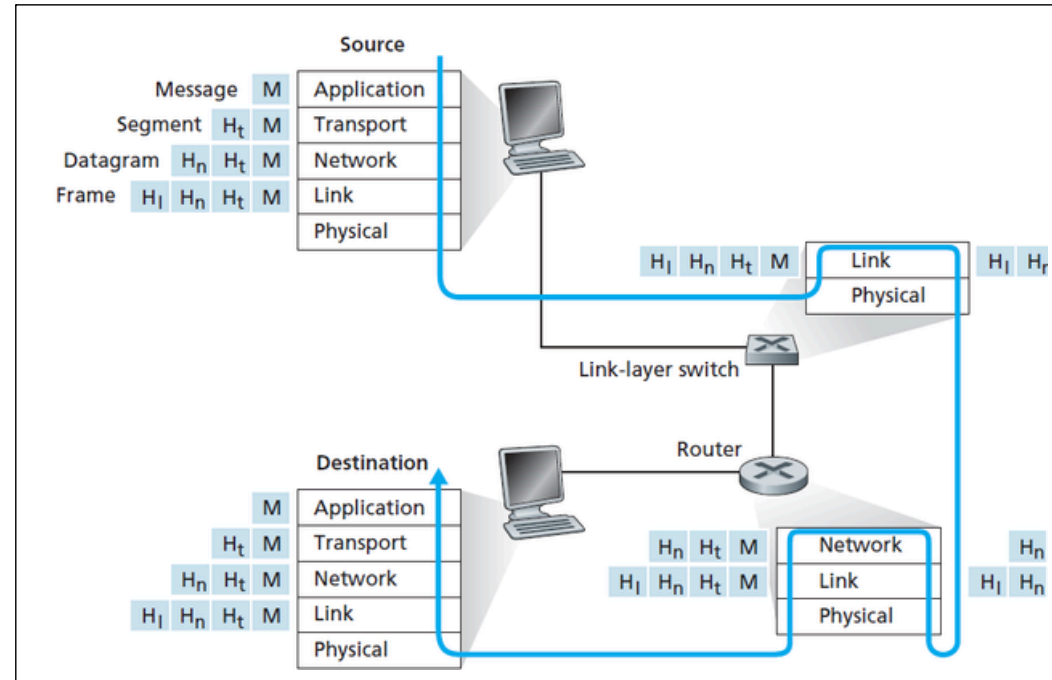
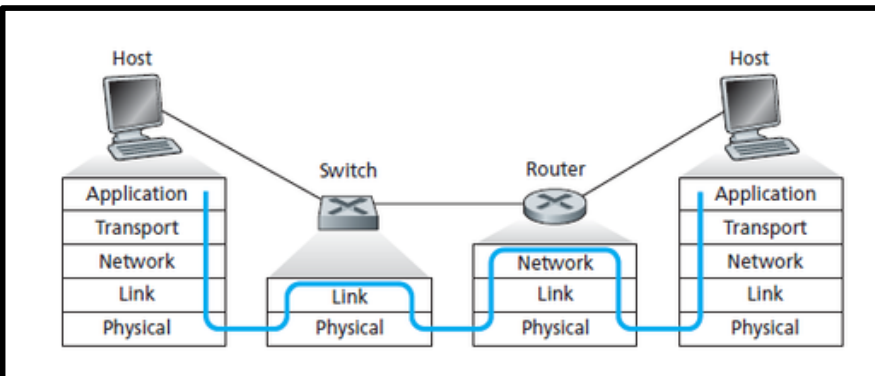
# Protocols and Standards



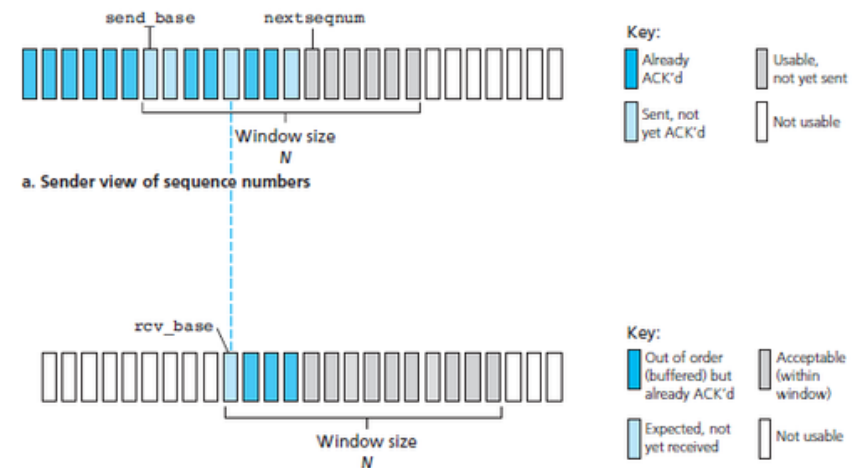
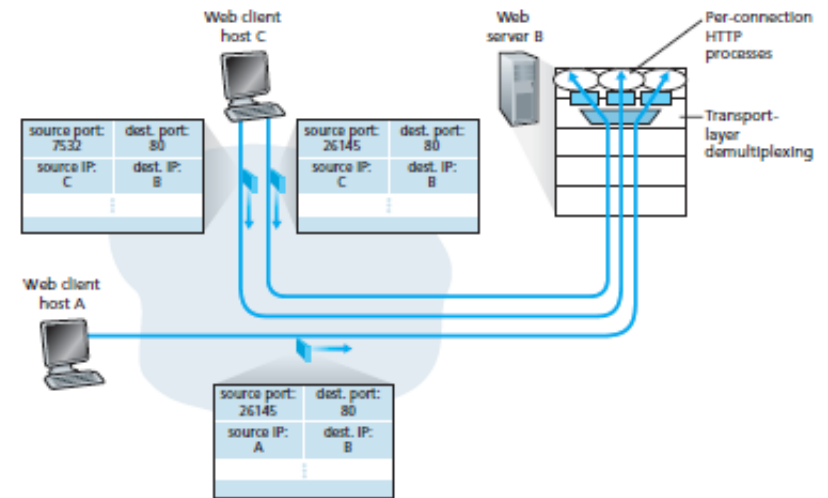
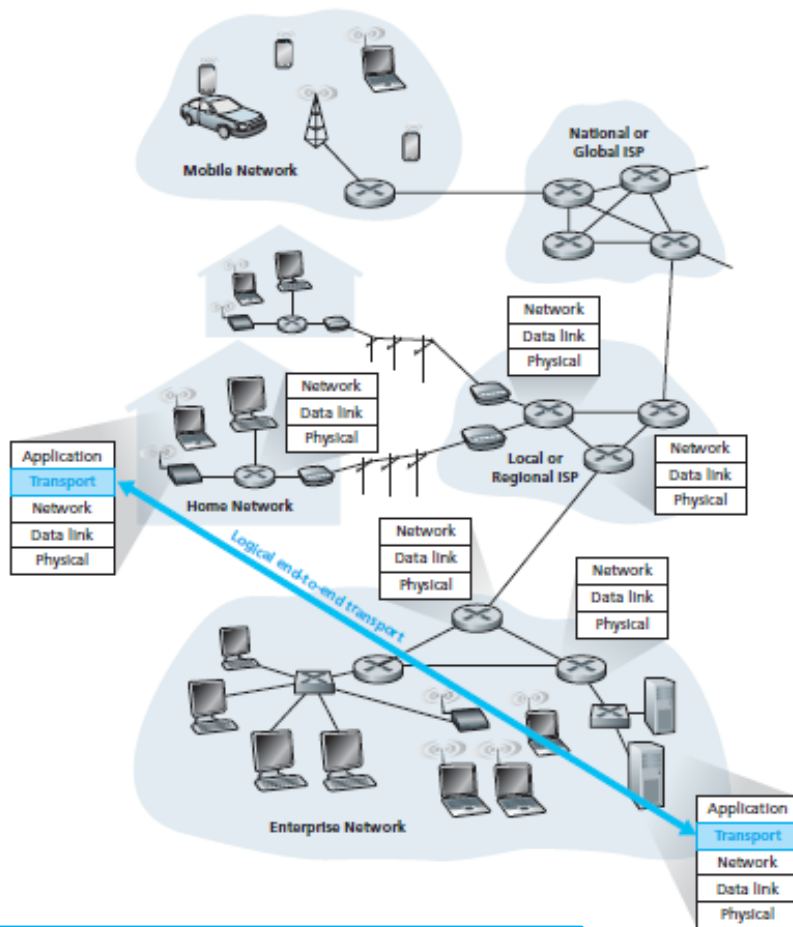
a. Five-layer Internet protocol stack



b. Seven-layer ISO OSI reference model



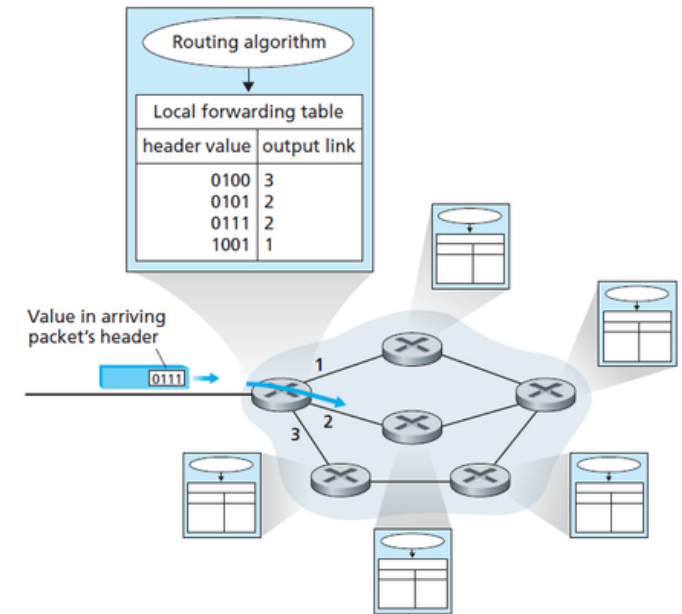
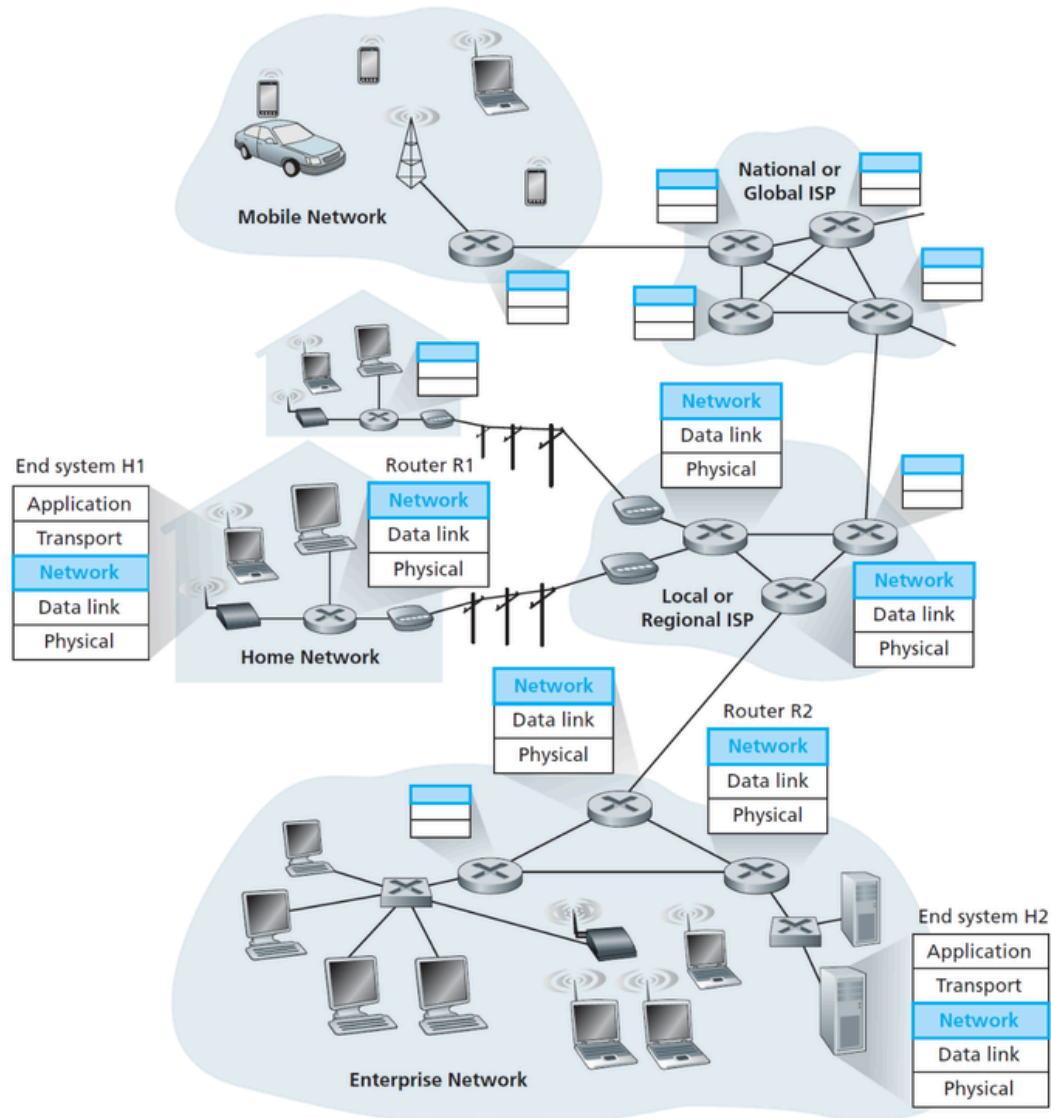
# The Transport Layer



- Multiplexing / de-multiplexing
- Reliability & in-order delivery
- Congestion control
- Flow Control



# The Network Layer



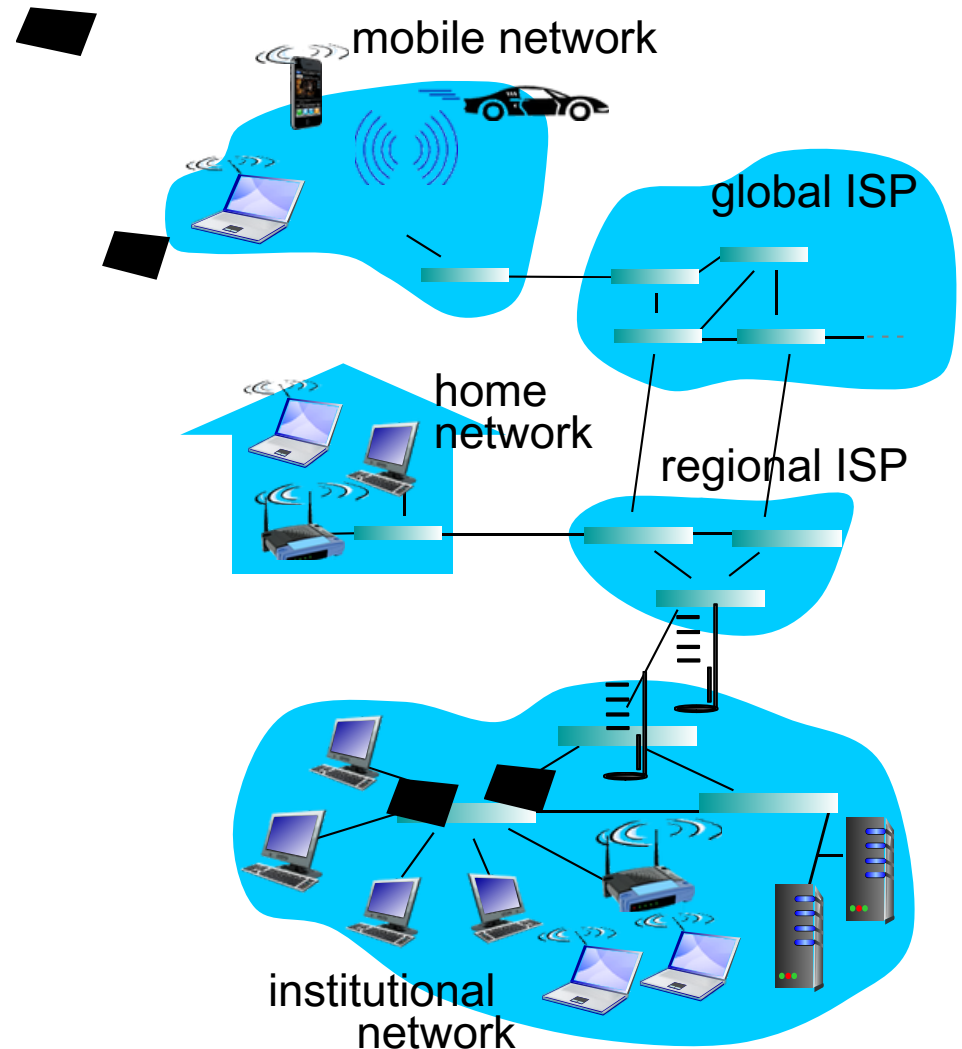
- Routing
  - Routing Metric
    - Hop-count, BW
    - ETX, ETT
- Routing Protocol
  - RIP, OSPF
  - BGP
- Packet Forwarding
- QoS

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# A closer look at network structure

# A closer look at network structure:

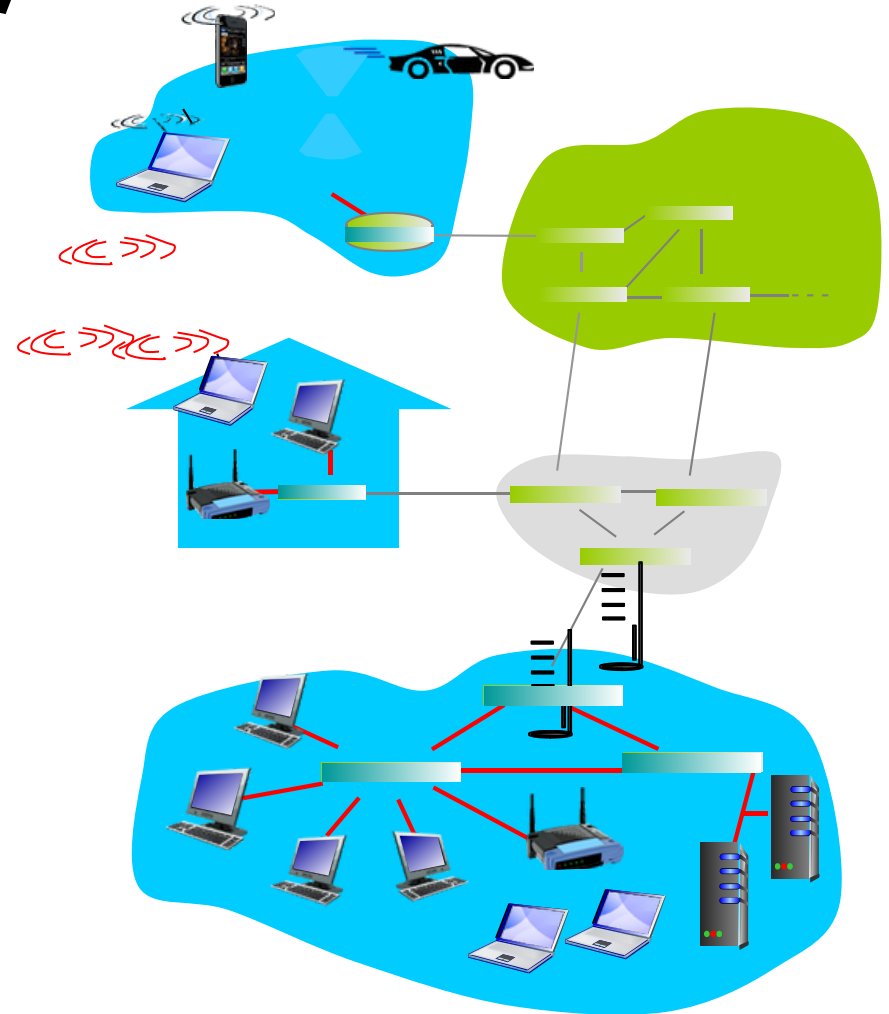
- *network edge:*
  - hosts: clients and servers
  - servers often in data centers
- *physical media:* wired, wireless communication links
- *network core:*
  - interconnected routers
  - network of networks



# Access networks and physical media

*Q: How to connect end systems to edge router?*

- residential access nets
- institutional access networks (school, company)
- mobile access networks



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# Access Networks

# Access Networks

## Nuts and Bolts of Internet



Host  
(= end system)



Server



Mobile



Router



Link-Layer  
switch



Modem



Base  
station



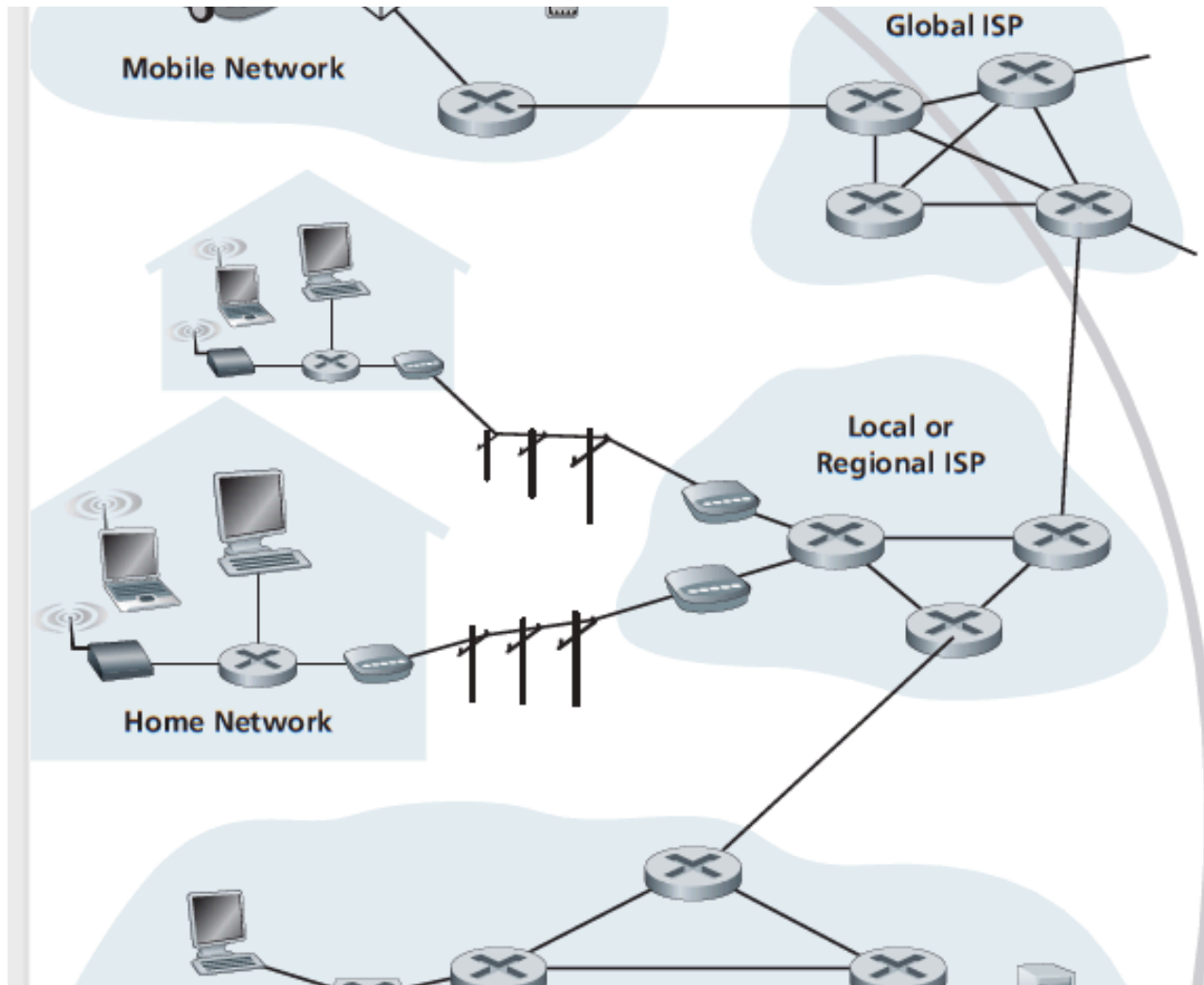
Smartphone



Cell phone  
tower

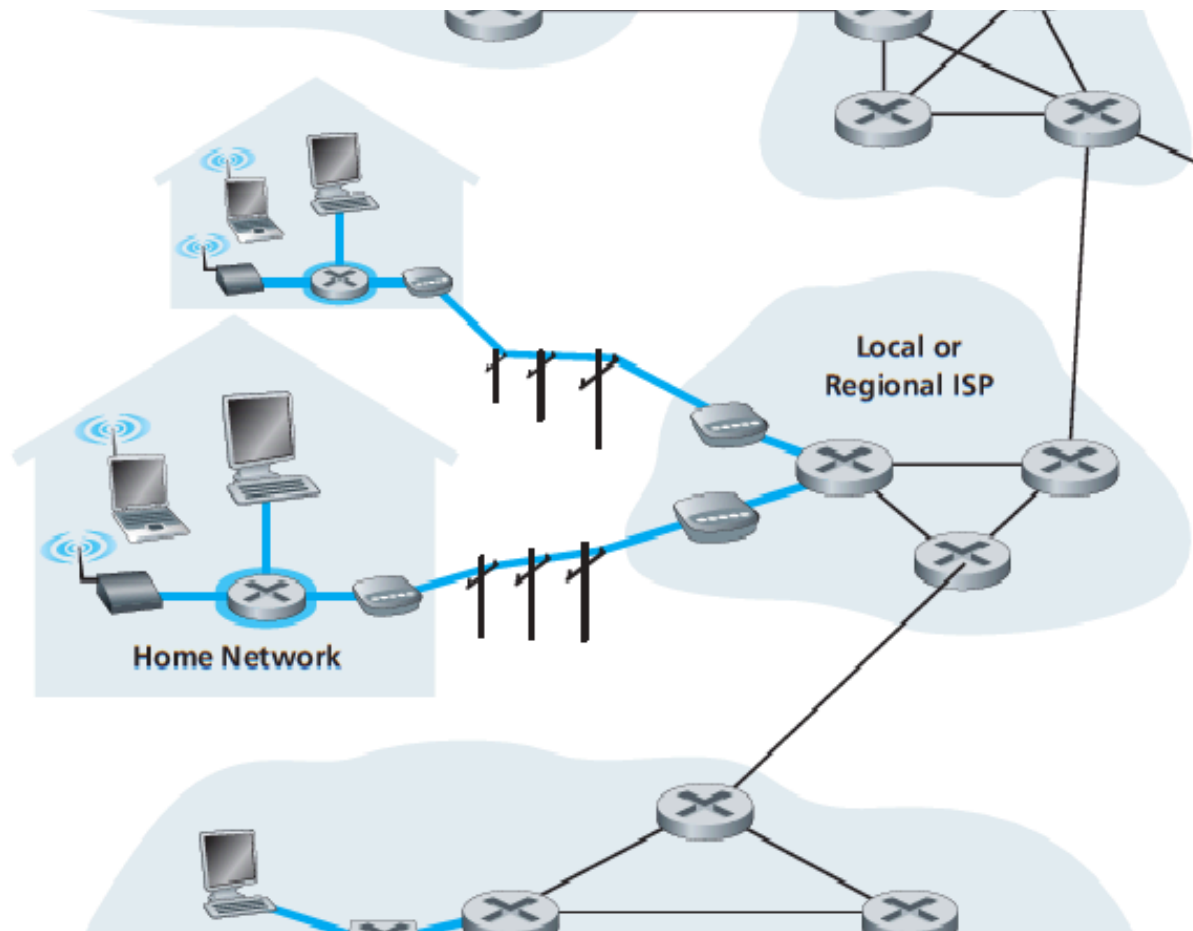
# Access Networks

## End System Interaction



# Access Networks

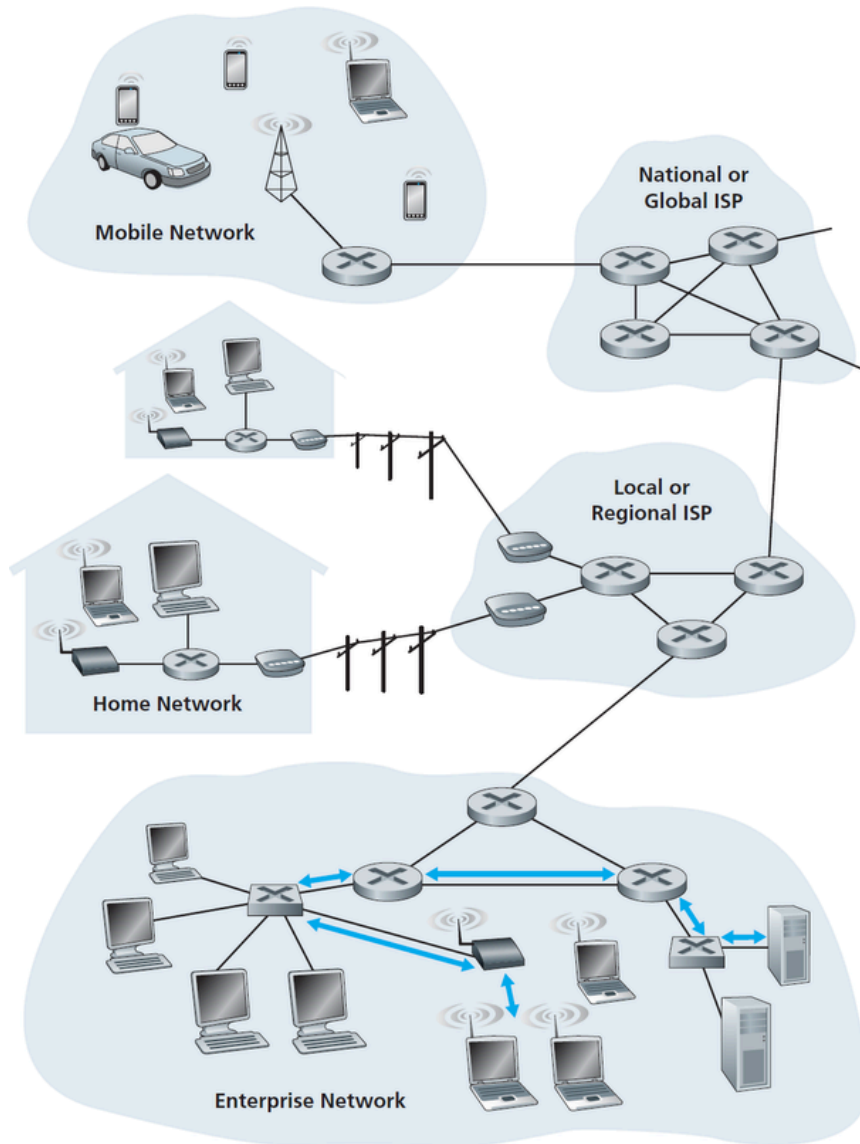
**The network that physically connects an end system to the first router (also known as the “edge router”) on a path from the end system to any other distant end system**





# Access Networks

## Taxonomy based on Technology Used



- Access Networks
  - Ethernet
  - WiFi-Based Access Networks
    - Wireless LAN
      - 802.11a/b/g
    - Ad-hoc Networks
      - Mesh Networks
      - DTN/ICN
      - VANET
      - 6LowPANs
        - WSNs
    - WiMAX-Based Access Net
    - Cellular Networks
      - 3G Cellular Networks and beyond
    - etc...

# Access Networks

## Taxonomy based on Technology Used

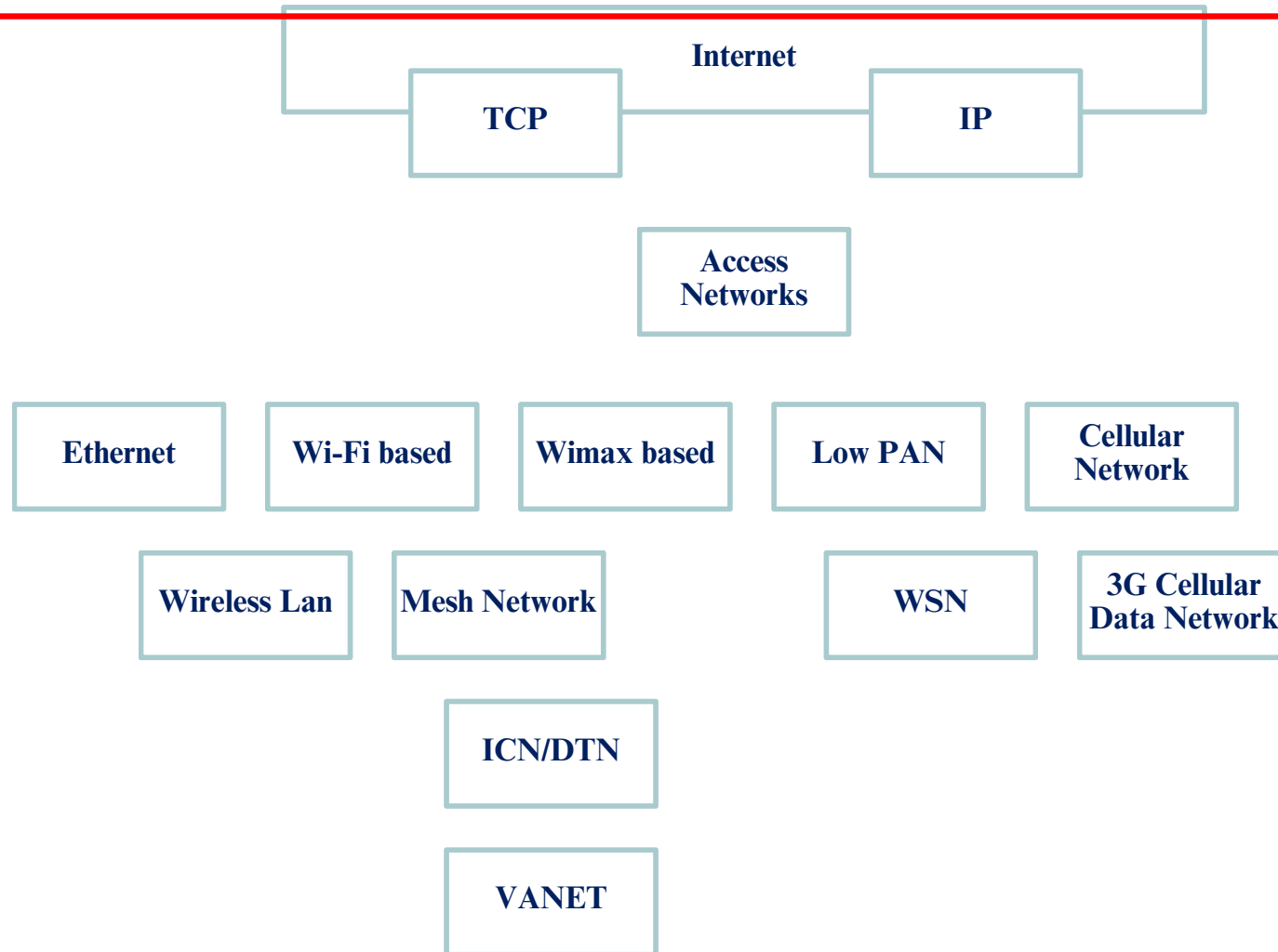


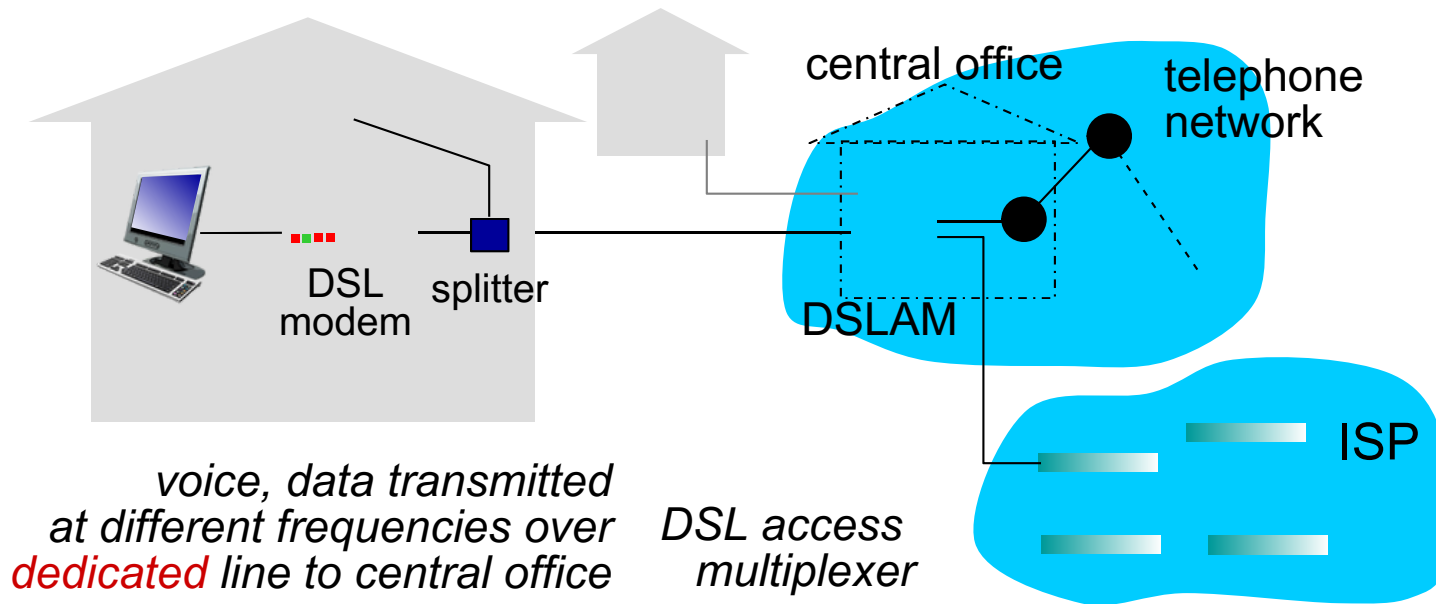
Figure: a taxonomy of access networks (approximate)

# Access Networks

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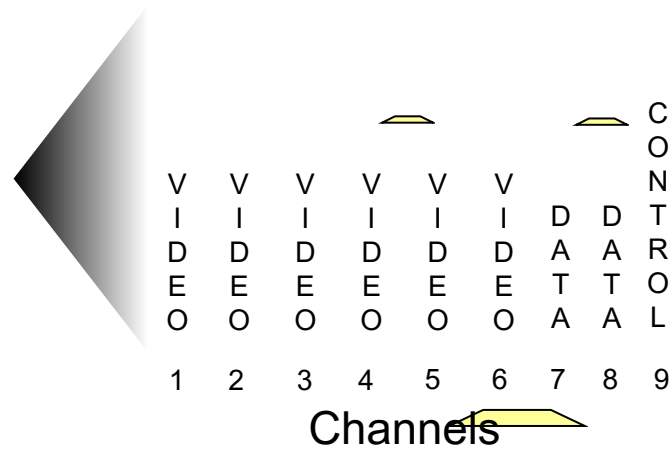
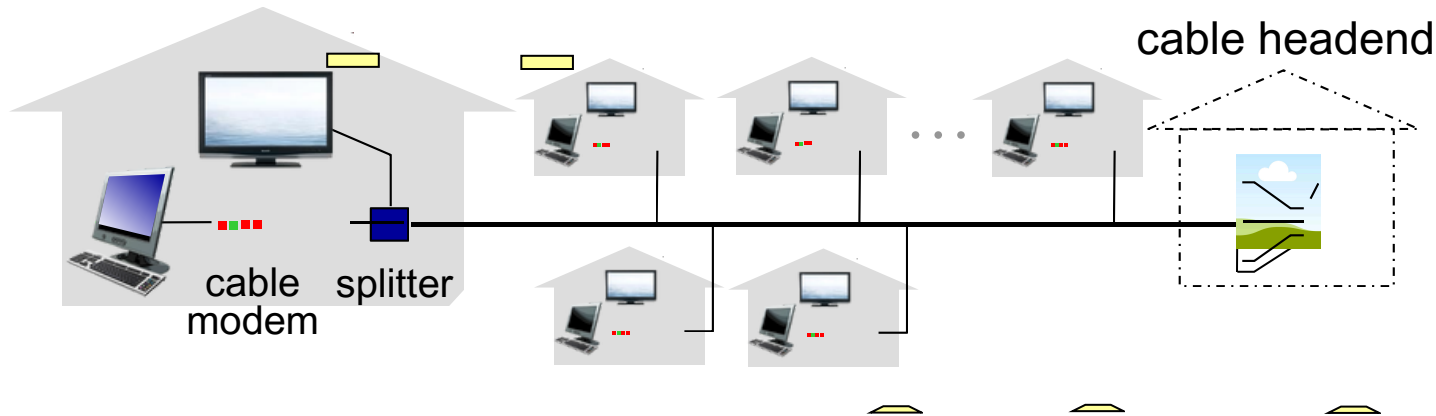
## Wired Access Networks

# Access net: digital subscriber line (DSL)



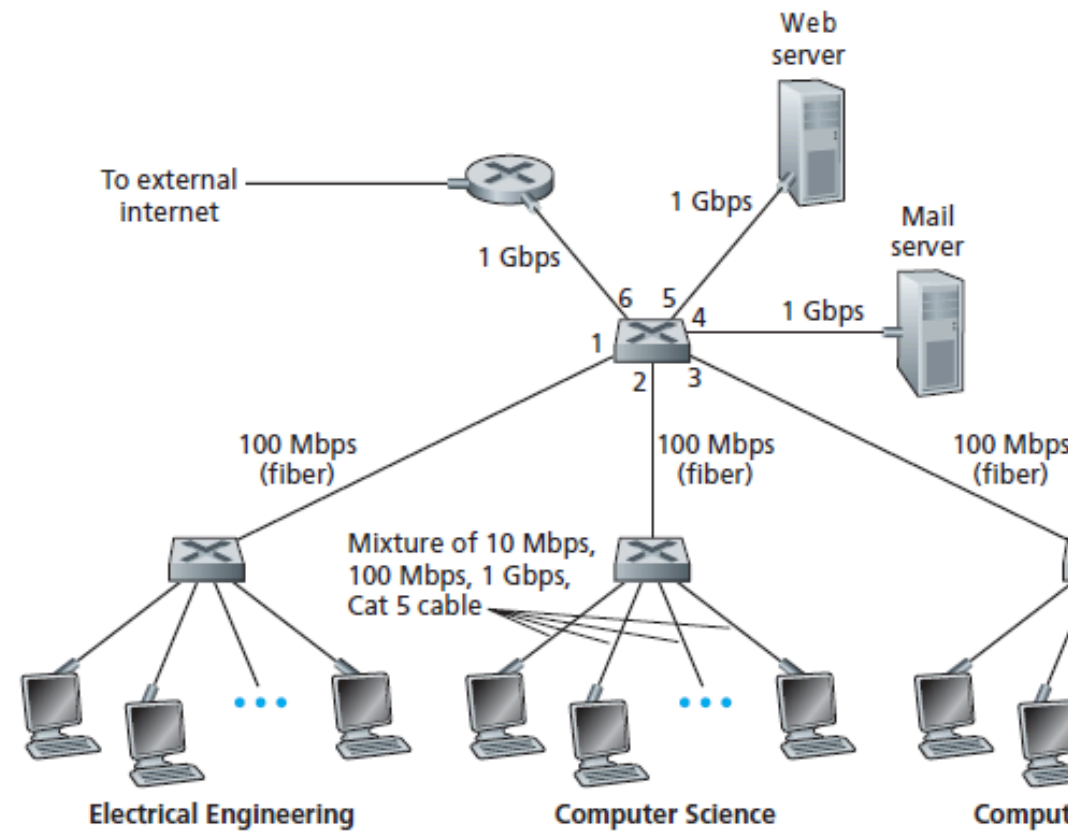
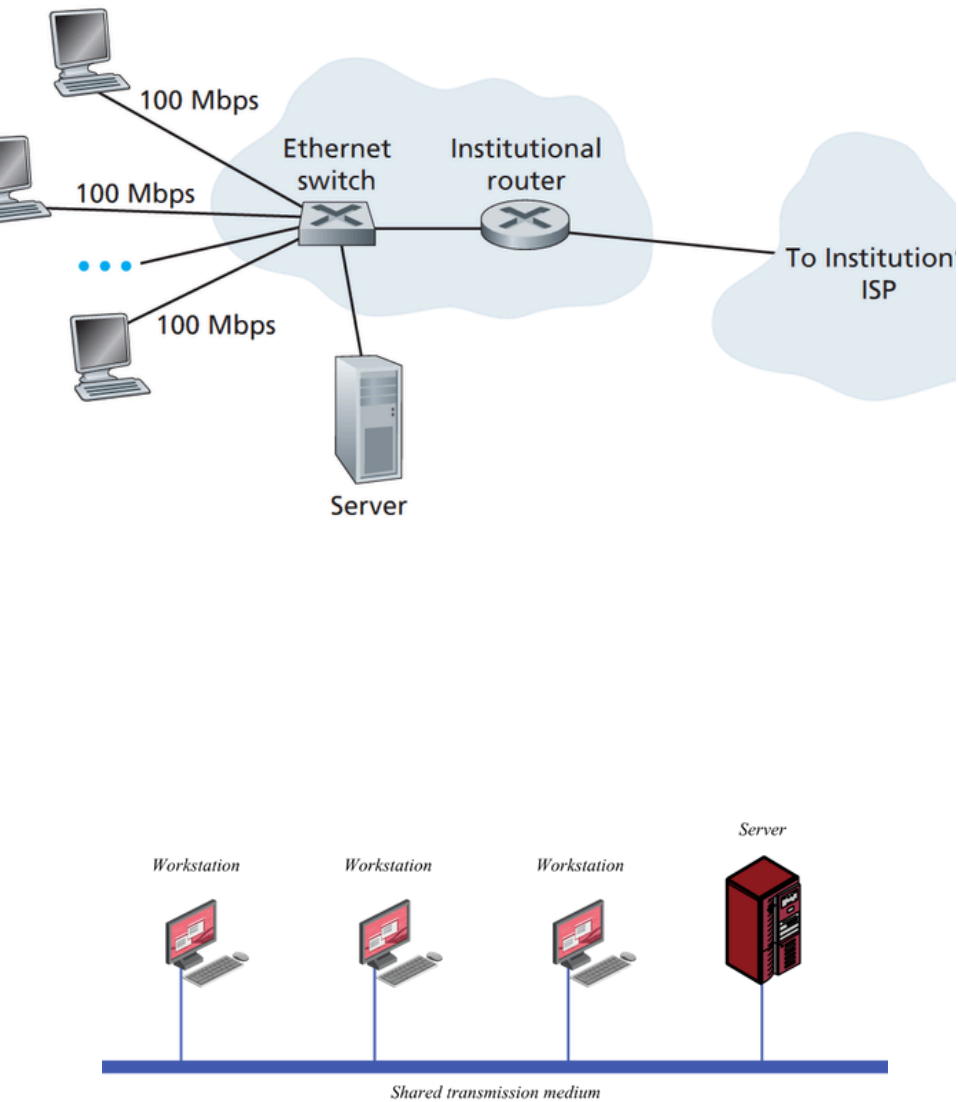
- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- < 24 Mbps downstream transmission rate (typically < 10 Mbps)

# Cable Network



*frequency division multiplexing:* different channels transmitted in different frequency bands

# Ethernet – Traditional Wired Access



# Access Networks

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Wireless Access Networks

# Wireless Access Networks

- shared *wireless* access network connects end system to router
  - via base station aka “access point”

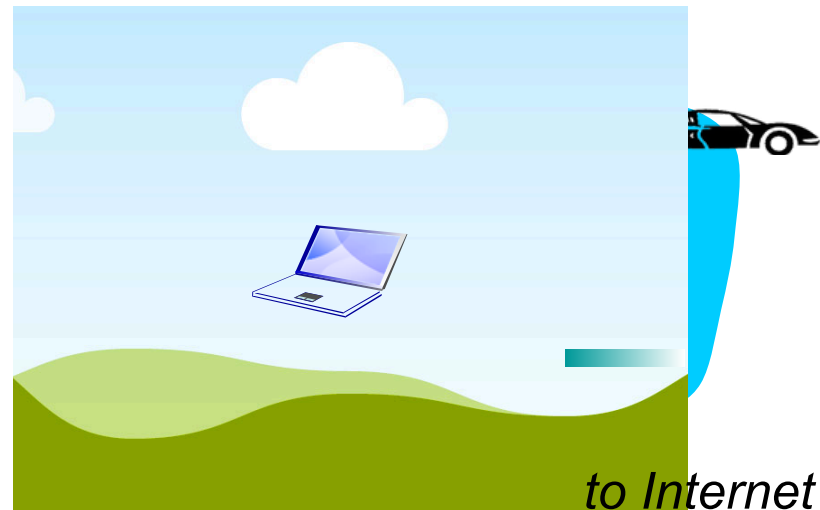
## wireless LANs:

- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate



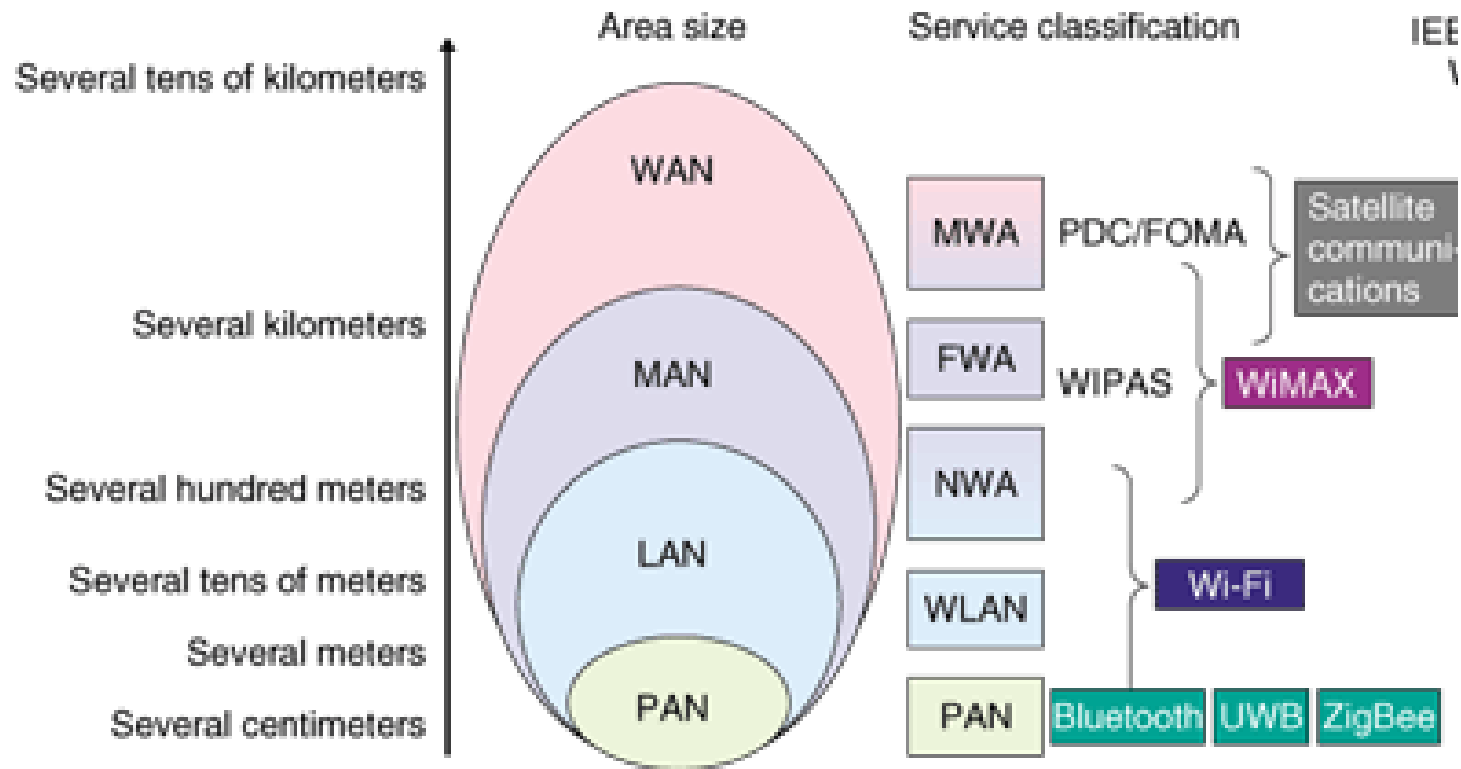
## wide-area wireless access

- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G: LTE





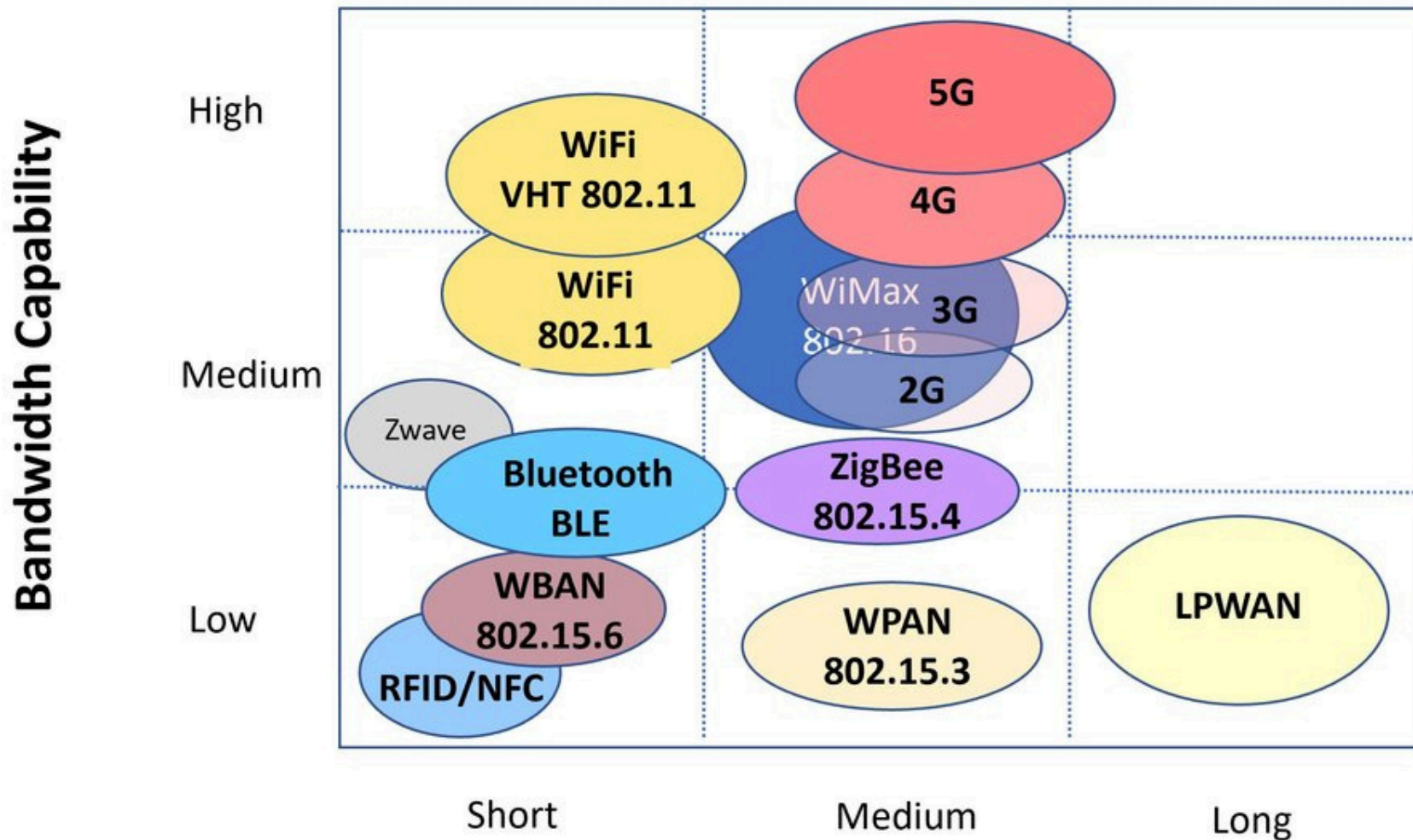
# Wireless Access Networks



WAN: wide area  
 MAN: metropoli  
 LAN: local area  
 PAN: personal ;

# WIRELESS ACCESS NETWORKS. Range Comparison

Comparison of Wireless Technologies

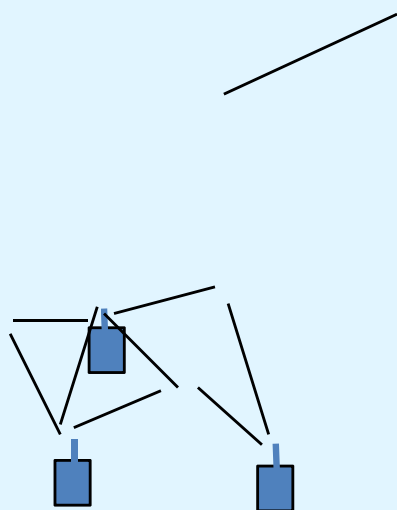


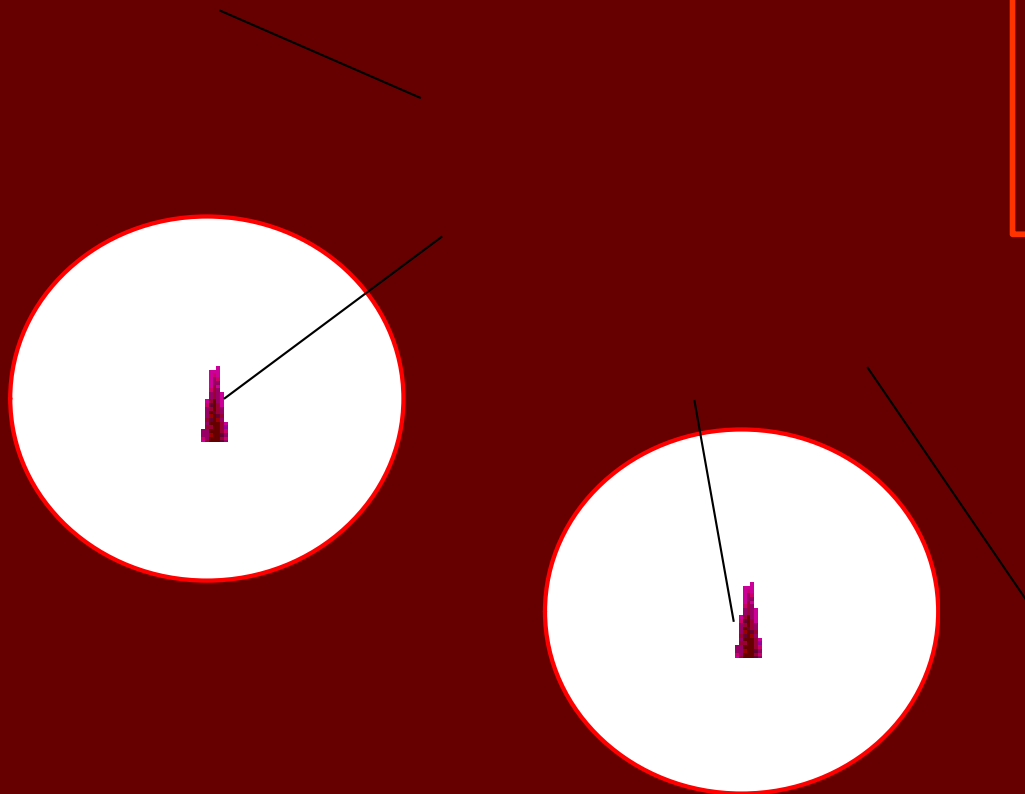
# Wireless Access Networks

| <i>Personal Area Network (PAN)</i>                  | <i>Local Area Network (LAN)</i>       | <i>Metropolitan Area Network (MAN)</i>                     |
|-----------------------------------------------------|---------------------------------------|------------------------------------------------------------|
| Bluetooth                                           | 802.11b<br>802.11a<br>802.11g         | 802.16<br>802.16a<br>802.16e                               |
| Medium data rates<br>1 Mbps to 2 Mbps               | High data rates<br>11 Mbps to 54 Mbps | Very high data rates<br>Quality of service, up to 280 Mbps |
| Very short range<br>3 m (~10 feet)                  | Short range<br>100 m (~300 feet)      | Medium range<br>50 km (~31 miles)                          |
| Notebook to PC to peripherals<br>devices to systems | Computer to computer and the internet | LAN or computer to high speed wire line internet           |

# Milestones in Wireless Communications

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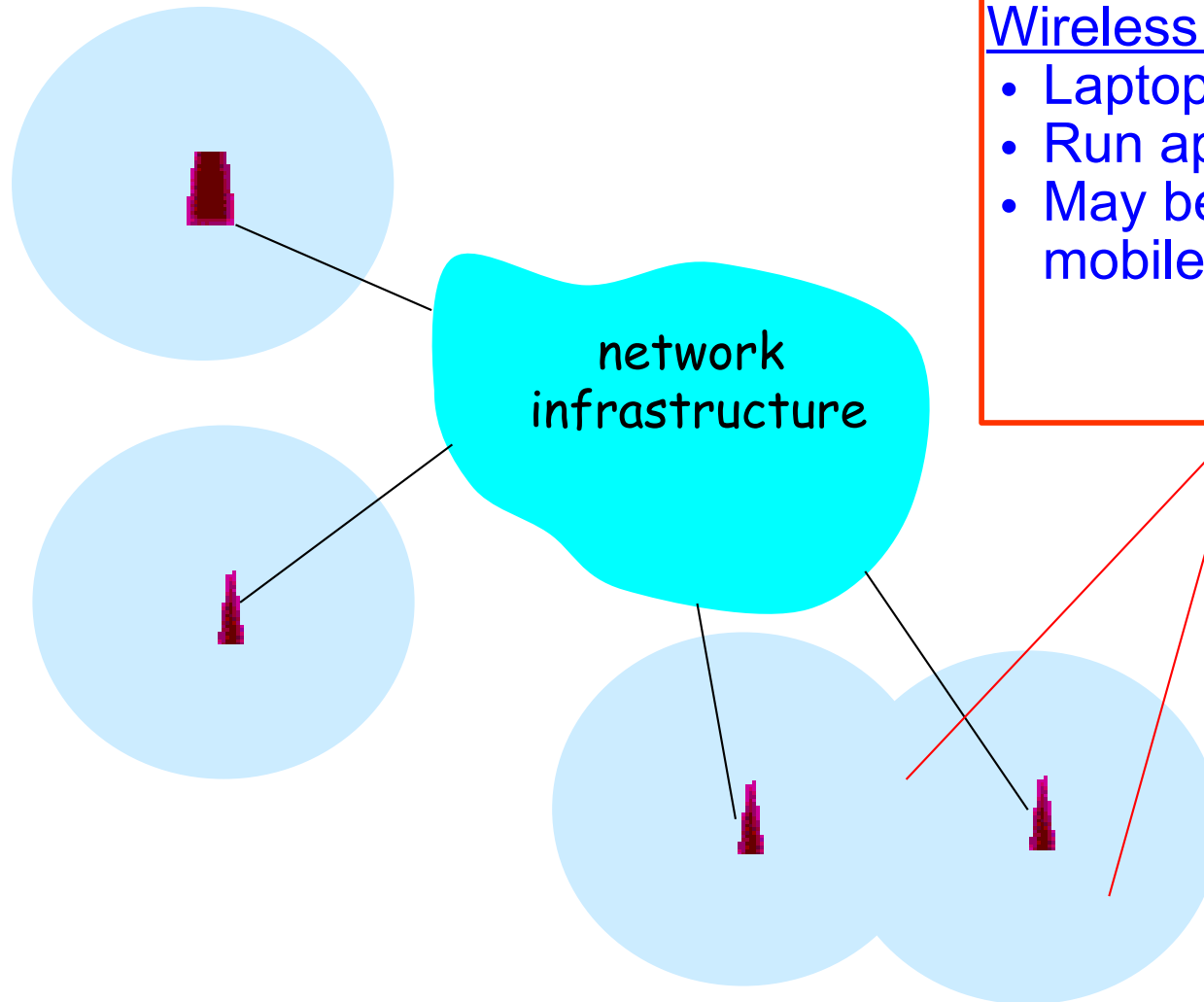




## Wireless link

- Typically used to connect mobile(s) to base station
- Also used as backbone link
- Multiple access protocol coordinates link access

# Wireless Network: Wireless **Hosts**



## Wireless host

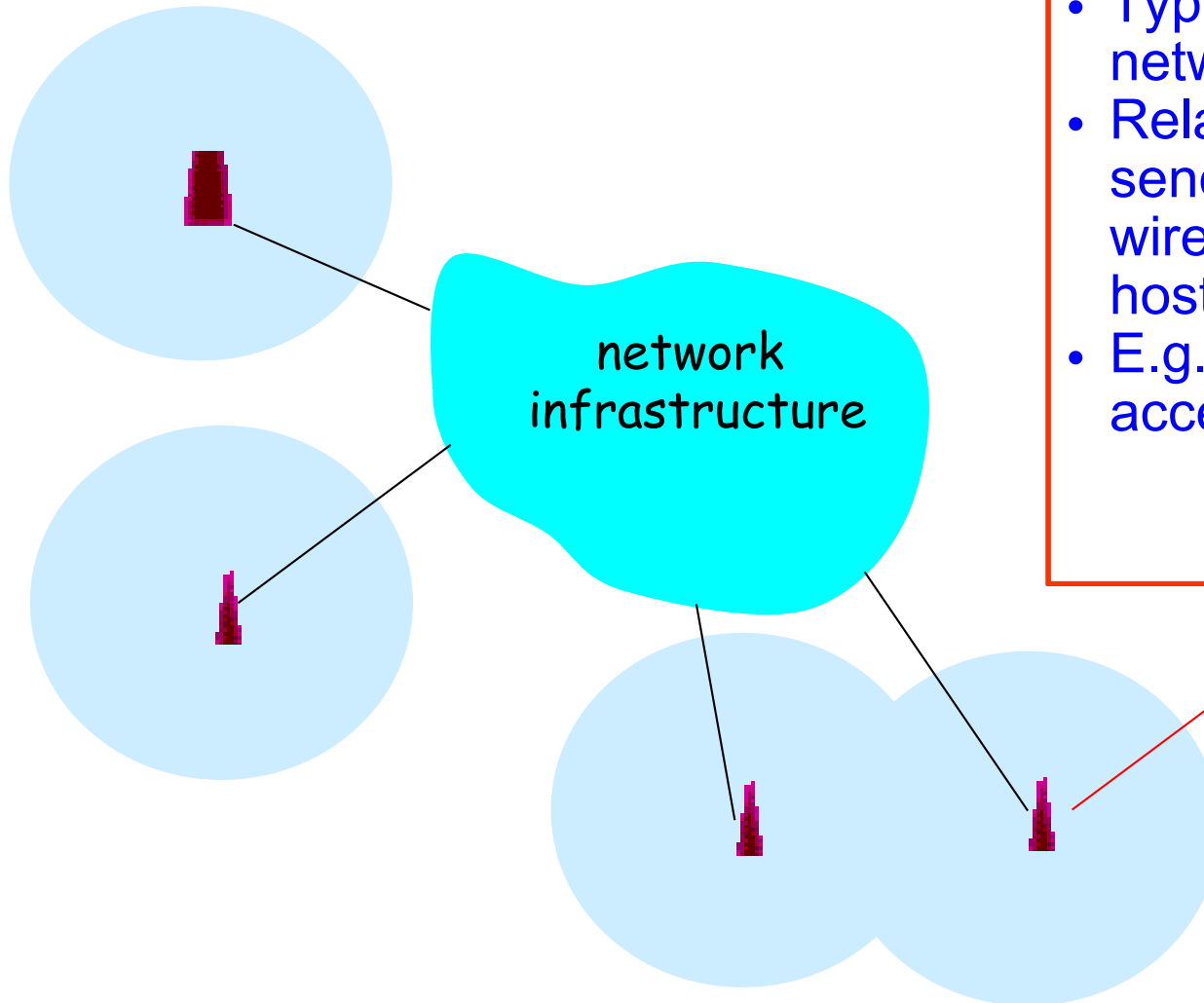
- Laptop, PDA, IP phone
- Run applications
- May be stationary (non-mobile) or mobile

# Wireless Network: **Base Station**



## Base station

- Typically connected to wired network
- Relay responsible for sending packets between wired network and wireless host(s) in its “area”
- E.g., cell towers, 802.11 access points



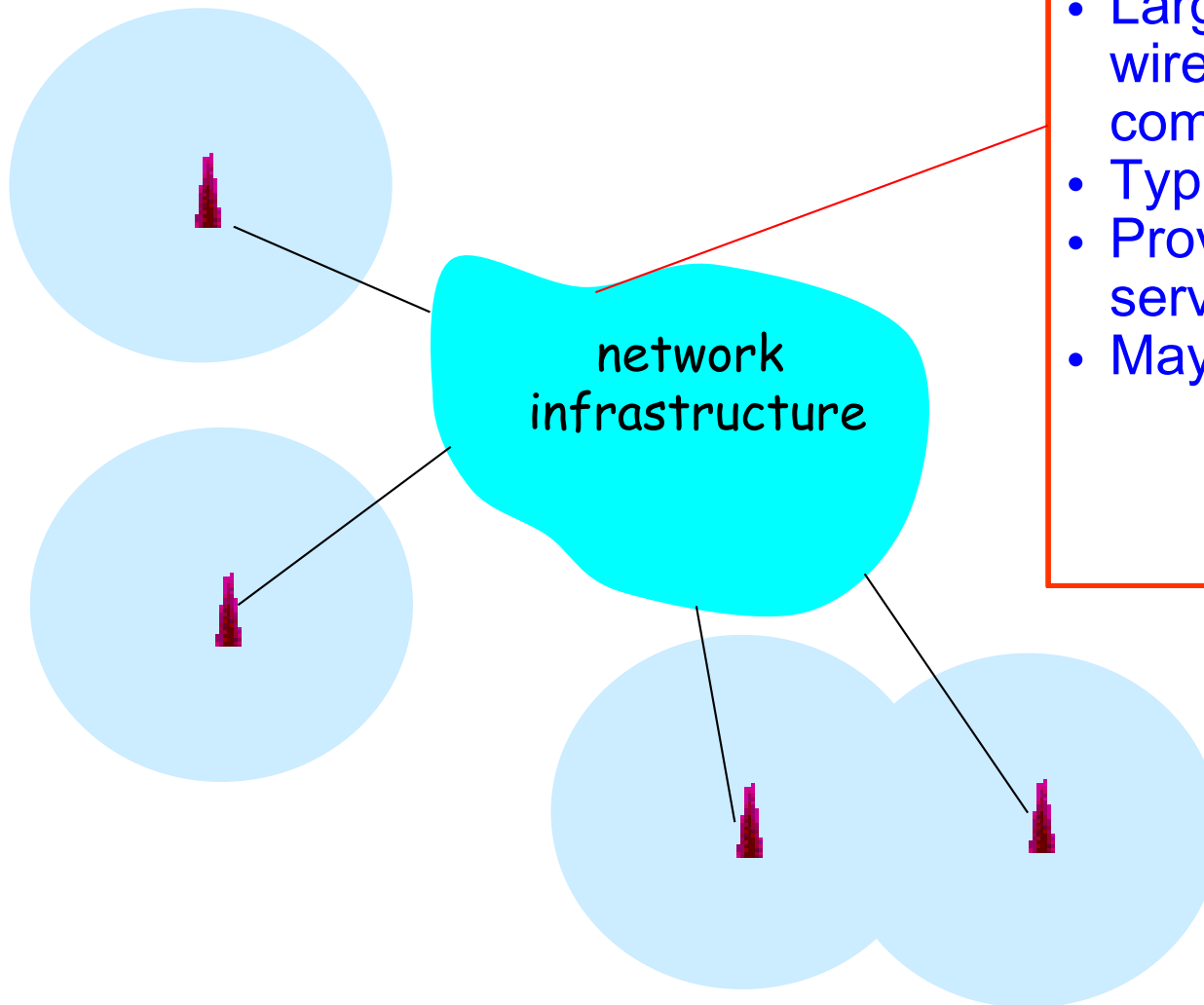


# Wireless Network: **Infrastructure**



## Network infrastructure

- Larger network with which a wireless host wants to communicate
- Typically a wired network
- Provides traditional network services
- May not always exist



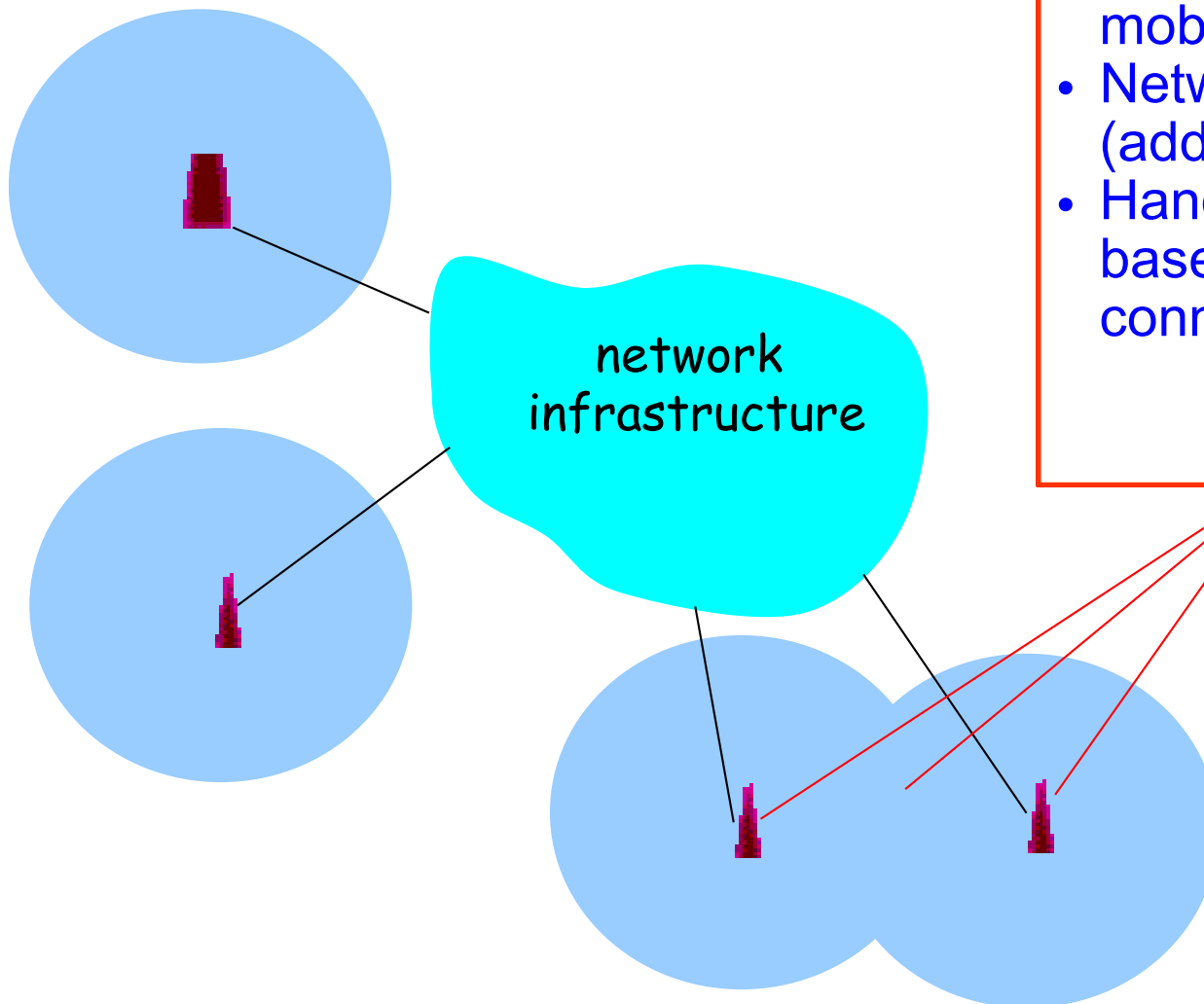
# Wireless Access Networks:: Infrastructure Modes

## Scenario #1: Infrastructure Mode



### Infrastructure mode

- Base station connects mobiles into wired network
- Network provides services (addressing, routing, DNS)
- Handoff: mobile changes base station providing connection to wired network



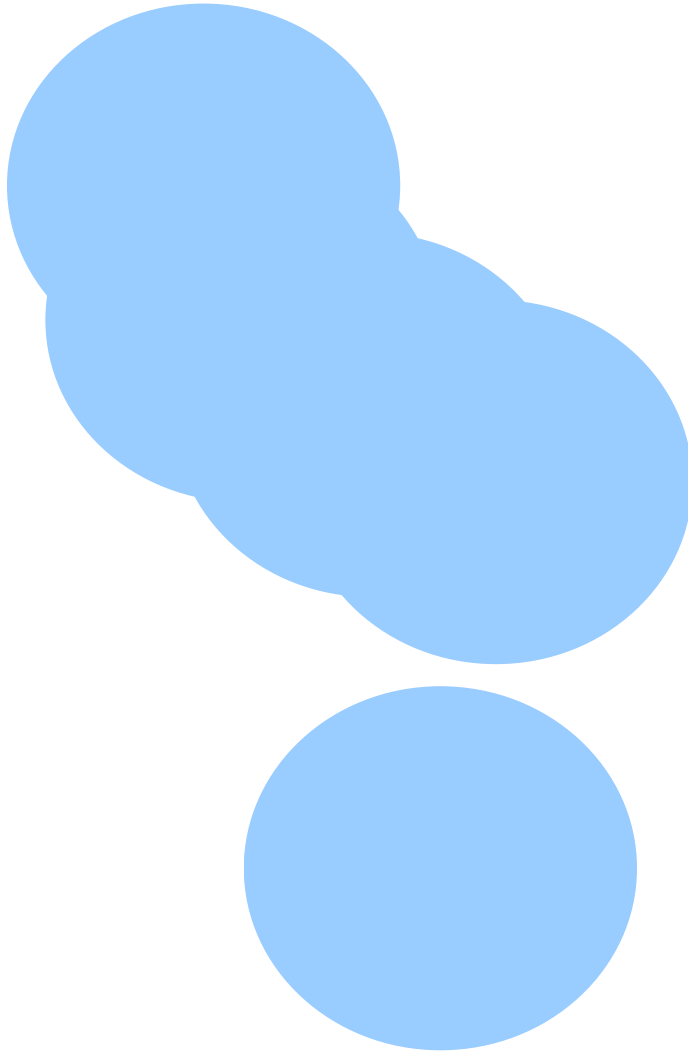
Client-server communication

# Wireless Access Networks:: Infrastructure Modes

## Scenario #2: Infrastructure-less (Ad Hoc) Mode



Peer-to-peer communication



### Ad hoc mode

- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes self-organize and route among themselves

# Wireless Access Networks:: Infrastructure Modes



|            | Infrastructure-based                                                                                                                                                                                                            |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single-hop | <ul style="list-style-type: none"><li>• Base station connected to larger wired network<ul style="list-style-type: none"><li>◦ WiFi based wireless LAN</li><li>◦ Cellular telephony networks</li><li>◦ WiMAX</li></ul></li></ul> |
|            |                                                                                                                                                                                                                                 |

# Wireless Access Networks:: Infrastructure Modes



|            | Infrastructure-based                                                                                                                                                                                                            | Infrastructure-less                                                                                                                                                                                                |
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|            |                                                                                                                                                                                                                                 |                                                                                                                                                                                                                    |

# Wireless Access Networks:: Infrastructure Modes



|            | Infrastructure-based                                                                                                                                                                                                            | Infrastructure-less                                                                                                                                                                                                |
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| Multi-hop  | <ul style="list-style-type: none"><li>• Base station exists</li><li>• Some nodes must relay through other nodes<ul style="list-style-type: none"><li>◦ Wireless Mesh Networks (semi-structured)</li></ul></li></ul>             |                                                                                                                                                                                                                    |

# Wireless Access Networks:: Infrastructure Modes



|            | Infrastructure-based                                                                                                                                                                                                            | Infrastructure-less                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Single-hop | <ul style="list-style-type: none"><li>• Base station connected to larger wired network<ul style="list-style-type: none"><li>◦ WiFi based wireless LAN</li><li>◦ Cellular telephony networks</li><li>◦ WiMAX</li></ul></li></ul> | <ul style="list-style-type: none"><li>• One node coordinates the transmissions of the others<ul style="list-style-type: none"><li>◦ Bluetooth</li><li>◦ Ad-hoc 802.11 (e.g., P2P, WiFi Direct)</li></ul></li></ul>                                                                                                                                                                                                   |
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# Wireless Access Networks

## Single-hop Infrastructure-based Wireless Systems

1. Cellular Networks
2. Wireless LANs
3. WiMAX

## Multi-hop Infrastructure-based Wireless Systems

5. Wireless Mesh Networks (semi-structured)

## Single-hop Infrastructure-less Wireless Systems

4. Bluetooth

## Multi-hop Infrastructure-less Wireless Systems

### 6. Ad hoc wireless Networks

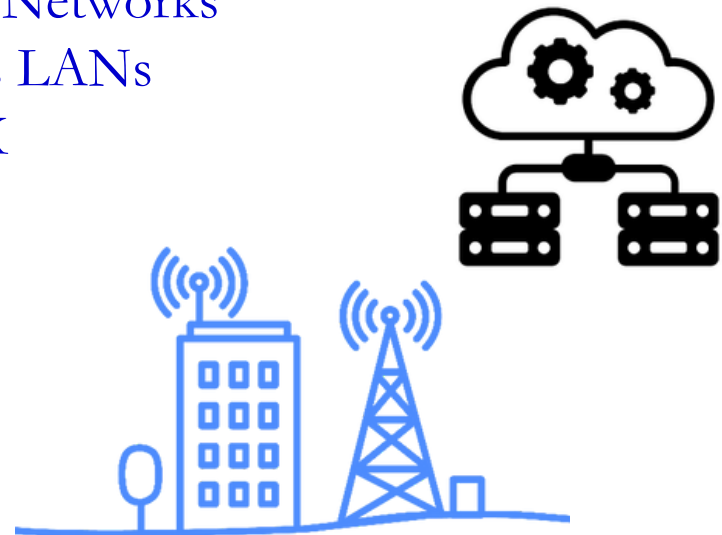
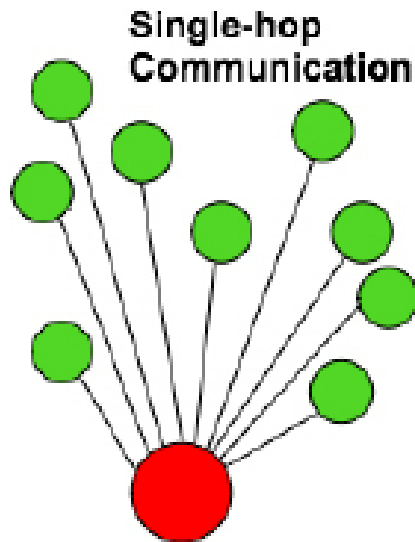
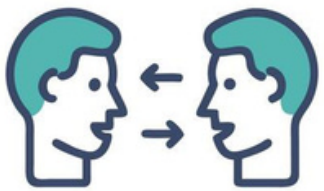
6.1. Wireless Sensor Networks (Structured / Semi-Structured / Un-Structured ::  
Depending on Application Nature)

- 6.2. Vehicular Ad-hoc Networks
- 6.3. Delay Tolerant Networks



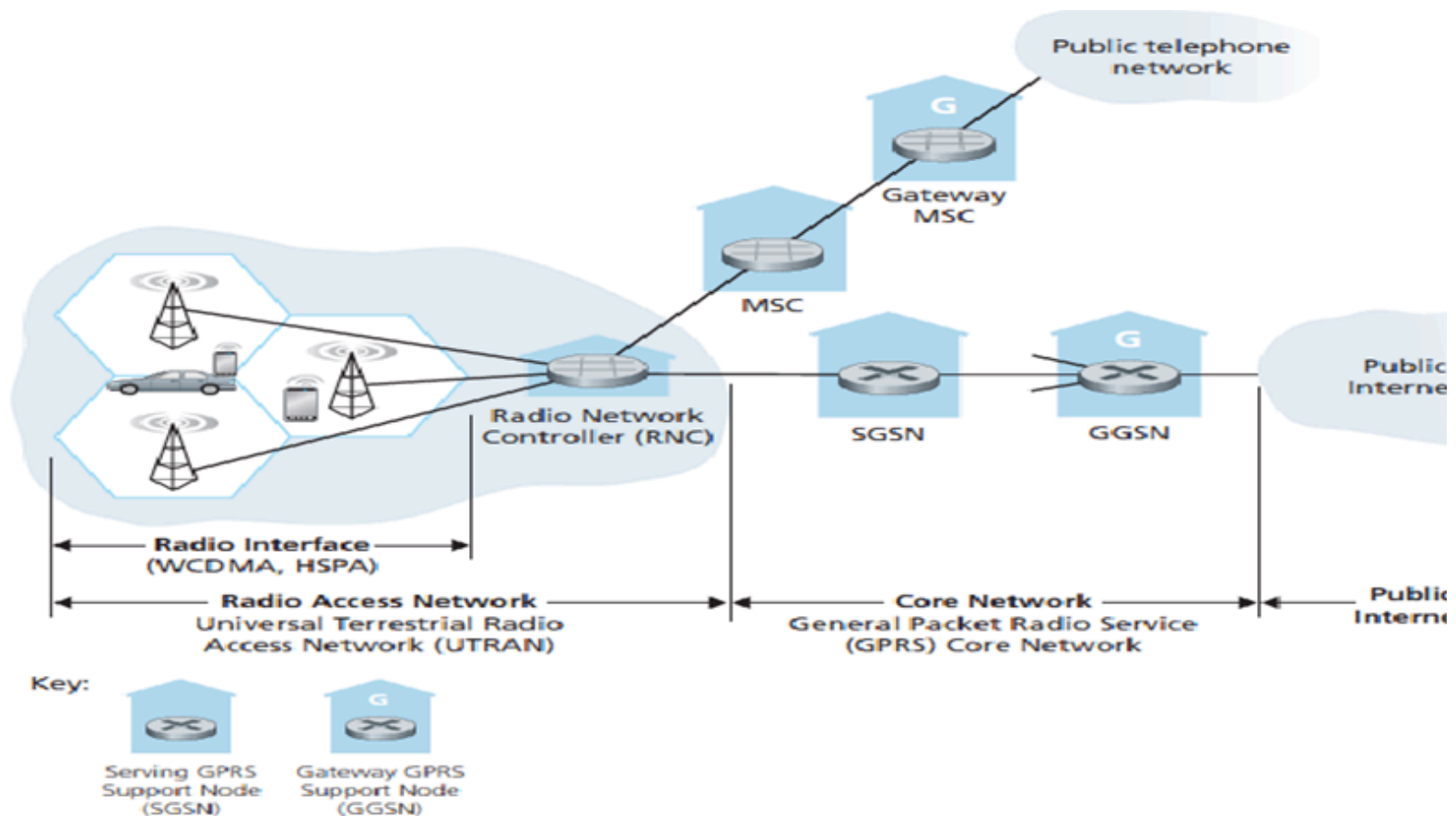
# Single-hop Infrastructure-base Wireless Access Networks

1. Cellular Networks
2. Wireless LANs
3. WiMAX



# Single-hop Infrastructure-based:: Cellular Networks

- 3G cellular data network connects radio access networks to the public Internet.



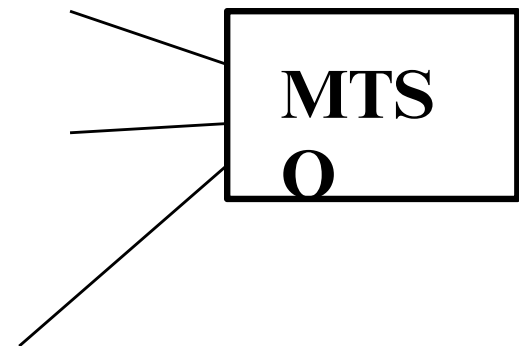
**Figure:** 3G cellular data network architecture

# Cellular Networks:

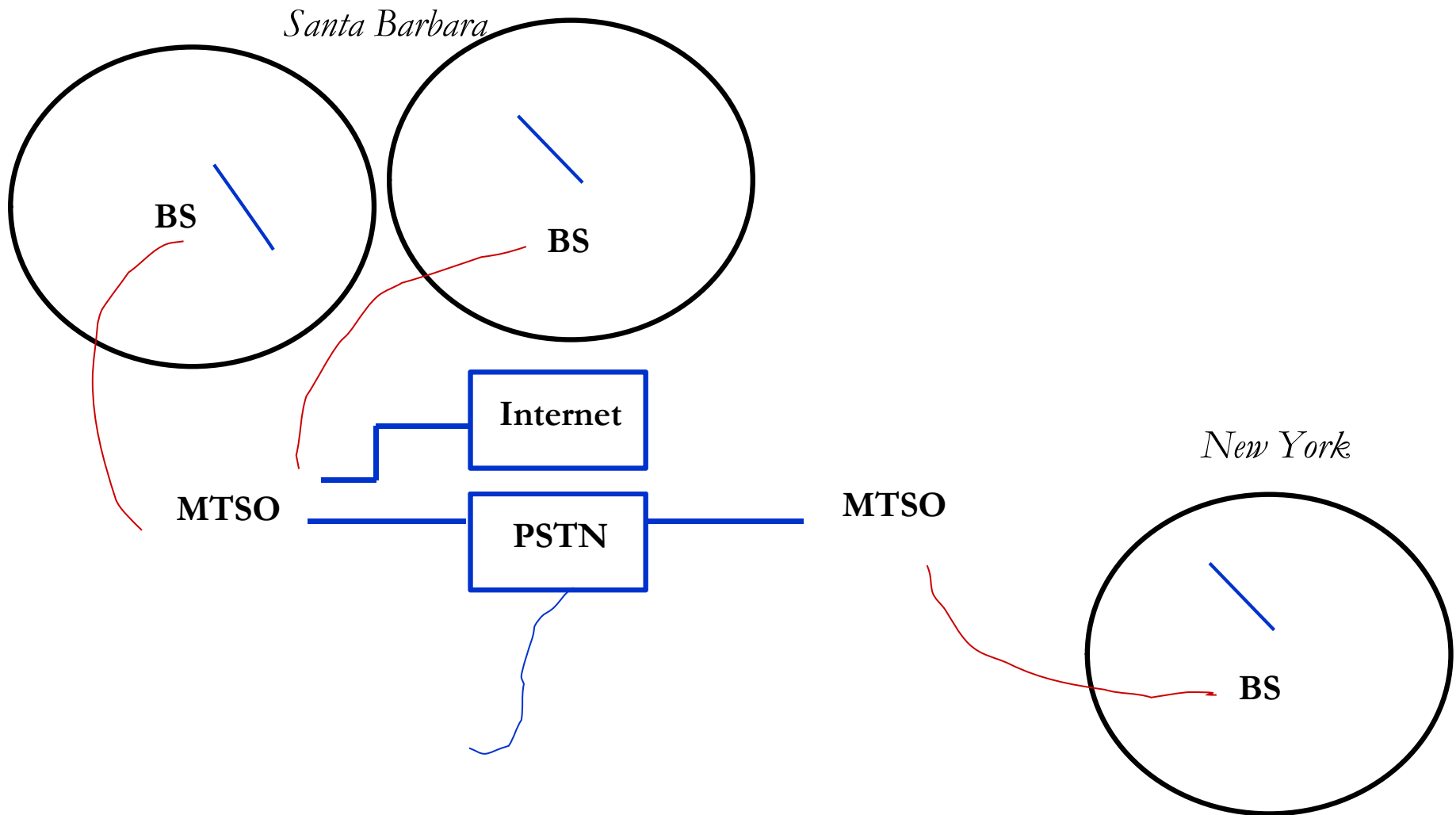
## Reuse channels to maximize capacity

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- Geographic region divided into cells
- Frequencies/timeslots/codes reused at spatially-separated locations.
- Co-channel interference between same color cells.
- Base stations/MTSOs coordinate handoff and control functions
- Shrinking cell size increases capacity, as well as networking burden



# Cellular Networks



# Cellular Design:

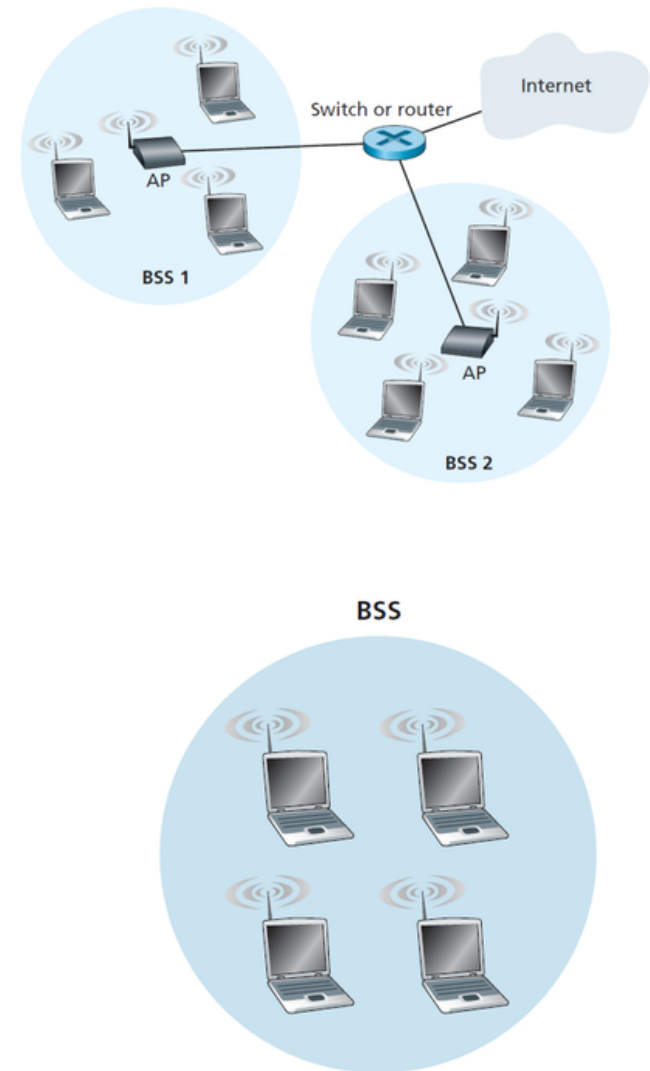
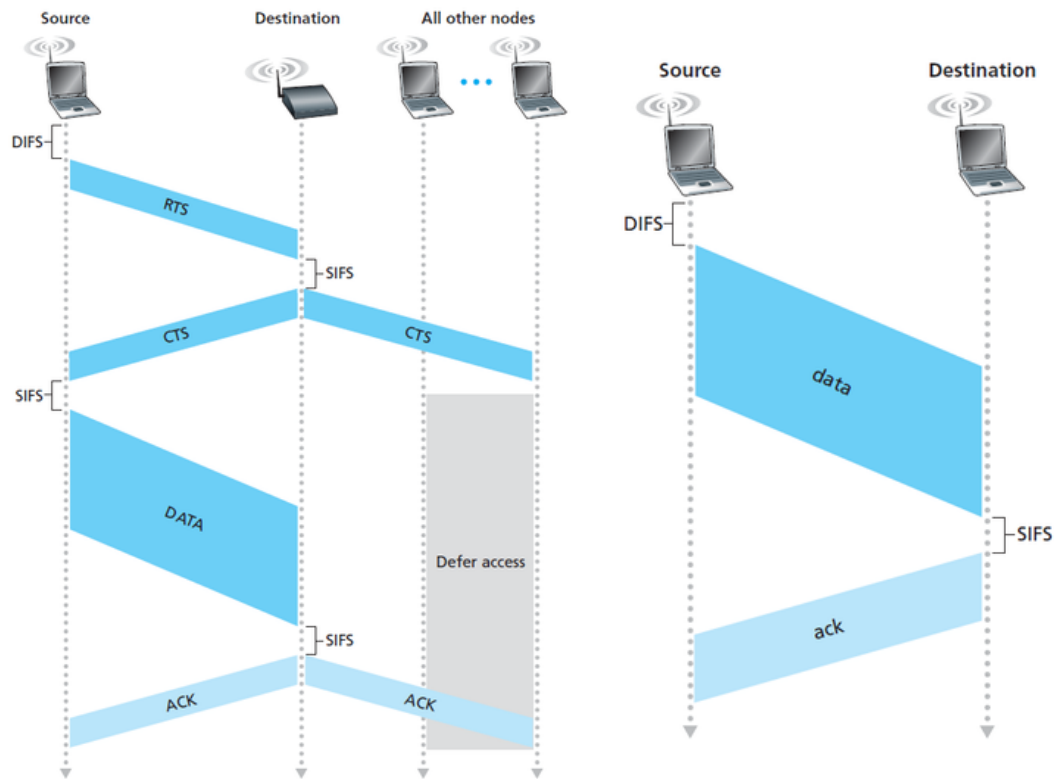
## Voice and Data

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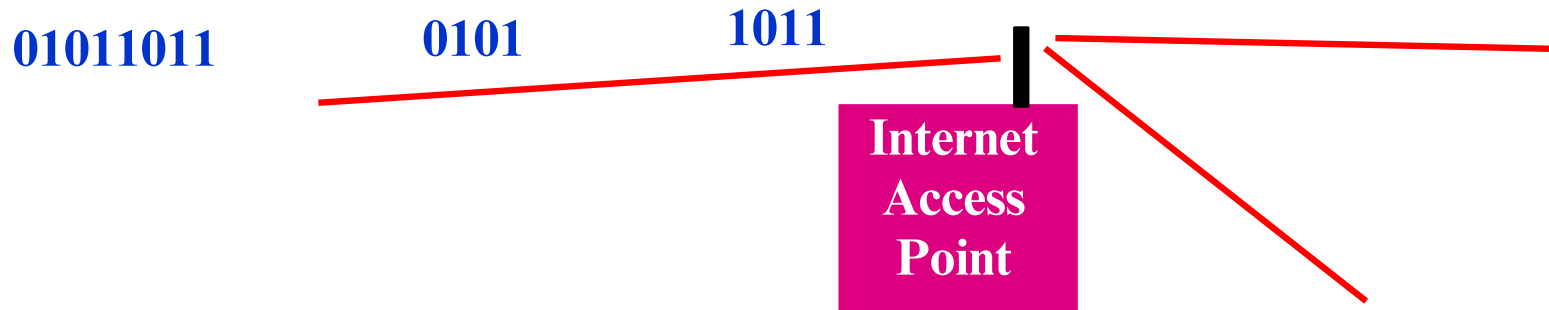
- Data is bursty, whereas voice is continuous
  - Typically require different access and routing strategies
- 3G “widens the data pipe”:
  - Standard based on wideband CDMA
  - Packet-based switching for both voice and data
- Evolution of wireless networks:
  - GSM
  - IS-95(CDMA)
  - CDMA 2000 / W-CDMA // IEEE 802.11a/g/b

# Single-hop Infrastructure-based:: Wireless LANs

| Standard | Frequency Range | Data Rate     |
|----------|-----------------|---------------|
| 802.11b  | 2.4–2.485 GHz   | up to 11 Mbps |
| 802.11a  | 5.1–5.8 GHz     | up to 54 Mbps |
| 802.11g  | 2.4–2.485 GHz   | up to 54 Mbps |



# Single-hop Infrastructure-based: Wireless LANs



- WLANs connect “local” computers
- Channel access is shared (random access)
- Backbone Internet provides best-effort service

# Single-hop Infrastructure-based:: Wireless LAN Standards

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- 802.11b
  - Standard for 2.4GHz ISM band (80 MHz)
  - Frequency hopped spread spectrum
  - 1.6-10 Mbps, 500 ft range
- 802.11a
  - Standard for 5GHz NII band (300 MHz)
  - OFDM with time division
  - 20-70 Mbps, variable range
  - Similar to HiperLAN in Europe
- 802.11g
  - Standard in 2.4 GHz and 5 GHz bands
  - OFDM
  - Speeds up to 54 Mbps



# Single-hop Infrastructure-based:: Wireless LAN Standards

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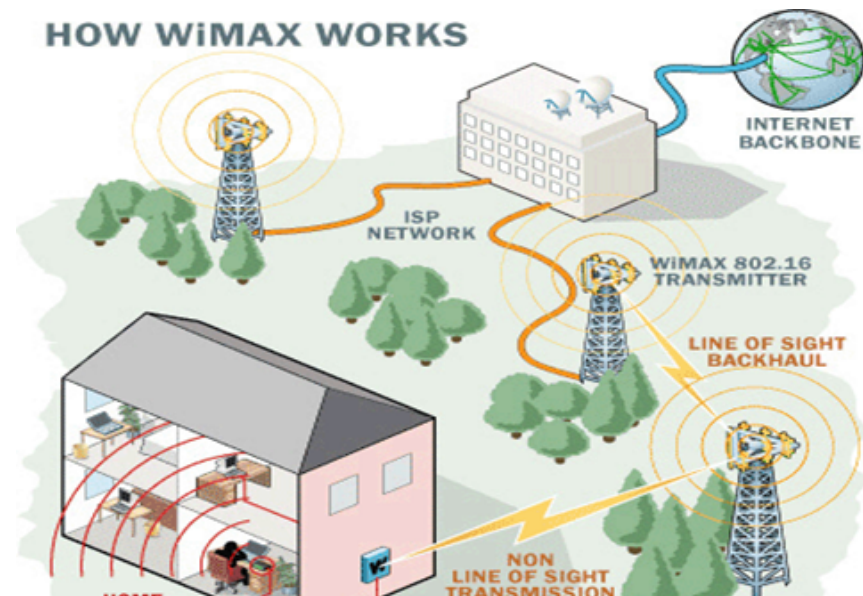
## Data rate :

- IEEE 802.11b: 11 Mbps
- IEEE 802.11g: 54 Mbps
- IEEE 802.11n (Wi-Fi 4): Up to 600 Mbps
- IEEE 802.11ac (Wi-Fi 5): Up to 6.9 Gbps
- IEEE 802.11ax (Wi-Fi 6/6E): Up to 9.6 Gbps
- IEEE 802.11be (Wi-Fi 7) (emerging): Targeting >30 Gbps

# Single-hop Infrastructure-based:: WiMAX

(Worldwide Interoperability for Microwave Access)

- A wireless communications standard - designed to provide 30 to 40 megabit-per-second data rates, with the 2011 update providing up to 1 Gbit/s for fixed stations.
- It refers to interoperable implementations of the IEEE 802.16 family of wireless-networks standards.



**Figure:** Working Procedure of WiMAX

# Single-hop Infrastructure-based::

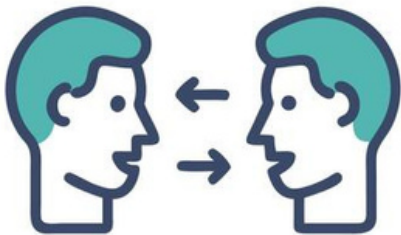
## WiMAX

- Provide portable mobile broadband connectivity across cities and countries
- Wireless alternative to cable and digital subscriber line (DSL) for "last mile" broadband access
- Provide data, telecommunications (VoIP) and IPTV services (triple play)
- A source of Internet connectivity as part of a business continuity plan

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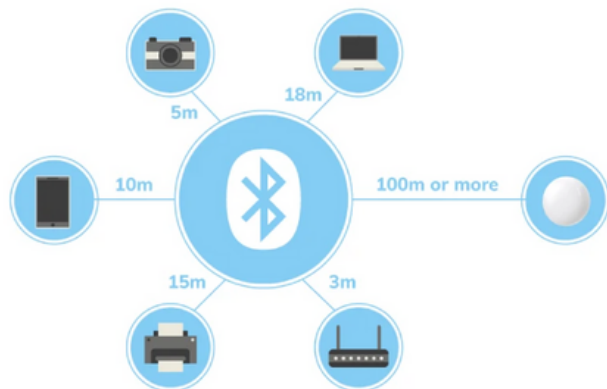
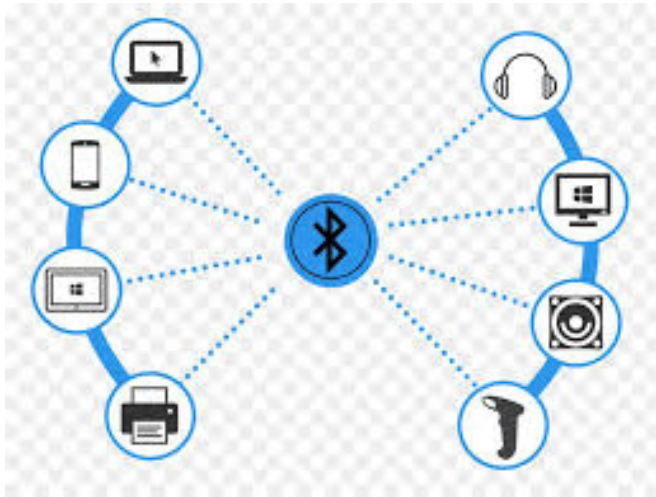
# Single-hop Infrastructure-less Wireless Access Networks

## 4. Bluetooth



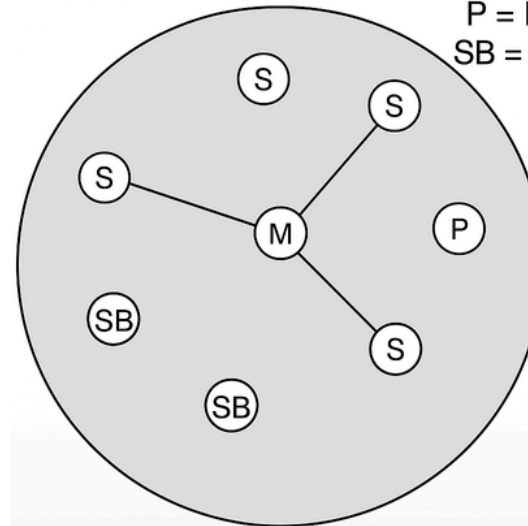
# Single-hop Infrastructure-less::

## Bluetooth Networks



Piconet

M = Master  
S = Slave  
P = Parked  
SB = Standby



### Master – Slave Communication

only the Master can initiate and control communication within a piconet.

Parked Node:  
actively participating

still synchronized

not

Standby Node:

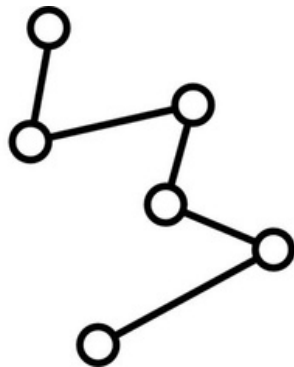
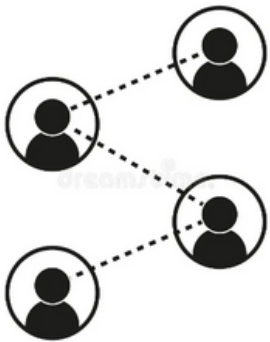
not connected

# Single-hop Infrastructure-less: Bluetooth Networks

| Property         | Description                                                             |
|------------------|-------------------------------------------------------------------------|
| Frequency Band   | 2.4 GHz ISM band (2.402 – 2.480 GHz)                                    |
| Technology       | Frequency Hopping Spread Spectrum (FHSS), 1600 hops/sec                 |
| Network Topology | Piconet (1 master + up to 7 active slaves)                              |
| Data Rate        | Up to 721 Kbps (Classic)                                                |
| Range            | ~10 meters (Class 2), up to 100 meters (Class 1)                        |
| Roles            | Master (initiates & controls communication), Slave (responds to master) |
| Security         | Authentication, Authorization, Encryption                               |
| Voice Support    | Up to 3 simultaneous SCO voice channels                                 |
| Device States    | Active, Parked, Standby                                                 |

# Multi-hop Infrastructure-base Wireless Access Networks

## 5. Wireless Mesh Networks (semi-structured)



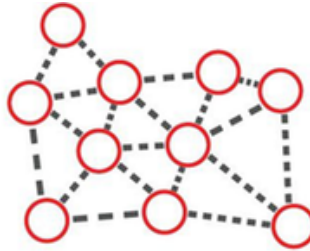
# Multi-hop Infrastructure-based: Wireless Mesh Networks



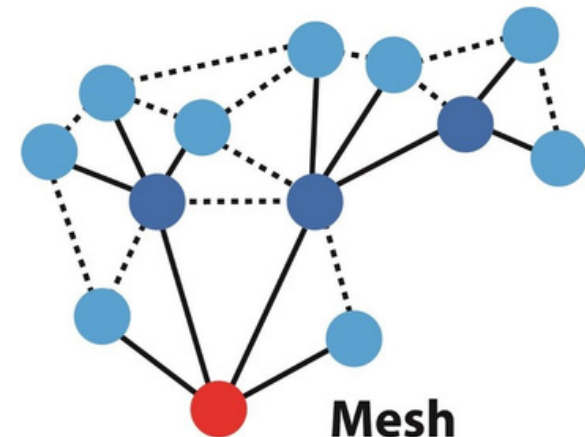
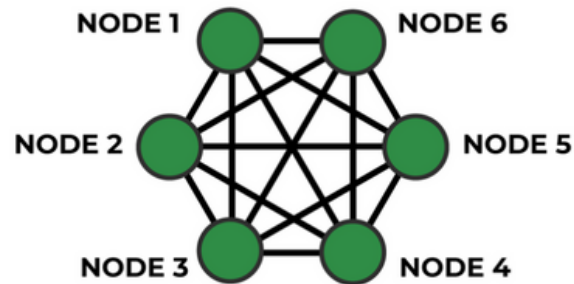
Point to point



Star



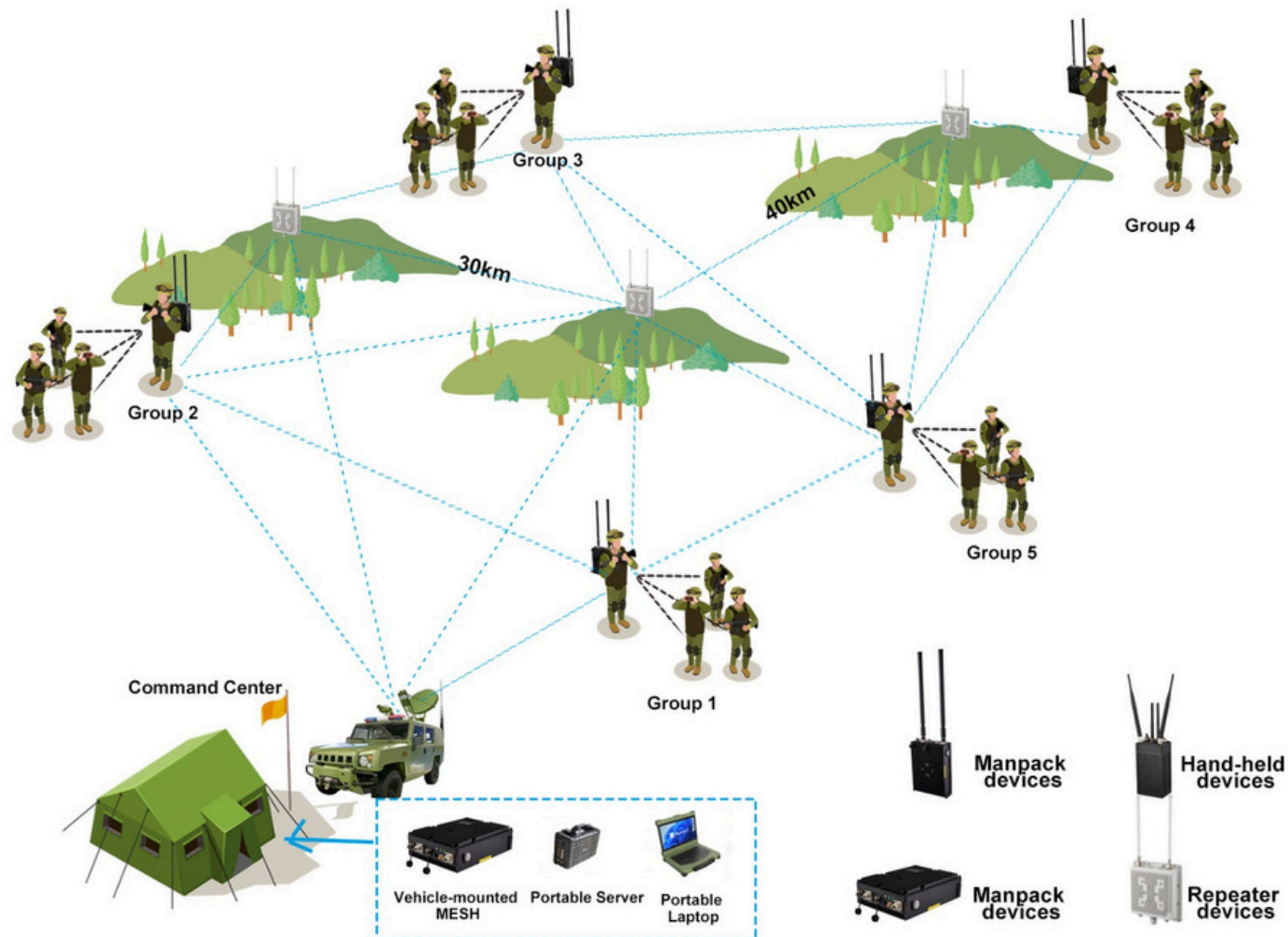
Mesh





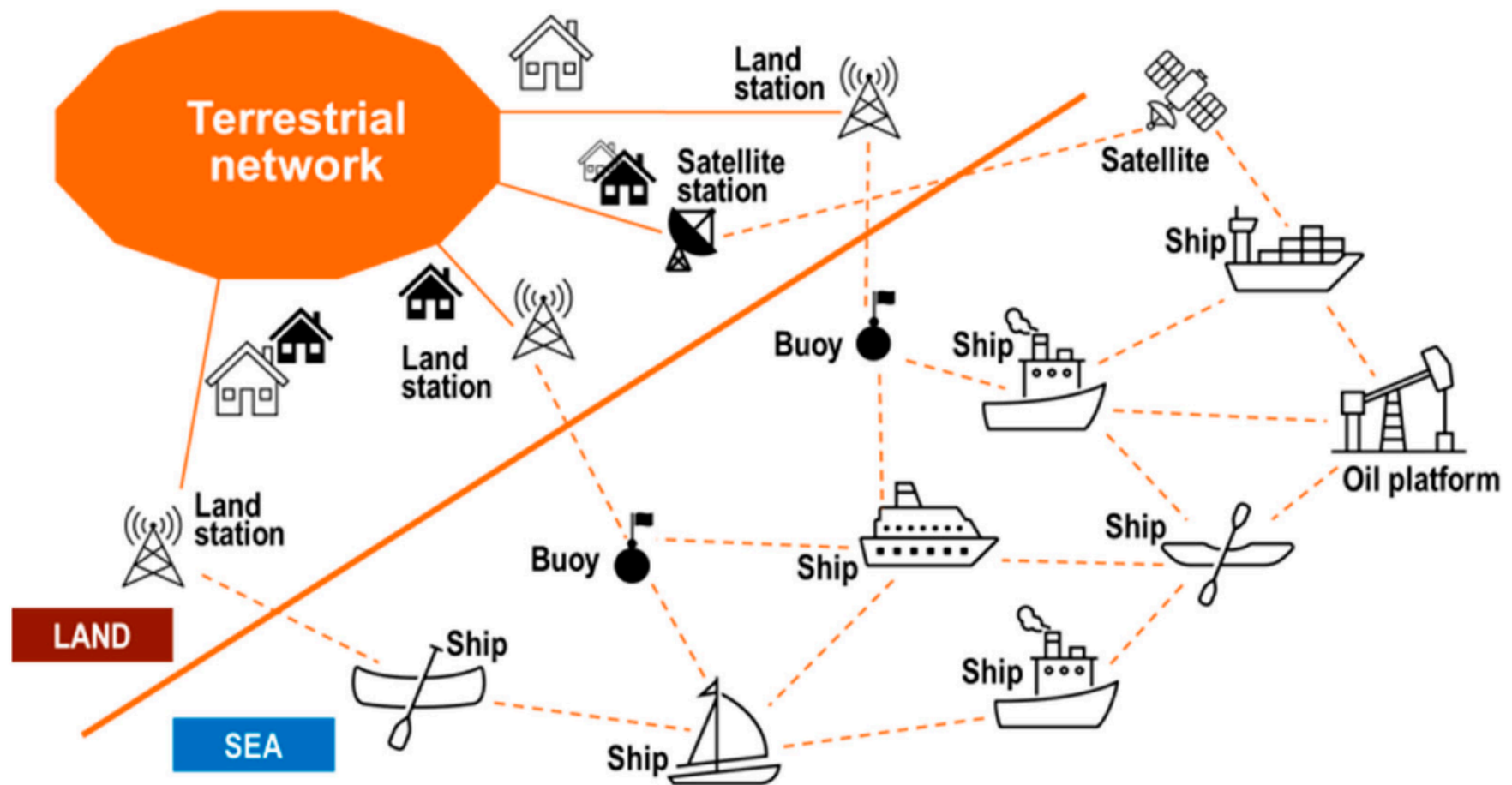
# Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Military battlefield communication



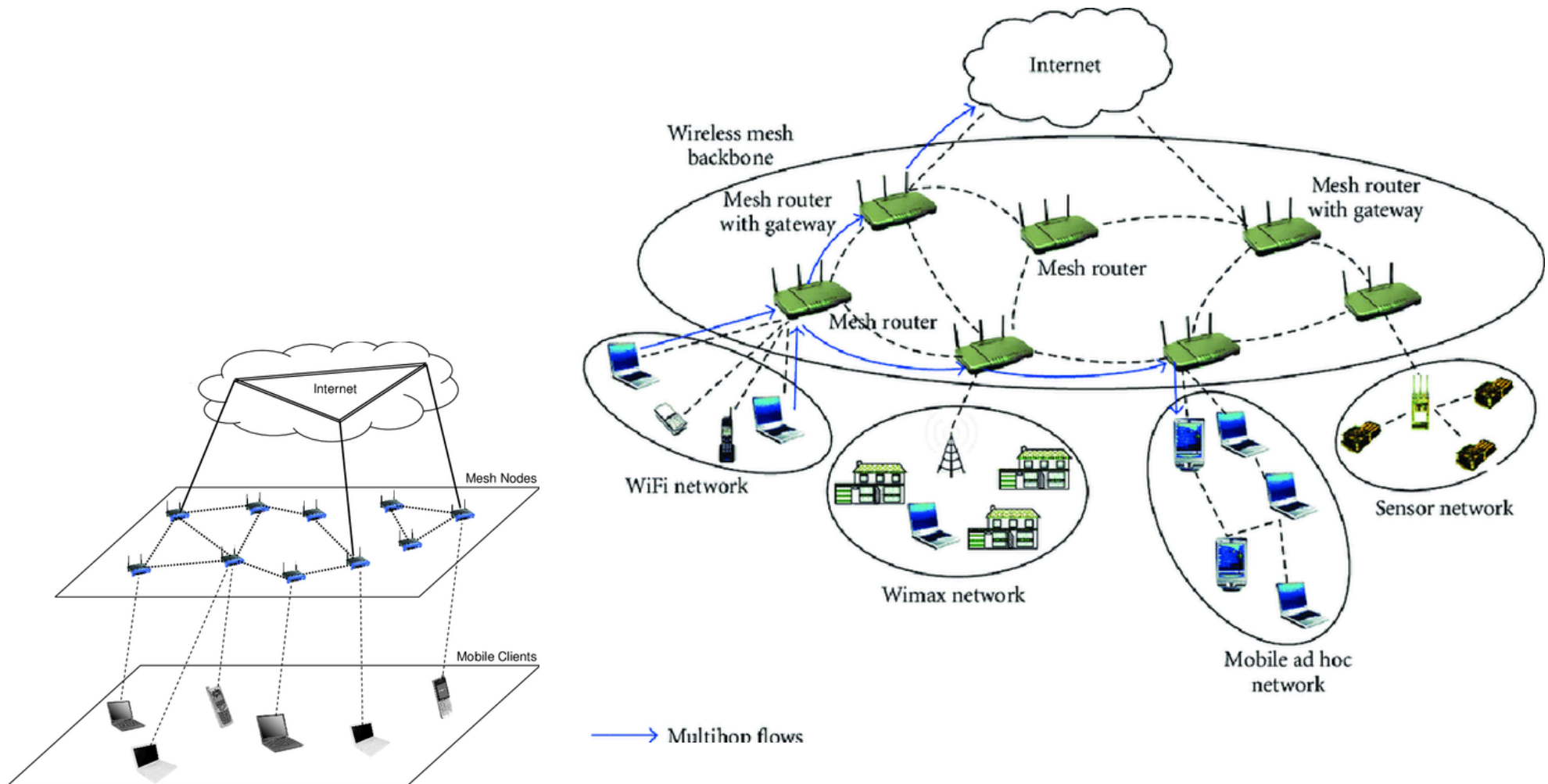
# Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Maritime Mesh Network



# Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Hybrid Wireless Mesh Network for Backbone for Community Broadband Connectivity



# Multi-hop Infrastructure-based: Wireless Mesh Networks

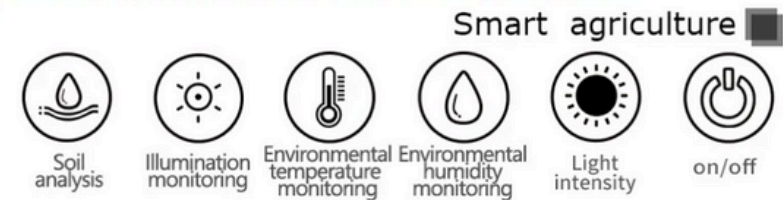
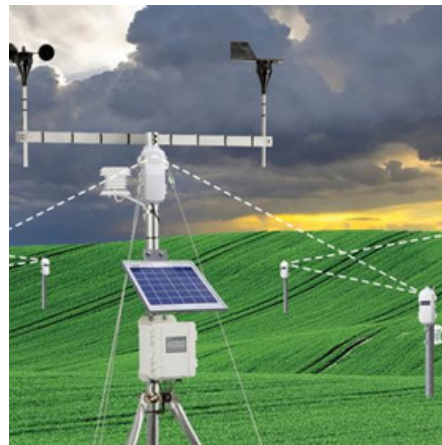
Application: Drone Mesh Networks (UAV Mesh Networks)





# Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Very Large-Scale Agricultural Monitoring with Mobile Machineries



# Multi-hop Infrastructure-based: Wireless Mesh Networks

Application: Smart & Connected City with Mobile & Static Devices



# Multi-hop Infrastructure-based: Wireless Mesh Networks

## How WMNs Work

### 1. Network Structure

WMNs consist of two types of devices:

- **Mesh Routers (MRs):**

- Fixed nodes that form the backbone of the mesh.
- Often connected to the internet.

- **Mesh Clients (MCs):**

- End-user devices like laptops, phones, sensors.
- Connect to the mesh routers or other clients.

### 2. Multi-hop Routing

- Data travels **hop-by-hop** through multiple mesh routers to reach its destination.
- If a direct path is not available, packets are forwarded through **intermediate nodes**.

### 3. Self-Healing Capability

- If a node fails or a link breaks, the network **automatically reroutes** traffic through alternative paths.
- Ensures **high reliability and robustness**.

### 4. Self-Configuration

- New nodes can join the network and start routing **without manual configuration**.
- Each node participates in the routing process and **discovers neighbors dynamically**.



# Multi-hop Infrastructure-based: Wireless Mesh Networks



|  |  |
|--|--|
|  |  |
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|  |  |
|  |  |



# Multi-hop Infrastructure-based::

## Wireless Mesh Networks



### 1. Interference and Shared Medium

**Problem:** Nodes use a shared wireless medium, leading to co-channel interference, hidden terminal, and exposed terminal problems.

**Impact:** Reduces throughput and increases collisions.

### 2. Scalability

**Problem:** As the number of nodes increases, routing overhead and contention grow.

**Impact:** Performance degrades due to increased latency and reduced bandwidth per node.

### 3. Dynamic Topology & Routing Complexity

**Problem:** Nodes may join, leave, or move, causing frequent route breaks and topology changes.

**Impact:** Requires adaptive routing protocols and fast convergence.

### 4. Limited Bandwidth and Throughput

**Problem:** Each additional hop in a path reduces available bandwidth due to medium reuse.

**Impact:** End-to-end throughput suffers, especially in large mesh networks.

# Multi-hop Infrastructure-based::

## Wireless Mesh Networks



### 5. Security Threats

**Problem:** WMNs are vulnerable to eavesdropping, spoofing, DoS attacks, and routing attacks.

**Impact:** Compromises confidentiality, integrity, and availability.

### 6. Quality of Service (QoS)

**Problem:** Hard to guarantee latency, jitter, and packet delivery ratio over unreliable, shared wireless links.

**Impact:** Affects performance of real-time applications like VoIP and video streaming

### 7. Synchronization and Coordination

**Problem:** Distributed nature of WMNs makes scheduling transmissions and synchronizing clocks challenging.

**Impact:** Causes collisions, inefficient channel use, and poor coordination.

### 8. Heterogeneity of Devices

**Problem:** WMNs may include devices with different capabilities, interfaces, and protocols.

**Impact:** Interoperability issues arise, complicating routing and communication.

# Multi-hop Infrastructure-based: Wireless Mesh Networks

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Watch: [How do Mesh Networks Works?](#)

Watch: [Mesh Networking Idea](#)

Watch: [Mesh WiFi Vs. WiFi Extenders](#)

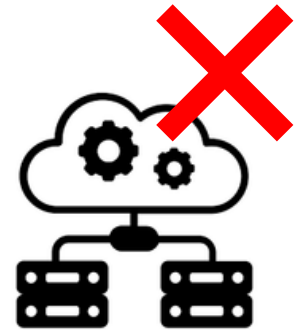
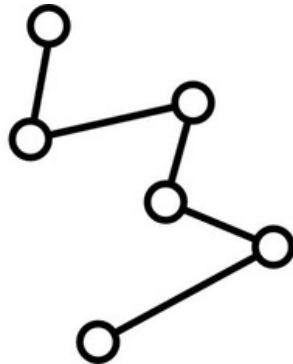
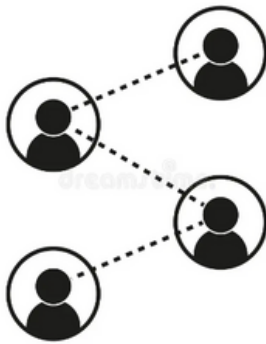
# Multi-hop Infrastructure-less Wireless Access Networks

## 6. Ad hoc wireless Networks

### 6.1. Wireless Sensor Networks

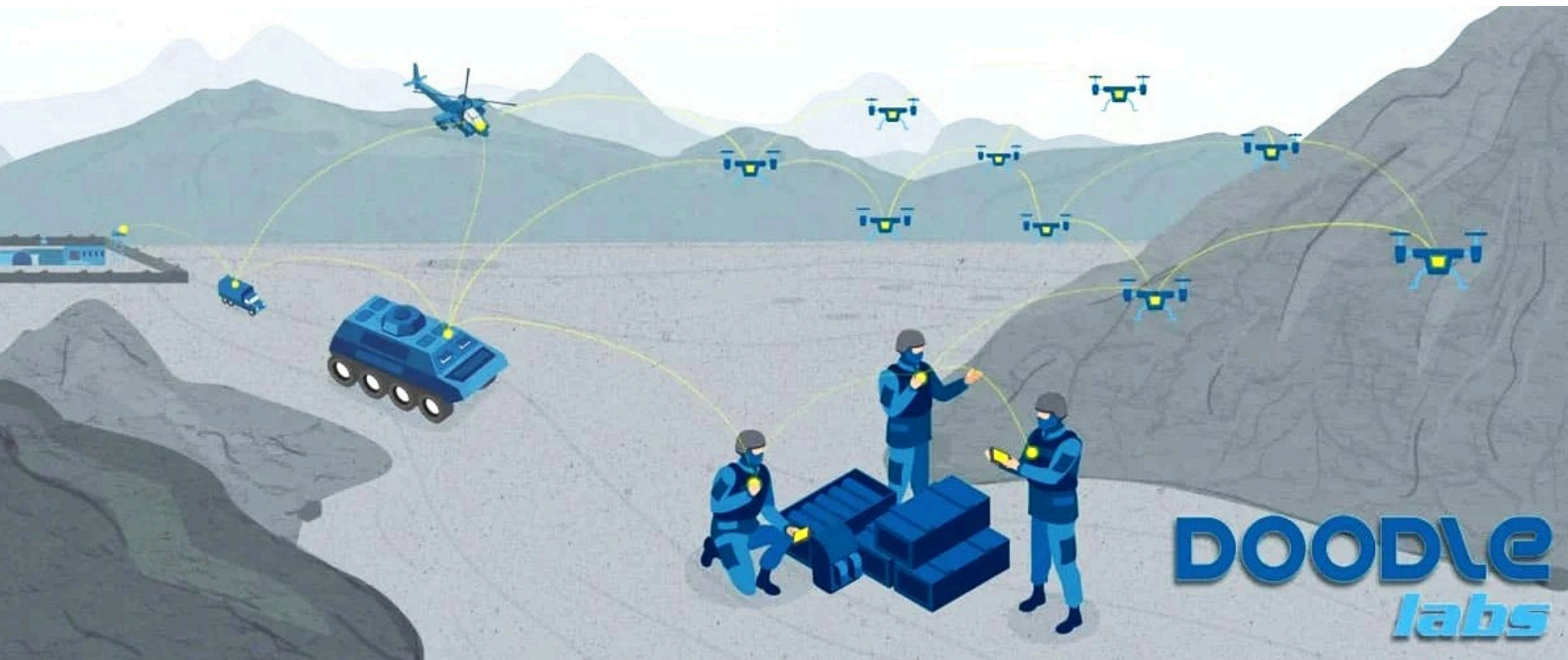
### 6.2. Mobile Ad-hoc Networks / Vehicular Ad-hoc Networks

### 6.3. Delay Tolerant Networks



# Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

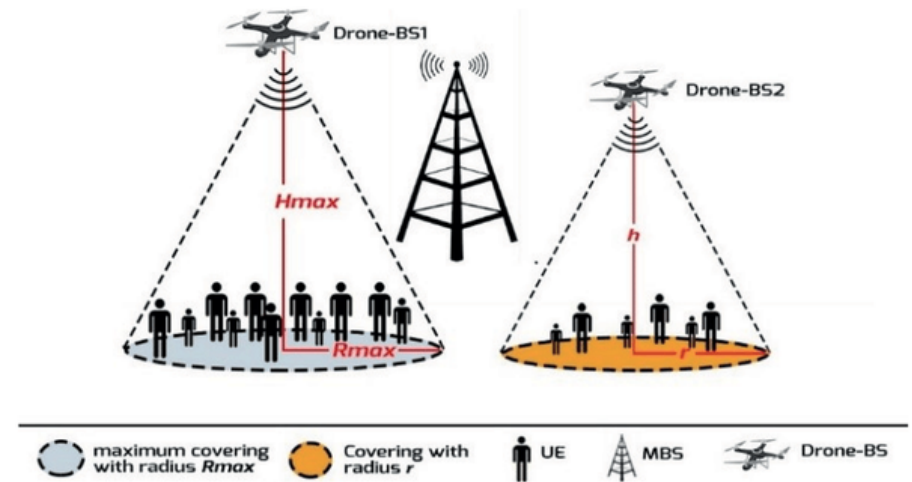
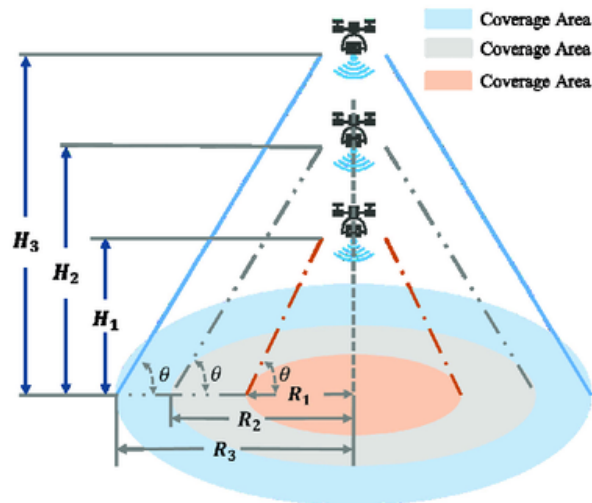
Application: Military battlefield communication in Ad-Hoc Mode



# Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

## Application: Drone Ad-Hoc Networks (UAV Ad-Hoc Networks)

- Temporary Internet Coverage



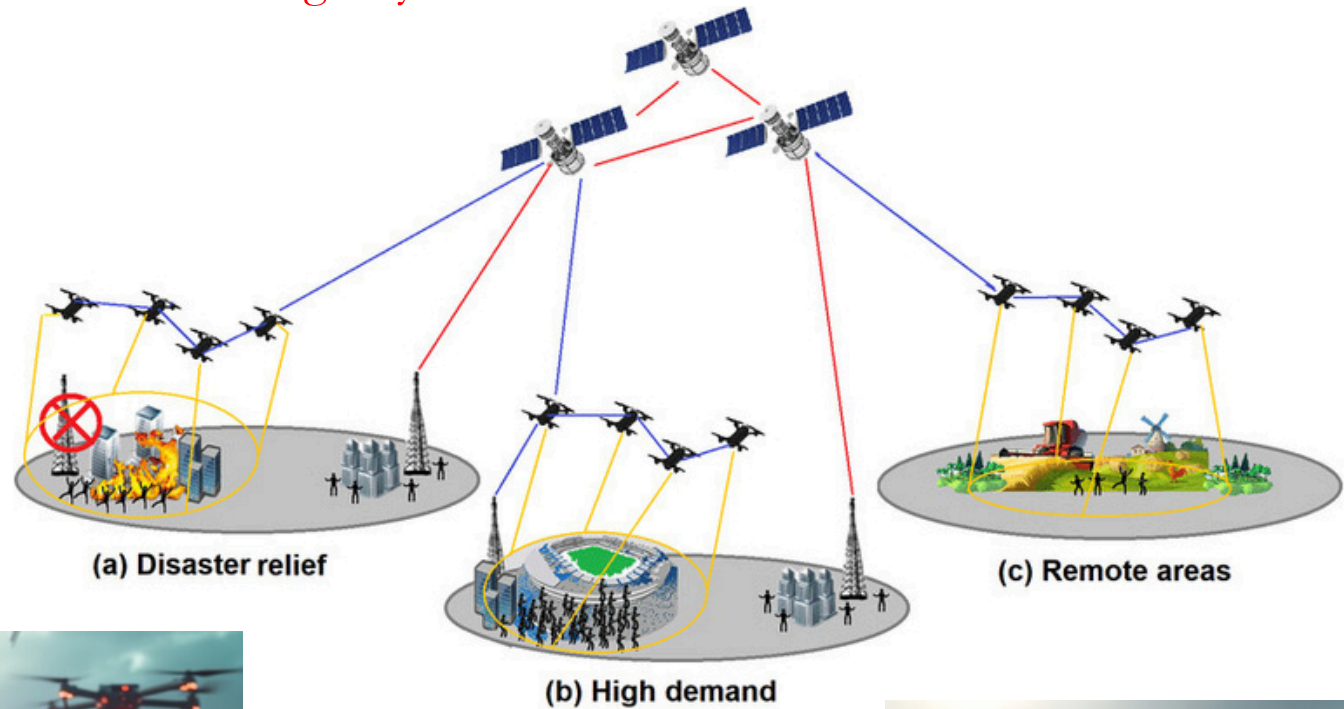
- Search and Rescue in Remote Areas
- Environmental and Wildlife Monitoring
- Disaster Response & Emergency Communications





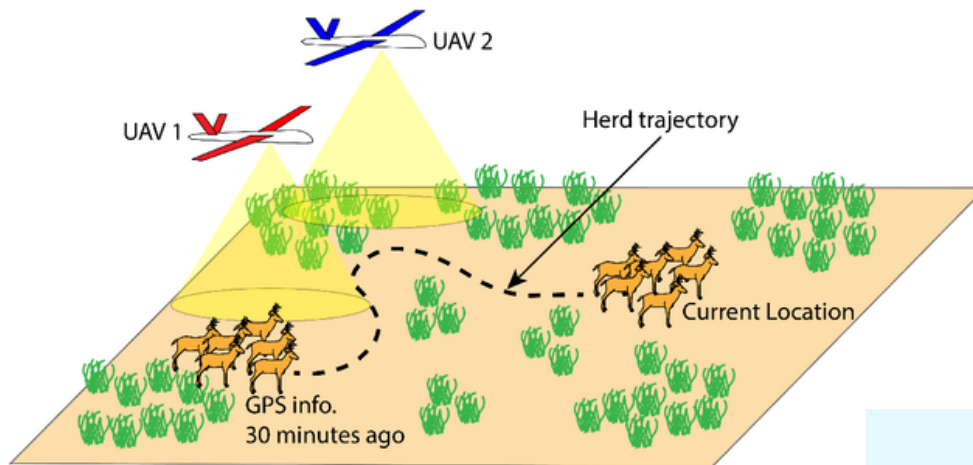
# Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

Application: Search and Rescue & Emergency Communications in Remote Areas



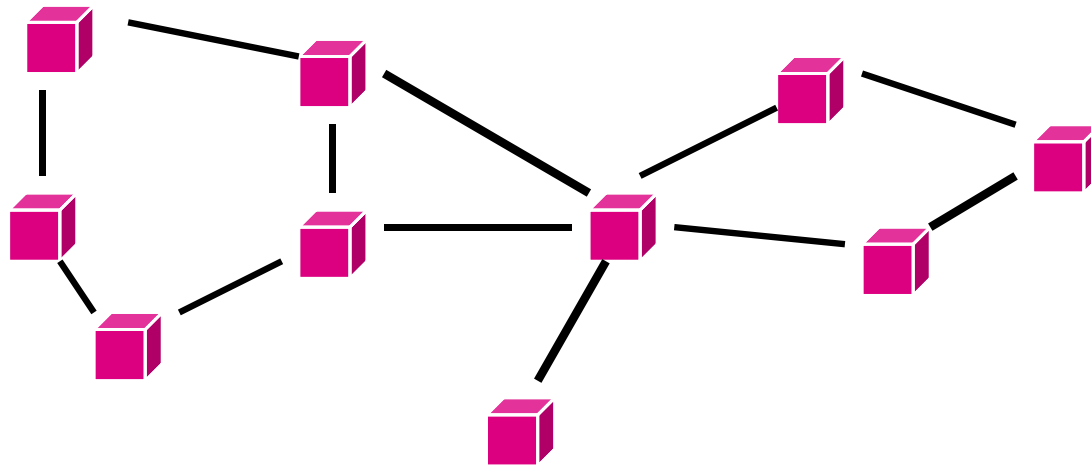
# Multi-hop Infrastructure-less: Wireless Ad-Hoc Networks

Application: Environmental and Wildlife Monitoring





# Multi-hop Infrastructure-less: Ad-Hoc Networks



- Peer-to-peer communications.
- No backbone infrastructure.
- Routing can be multi-hop.
- Topology is dynamic.
- Fully connected with different link SINRs

# Multi-hop Infrastructure-less::

## Ad-Hoc Networks

- Ad-hoc networks provide a flexible network infrastructure for many emerging applications.
- The capacity of such networks is generally unknown.
- Transmission, access, and routing strategies for ad-hoc networks are generally ad-hoc.
- Cross layer design critical are very challenging.
- Energy constraints impose interesting design tradeoffs for communication and networking.

# Multi-hop Infrastructure-less:: Wireless Mesh Network vs Wireless Ad-Hoc Networks

| Aspect         | Wireless Mesh Network (WMN)                                               | Wireless Ad-hoc Network (WAHN)                           |
|----------------|---------------------------------------------------------------------------|----------------------------------------------------------|
| Infrastructure | Semi-infrastructure (uses fixed mesh routers as backbone)                 | Purely infrastructure-less                               |
| Node Roles     | Distinct roles: <b>Mesh routers</b> (static) and <b>clients</b> (dynamic) | All nodes are equal; any node can route data             |
| Deployment     | Pre-planned, often permanent                                              | Temporary, spontaneous setup                             |
| Routing        | Stable and optimized (e.g., HWMP, OLSR)                                   | Dynamic and reactive (e.g., AODV, DSR)                   |
| Scalability    | Highly scalable due to backbone support                                   | Limited scalability due to overhead and congestion       |
| Mobility       | Clients may be mobile; routers are often fixed                            | All nodes may be mobile                                  |
| Reliability    | More reliable due to fixed routers and redundant paths                    | Less reliable, frequent topology changes                 |
| Use Cases      | <b>Community networks, smart cities, campus-wide networks</b>             | <b>Military ops, disaster relief, emergency response</b> |

# Multi-hop Infrastructure-less:: Wireless Mesh Network vs Wireless Ad-Hoc Networks

| Aspect             | Wireless Mesh Network (WMN)                                     | Wireless Ad-hoc Network (WAHN)                                 |
|--------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| Radio Interfaces   | Often multi-radio and multi-channel (e.g., 802.11s, WiMAX, LTE) | Usually single-radio, single-channel (e.g., 802.11, Bluetooth) |
| Backhaul Support   | Yes, mesh routers may connect to the Internet or backbone       | No backhaul, only peer-to-peer communication                   |
| Gateway Capability | Mesh routers often serve as gateways to external networks       | No defined gateway; all nodes are peers                        |
| Energy Source      | Often powered (e.g., fixed routers)                             | Often battery-operated (mobile nodes)                          |

# Multi-hop Infrastructure-less:: Wireless Mesh Network vs Wireless Ad-Hoc Networks

| Aspect            | Wireless Mesh Network (WMN)                                                  | Wireless Ad-hoc Network (WAHN)                                                   |
|-------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Primary Goal      | Provide last-mile connectivity and reliable internet access over large areas | Enable peer-to-peer communication in infrastructure-less and mobile environments |
| Design Focus      | Stability, scalability, and long-term connectivity                           | Quick deployment, flexibility, and dynamic adaptability                          |
| Application Focus | Community WiFi, smart cities, industrial IoT, public safety                  | Military operations, disaster relief, vehicular networks                         |
| Traffic Pattern   | Typically many-to-one or client-gateway                                      | Dynamic patterns; often all-to-all or many-to-many                               |
| Performance Goal  | High throughput, fault tolerance, consistent QoS                             | Fast topology adaptation, minimal delay in dynamic environments                  |

# Multi-hop Infrastructure-less:: Wireless Ad-Hoc Networks

## 6. Ad hoc wireless Networks

### 6.1. Wireless Sensor Networks

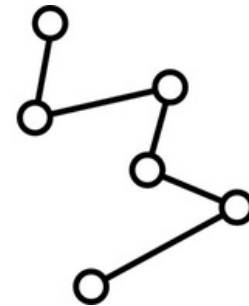
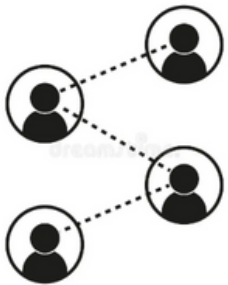
### 6.2. Mobile Ad-hoc Networks /Vehicular Ad-hoc Networks

### 6.3. Delay Tolerant Networks

| Feature                          | WSN                           | VANET                       | DTN                                  |
|----------------------------------|-------------------------------|-----------------------------|--------------------------------------|
| Infrastructure-less (or minimal) | ✓                             | ✓                           | ✓                                    |
| Multi-hop communication          | ✓                             | ✓                           | ✓                                    |
| Dynamic topology                 | ✓ (due to node failure/sleep) | ✓ (due to vehicle mobility) | ✓ (due to intermittent connectivity) |
| Self-organizing                  | ✓                             | ✓                           | ✓                                    |
| Decentralized control            | ✓                             | ✓                           | ✓                                    |

# Wireless Sensor Networks

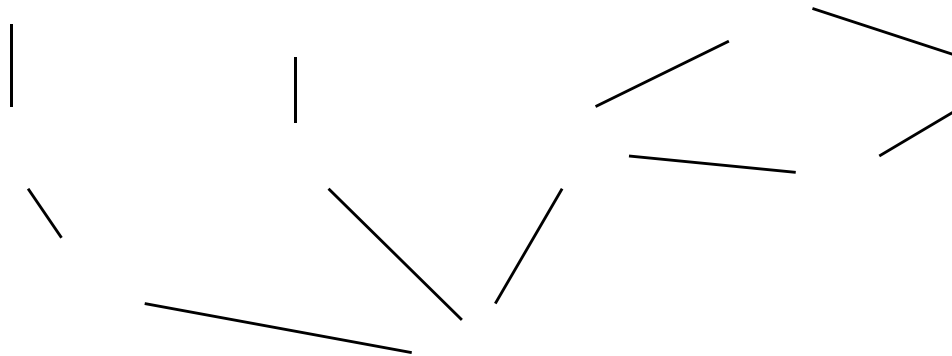
( Structured / Semi-Structured / Un-Structed :: Depending on Application Nature)



# Multi-hop Infrastructure-less: Wireless Sensor Networks

*Energy is the primary constraint*

Low power Wireless Sensor Network (Low PAN)[IEEE 802.15]



- Nodes powered by non rechargeable batteries
- Data flows usually to centralized location.
- Low per-node rates but up to 100,000 nodes.
- Data highly correlated in time and space.
- Nodes can cooperate in transmission, reception, compression, and signal processing.

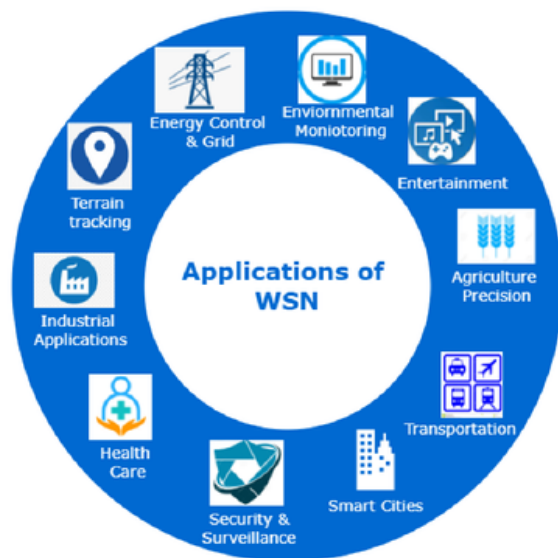


# Sensor Networks

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- Short-range networks must consider transmit, circuit, and processing energy.
  - Sophisticated techniques not necessarily energy-efficient.
  - Sleep modes save energy but complicate networking.
- Changes **everything** about the network design:
  - Delay vs. throughput vs. node/network lifetime tradeoffs.
  - Optimization of node cooperation.

# Sensor Networks :: Applications



# Sensor Networks :: Applications

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Constant monitoring & detection of specific events

Military, battlefield surveillance

Forest fire & flood detection

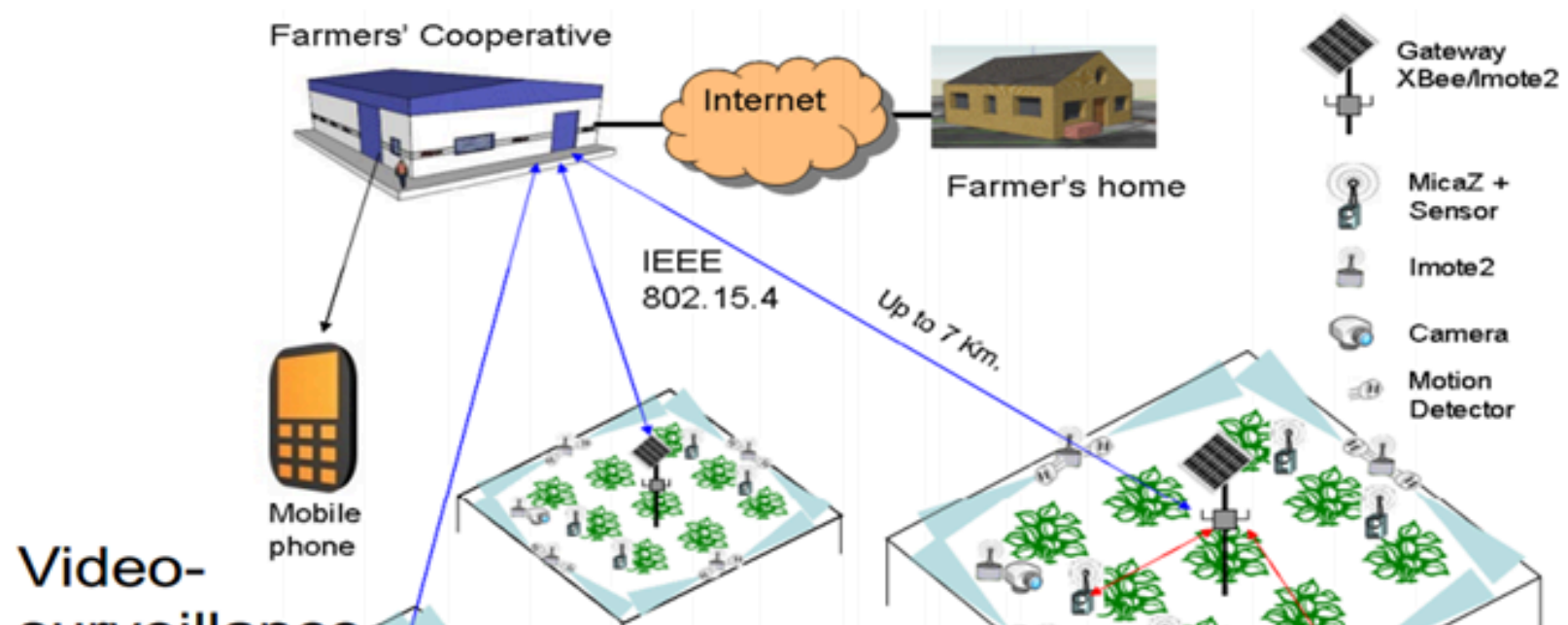
Habitat exploration of animals

Patient monitoring

Home appliances

# Sensor Networks :: Applications

## Precision Agriculture: (Structured / Semi-Structured )



**Figure:** An example of Low PAN Application

# Sensor Networks :: Applications

Remote deployment of  
sensors for **tactical monitoring**  
of enemy troop movements.

# Sensor Networks :: Applications

---

## Smart Home/Smart Office: (Structured)

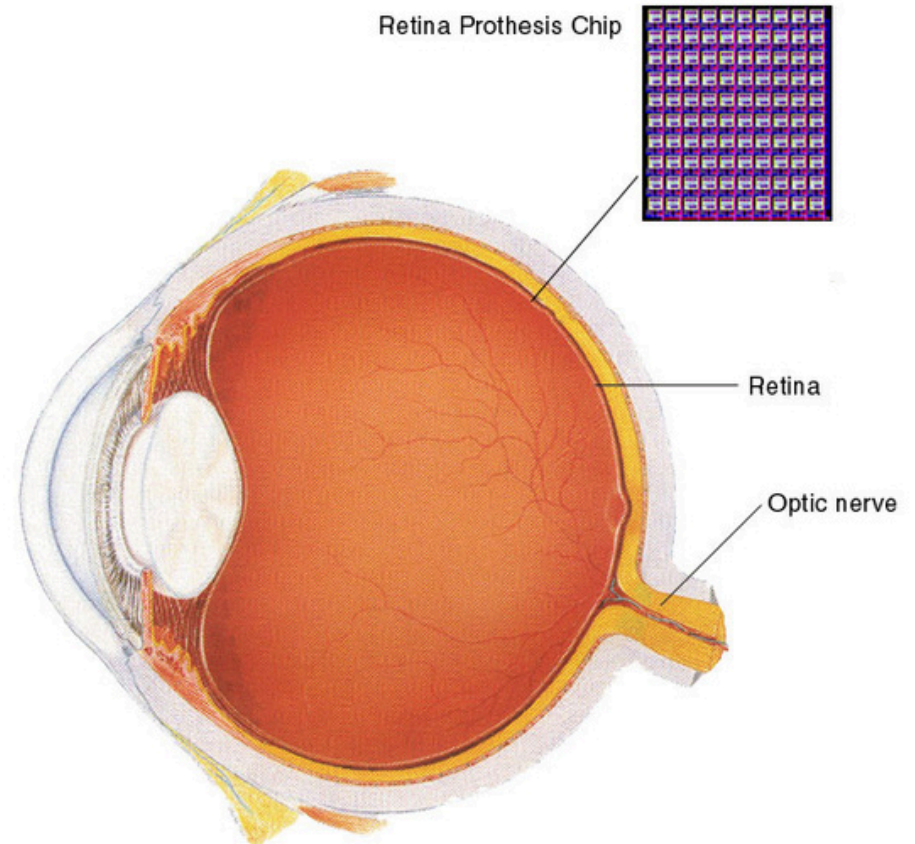


- Sensors controlling appliances and electrical devices in the house
- Better lighting and heating in office buildings

# Sensor Networks :: Applications

## Medical/Biomedical: Body Area Network: (Structured)

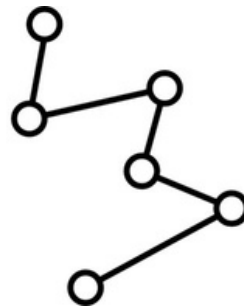
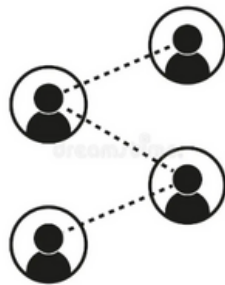
- Health Monitors
  - Glucose
  - Heart rate
  - Cancer detection
- Chronic Diseases
  - Artificial retina
  - Cochlear implants
- Hospital Sensors
  - Monitor vital signs
  - Record anomalies



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# Delay/Disruption Tolerant Network (DTN)

(Un-Structed )





# Delay/Disruption Tolerant Network (DTN)

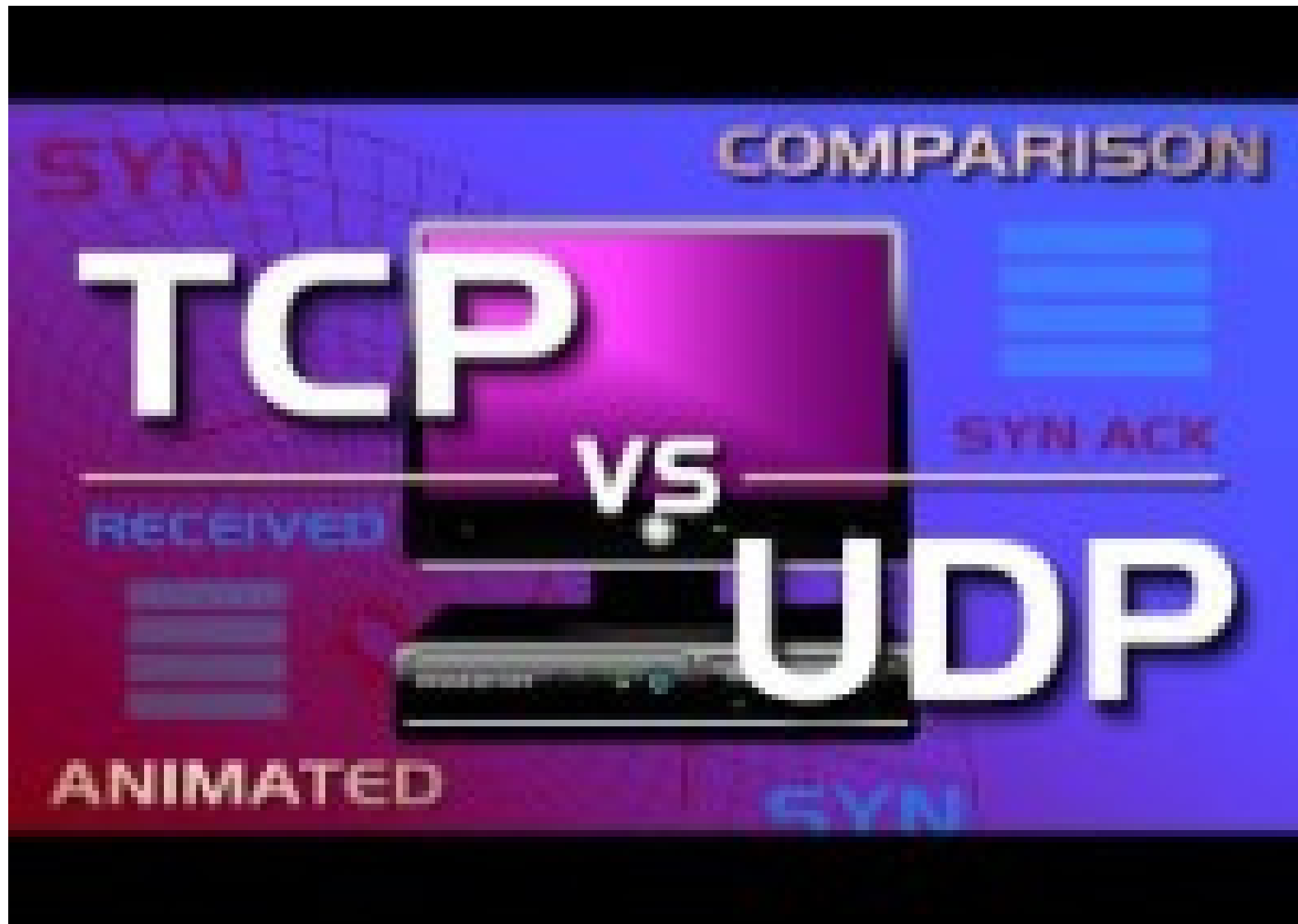
Standard data exchange over the internet



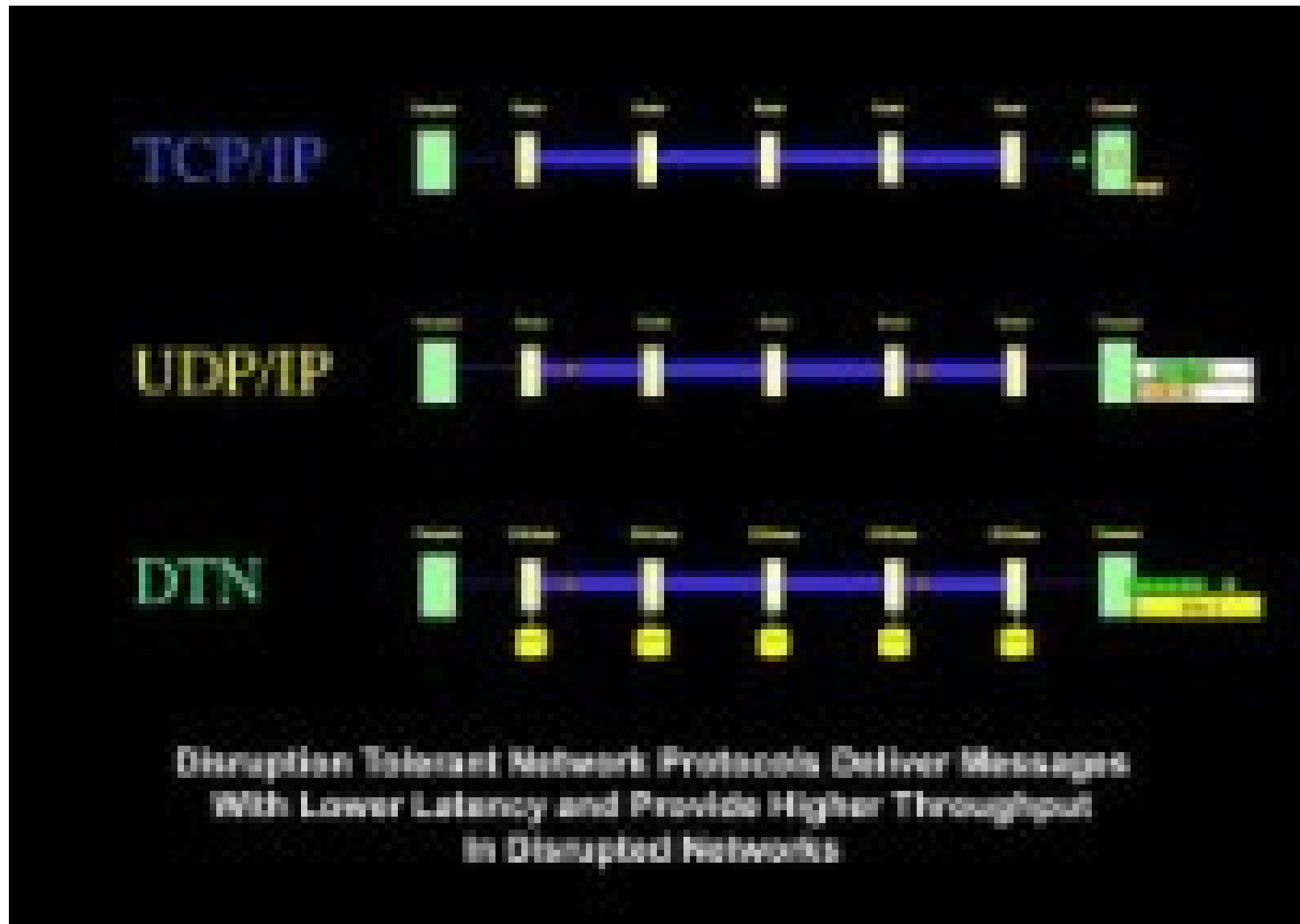
Data exchange using DTN



# Understanding the TCP and UDP



# TCP, UDP and DTN



# Delay Tolerant Network (DTN)

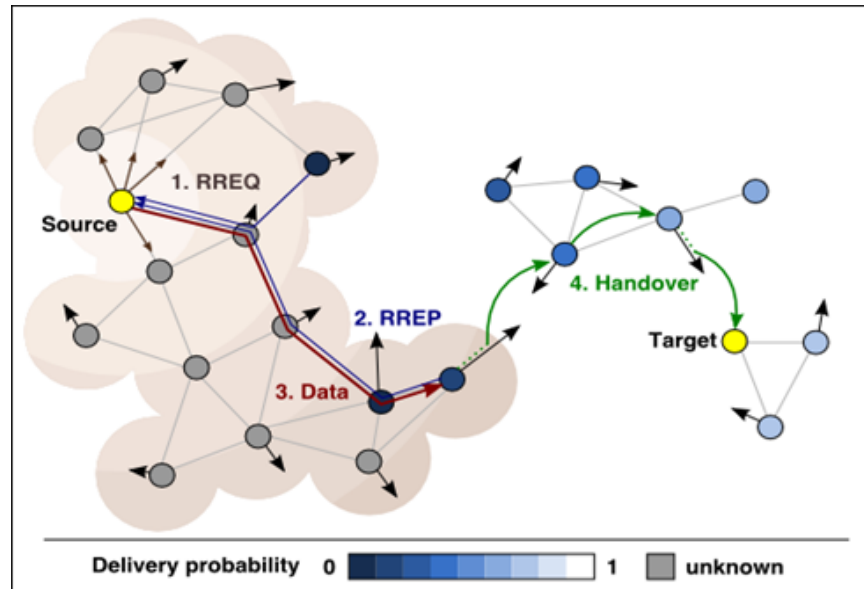


Figure:

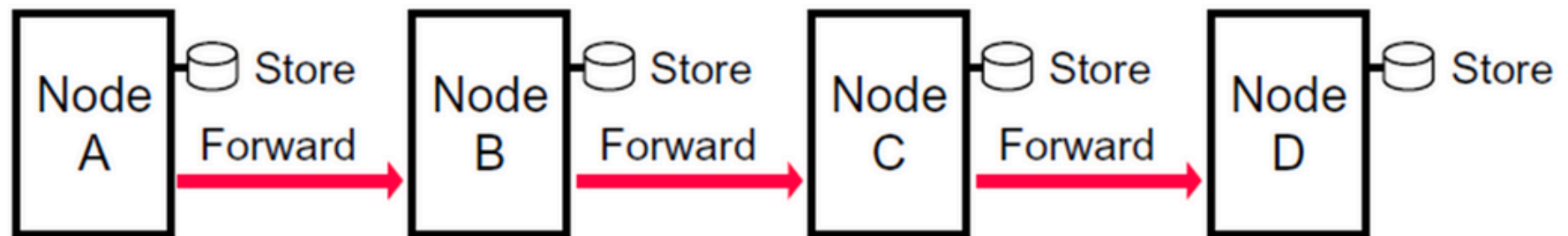


Figure: Store-And-Forward Message Switching

# Delay Tolerant Network

- DTN architecture expects nodes to store bundles for an extended period.
- Whenever possible, each received data packet is forwarded immediately.
- If forwarding is not currently possible but is expected to be possible in the future, the packet is stored for future transmission.

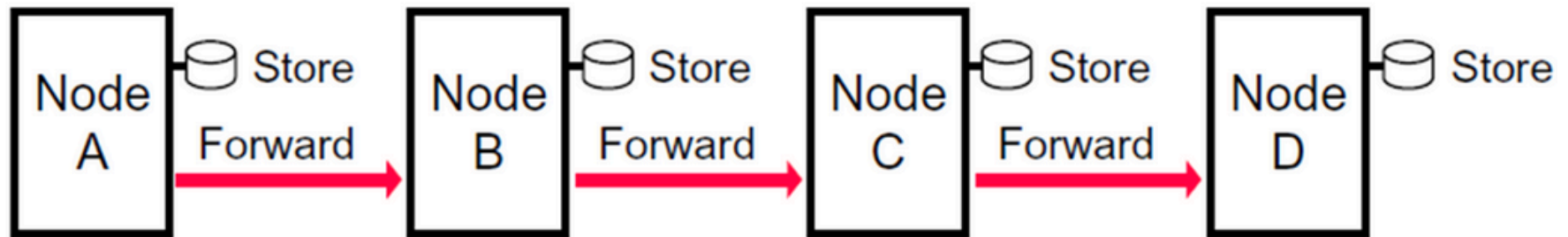


Figure: Store-And-Forward Message Switching

# Delay Tolerant Network

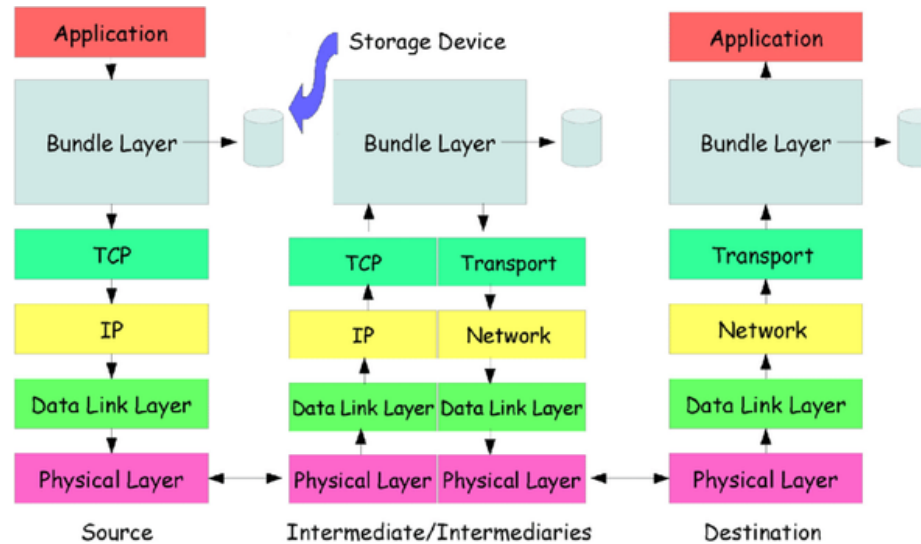


Figure:

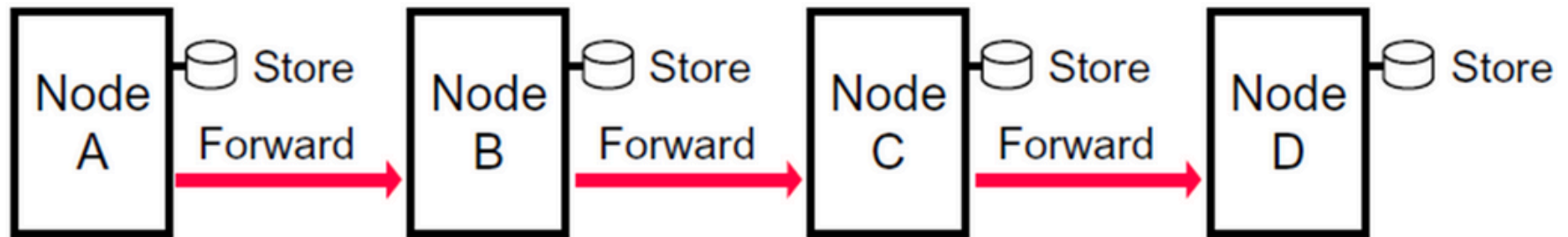


Figure: Store-And-Forward Message Switching

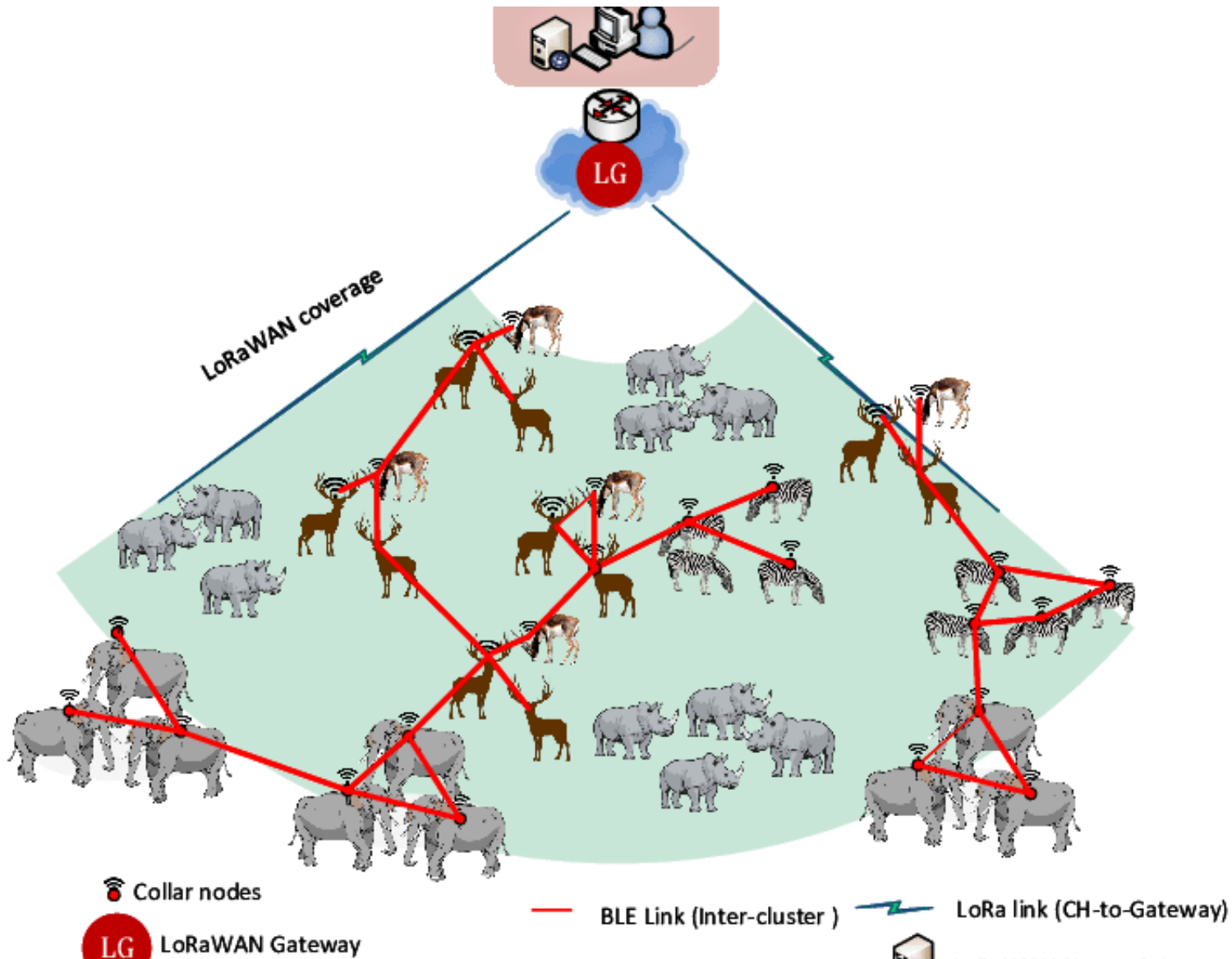
# Delay Tolerant Network (DTN):: Applications

## Mobile-enabled delay tolerant networking in rural developing regions





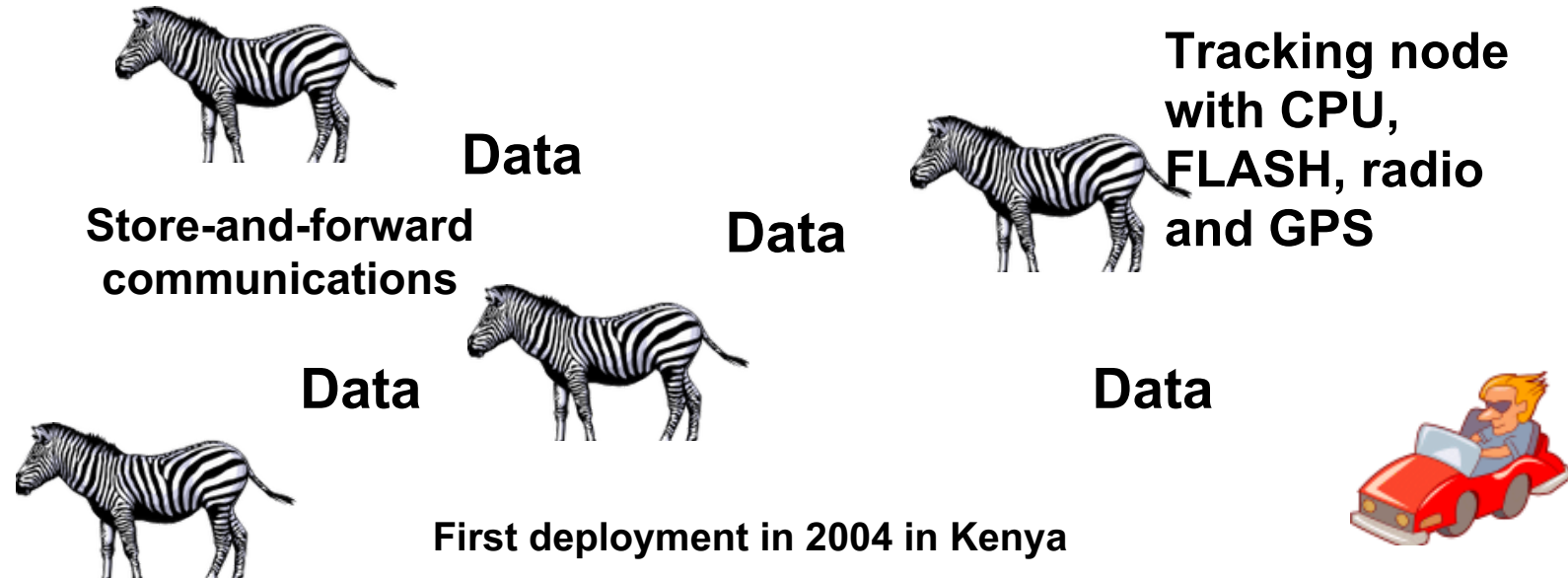
# Delay Tolerant Network (DTN):: Applications





# Delay Tolerant Network (DTN):: Applications

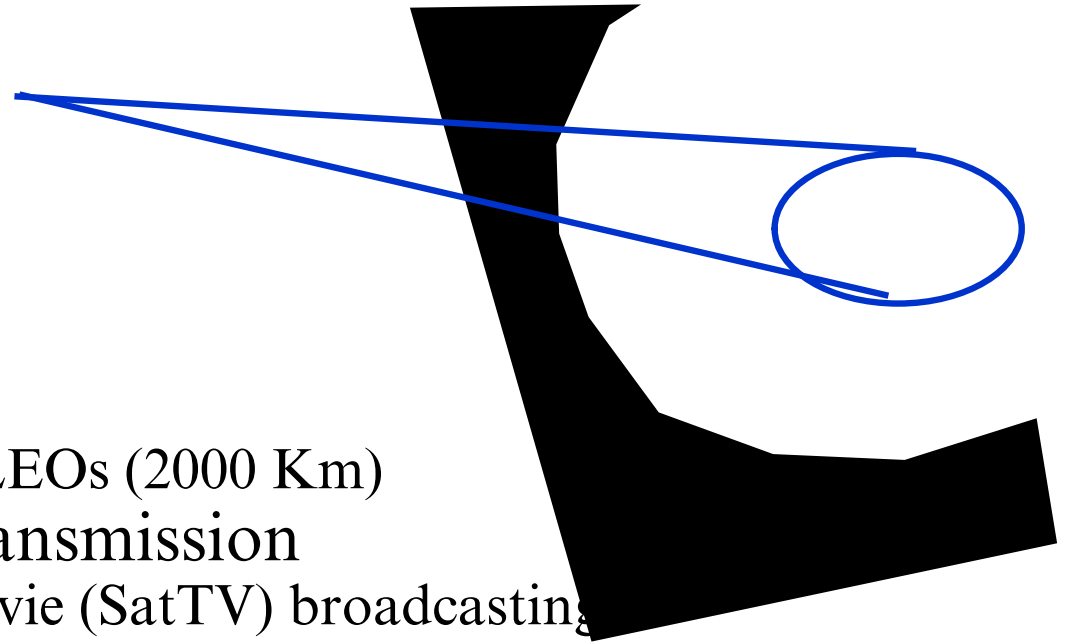
## ZebraNet (a real life application)



# Delay Tolerant Network (DTN):: Applications

## Satellite Systems

- Cover very large areas
- Different orbit heights
  - GEOs (39000 Km) versus LEOs (2000 Km)
- Optimized for one-way transmission
  - Radio (XM, DAB) and movie (SatTV) broadcasting



# DTN :: Characteristics

- Internet environment
  - End-to-end RTT is not large.
  - Some path exists between endpoints.
  - E2E reliability using ARQ works well.
  - Packet-switching is the right abstraction.
- DTN characteristics
  - Very large delays.
  - Intermittent and scheduled links.
  - Different network architectures.
  - Conversational protocols fail.
  - No ARQ.

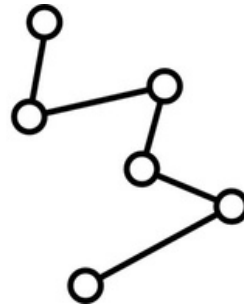
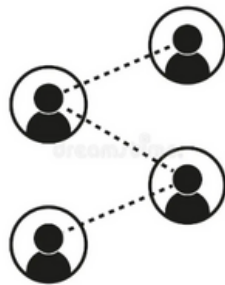
# DTN :: Challenges

- Applications assume e2e ‘connections’
- Routing does not understand schedules
- Long-term disconnection causes failures
- End-to-end reliability is poor with high loss rates
- E2E performance is poor with high delays

---

# Vehicular Ad-hoc Networks

(Un-Structured )



# Vehicular Ad-hoc Networks

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## VANET

- a form of Mobile ad-hoc network
- provide communication
  - among nearby vehicles
  - nearby fixed equipment

# Vehicular Ad-hoc Networks

How vehicular communications work:

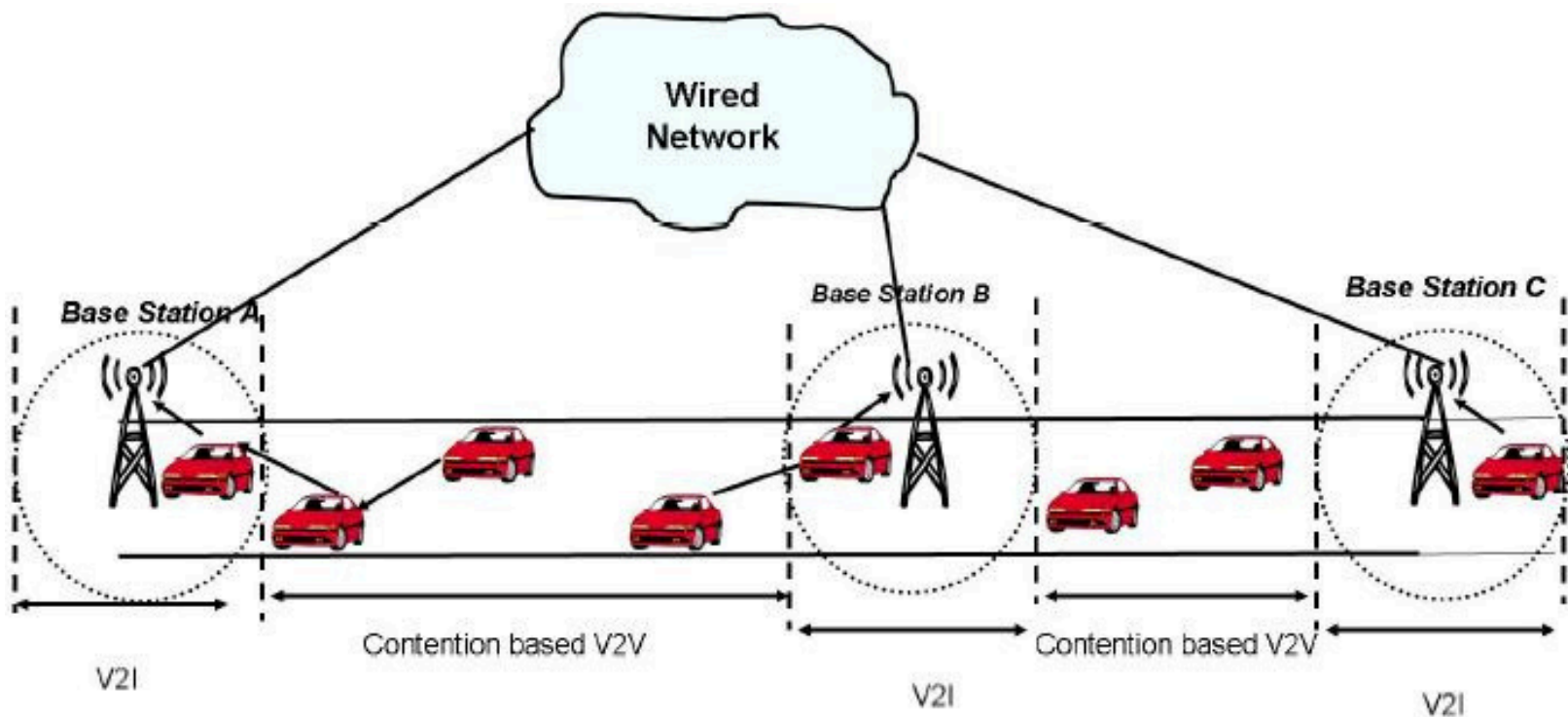
**Road-Side infrastructure Units (RSUs)**, are equipped with **on-board processing and wireless communication modules**



# Vehicular Ad-hoc Networks

How vehicular communications work:

- Vehicle-to-Vehicle ( $V2V$ )
- Vehicle-to-Infrastructure ( $V2I$ ) communication will be possible





# Car-to-X (C2X) Communication Patterns

- Vehicle-to-X (V2X)/ Car-to-X (C2X),
- Inter-Vehicle Communication (IVC),
- Vehicular ad-hoc network (VANET),

• ...

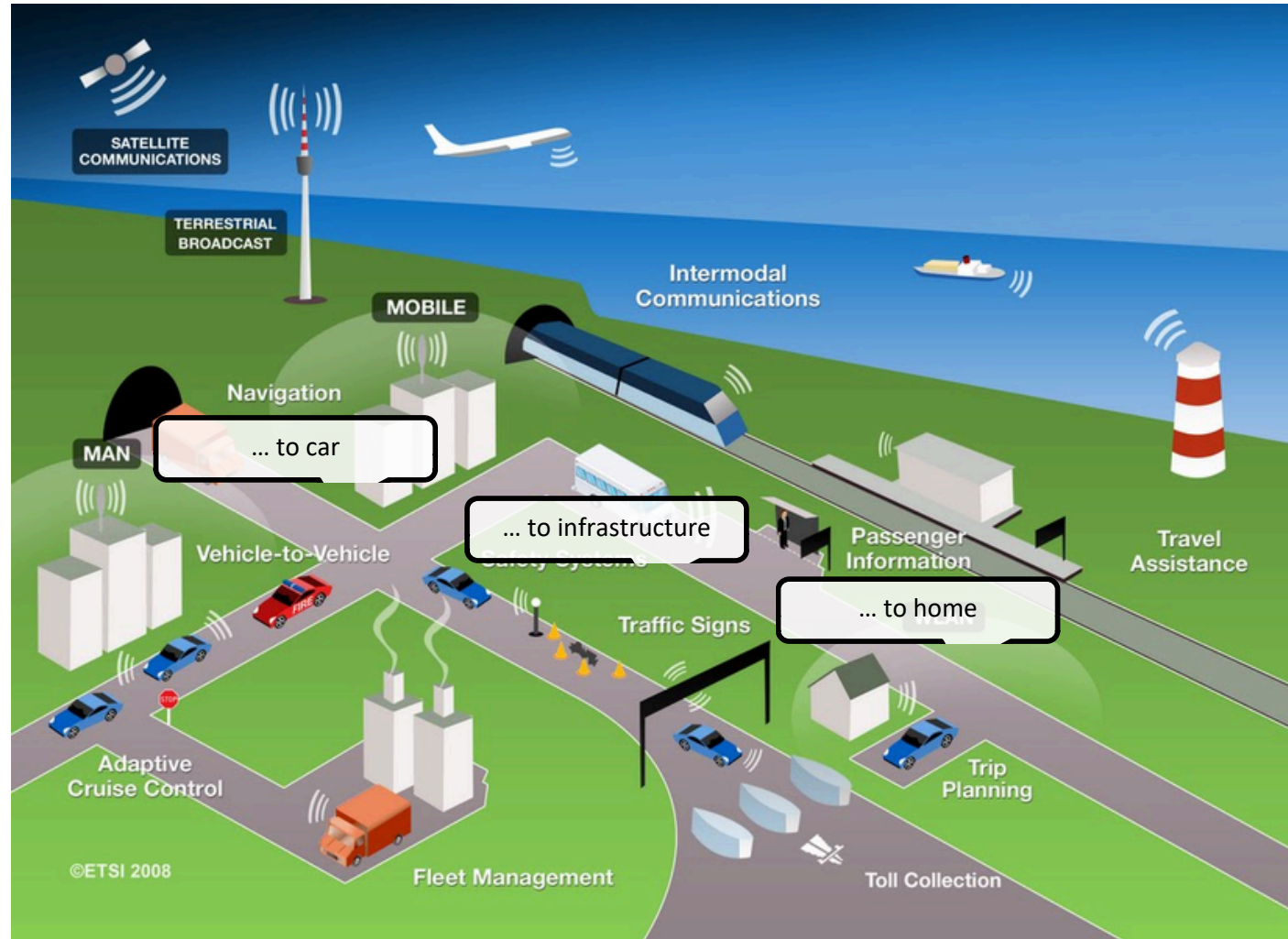
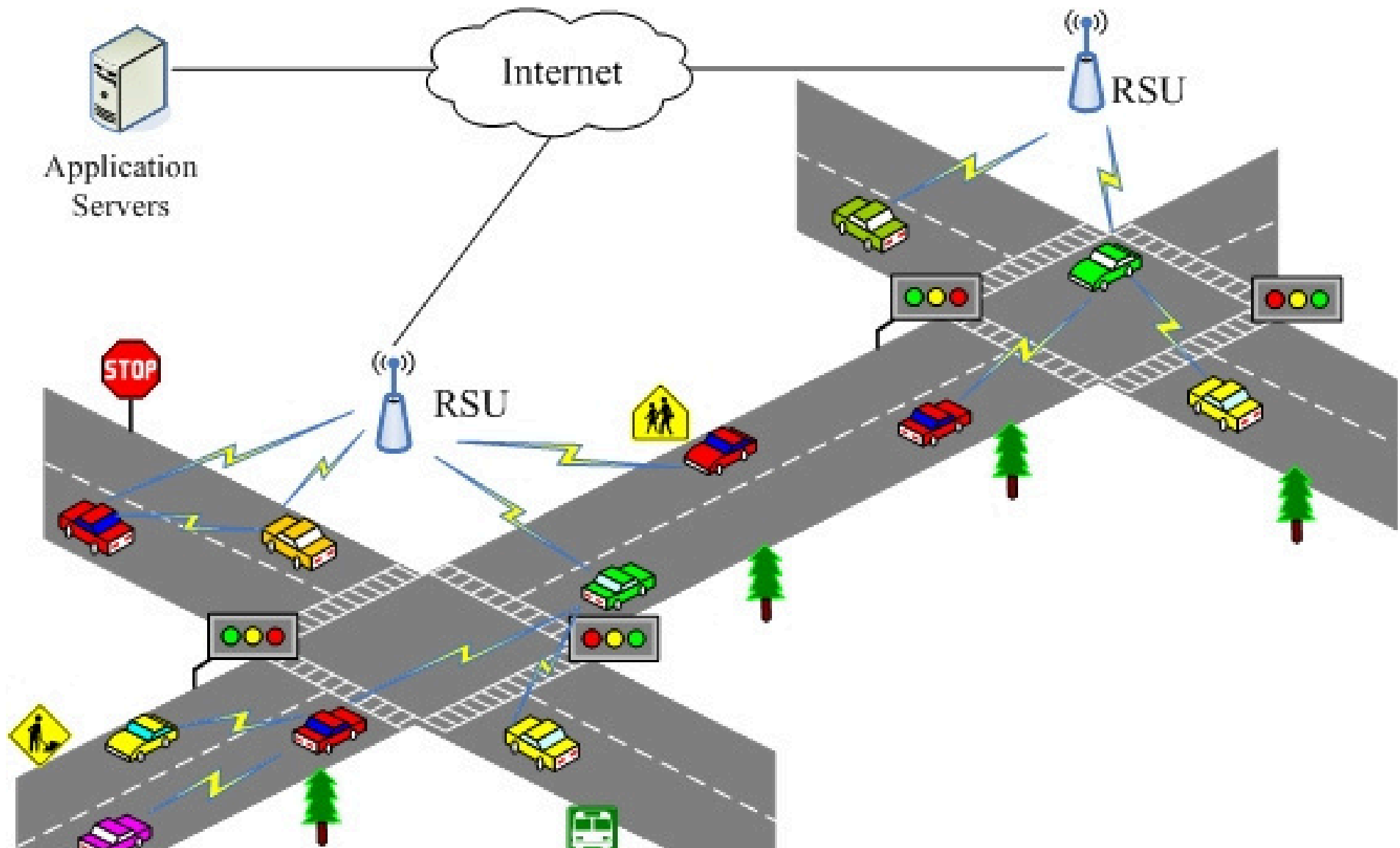


Illustration: ETSI

# Car-to-X (C2X) Communication



# Intelligent Transportation Systems (ITS)





# Intelligent Transportation Systems (ITS)



# Intelligent Transportation Systems (ITS)

---

What can ITS provide ?

- Safety
  - Efficiency
  - Traffic and road conditions
  - Road signal alarm
  - Local information
-

# Intelligent Transportation Systems (ITS)

---

Few Emerging Applications that ITS provide ?

- Autonomous Intersection Management (AIM)
  - Vehicle Platooning for Fuel and Space Efficiency
  - Eco-Driving and Green Routing
  - Cooperative Perception and Safety Sharing
  - Autonomous On-Demand Public Transit (Dynamic ride-pooling)
  - Automated Smart Parking
  - Real-Time Traffic Prediction and Control
  - Cooperative Highway Merging and Lane Changing
  - Dynamic Speed and Lane Recommendations
-

# Car-to-X (C2X) Communication

- Autonomous Intersection Management (AIM):



However, future AIM will adhere Strictly to Fundamental Traffic Rules, Prioritizing Safety as the Top Priority without Human Intervention

# Car-to-X (C2X) Communication

## • AUTONOMOUS INTERSECTION MANAGEMENT (AIM):

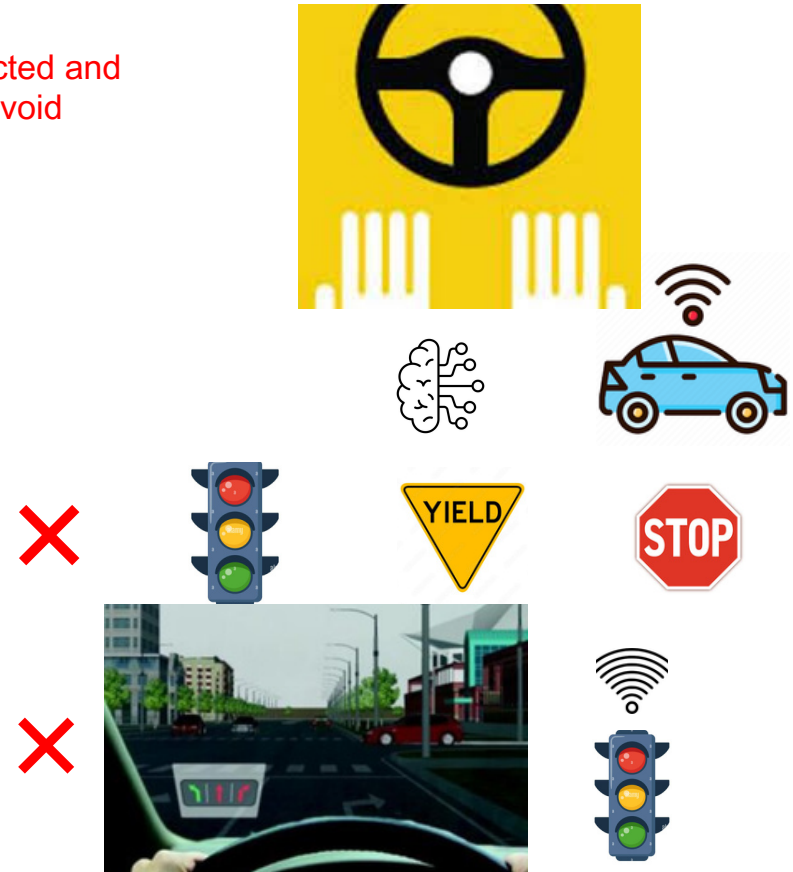
- Replaces traffic lights with autonomous intersection control, where Connected and Autonomous Vehicles (CAVs) communicate with a central coordinator to avoid collisions.

### Benefits:

- Minimal vehicle waiting time
- Reduced fuel consumption and emissions
- Elimination of signal delays
- Improved intersection throughput
- Enhanced safety

### Research Extensions:

- Robustness to Communication Delays and Packet Loss
- Human-Driven Vehicle (HDV) and CAV Coexistence Models
- Integration with Eco-Driving Strategies
- Scalability in High-Density Urban Grids

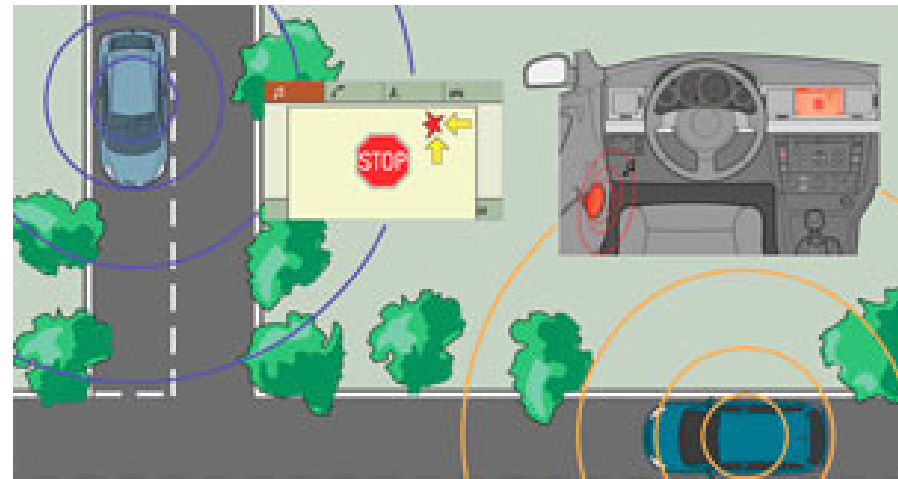
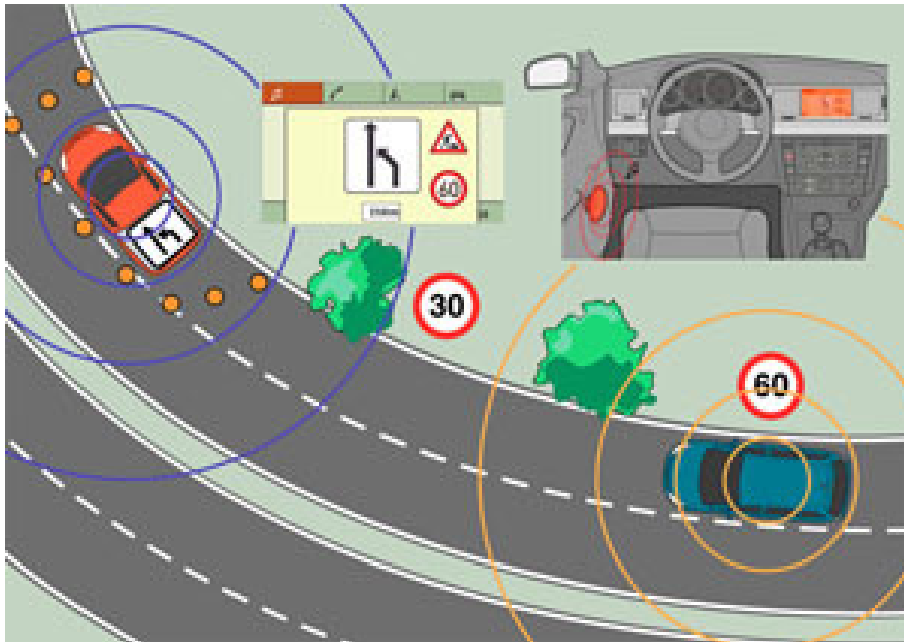




# Car-to-X (C2X) Communication

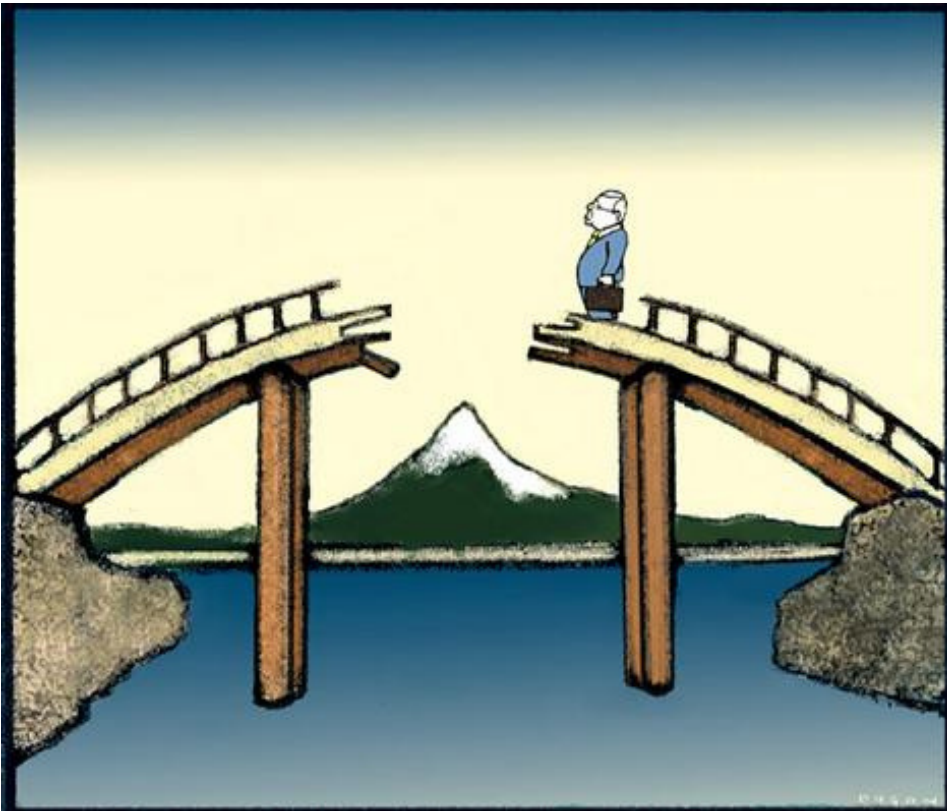
What can Car-to-X (C2X) communication provide ?

**Cooperative Perception and Safety Sharing  
Warning!!**



# Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

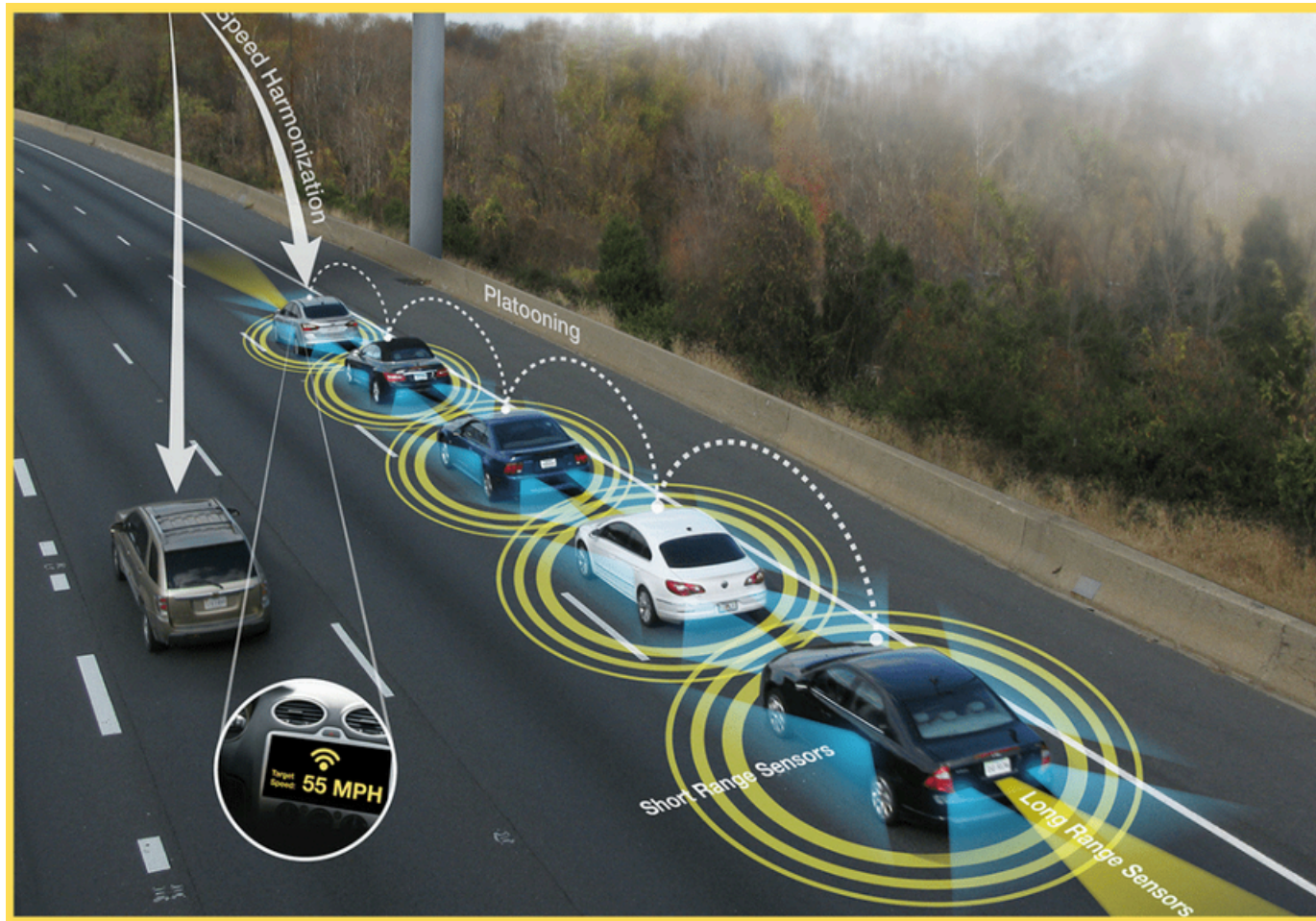


Warnings!!!  
Incident Detection and Reporting

# Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

**Vehicle Platooning :: Eco-Driving**



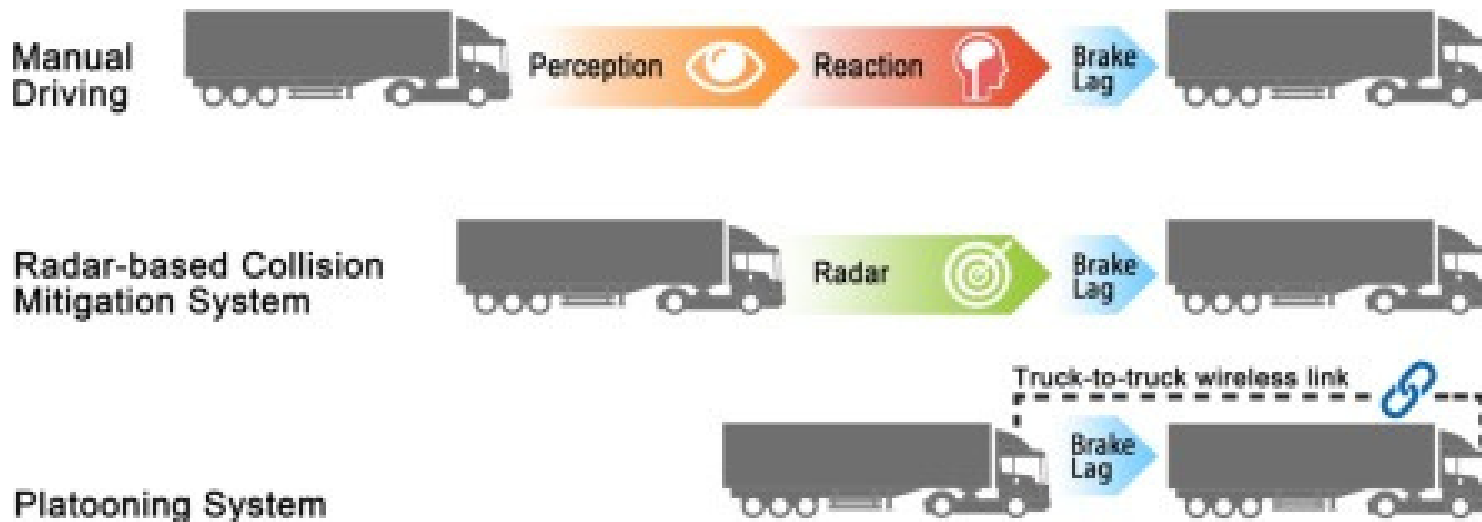
# Vehicle-to-Vehicle (V2V) Communication

Vehicle-to-Vehicle (V2V) communication enables direct wireless data exchange between vehicles, allowing them to share real-time information such as position, speed, and braking status.

## Vehicle Platooning:: Eco-Driving

### Faster and more synchronized braking response

- enhancing safety
- enabling closer following distances
- improving traffic flow efficiency



# Vehicle-to-Vehicle (V2V) Communication

## Manual Driving:

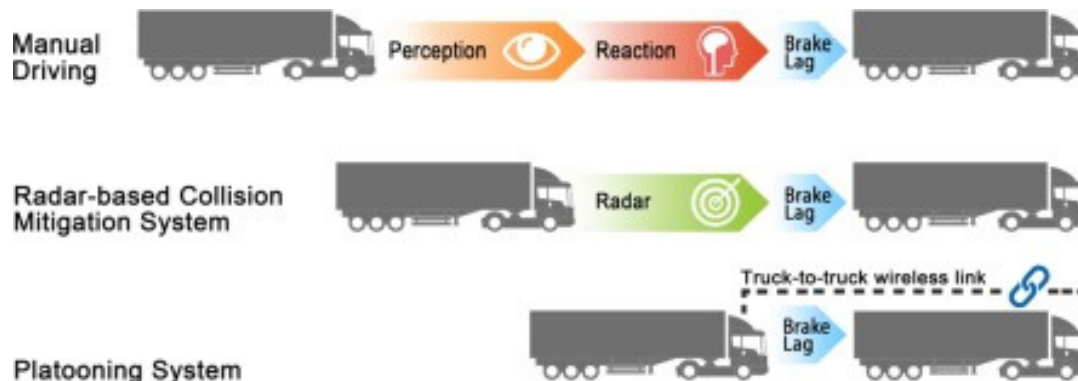
- Depends on **human perception and reaction**.
- Causes **significant brake lag** and larger following distances.

## Radar-Based Systems:

- Uses sensors to detect deceleration.
- Reduces reaction time but still has **processing delays**.

## V2V-Based Platooning:

- Shares braking information **wirelessly and instantly**.
- Enables **simultaneous braking**, minimizing brake lag and allowing **closer spacing**.



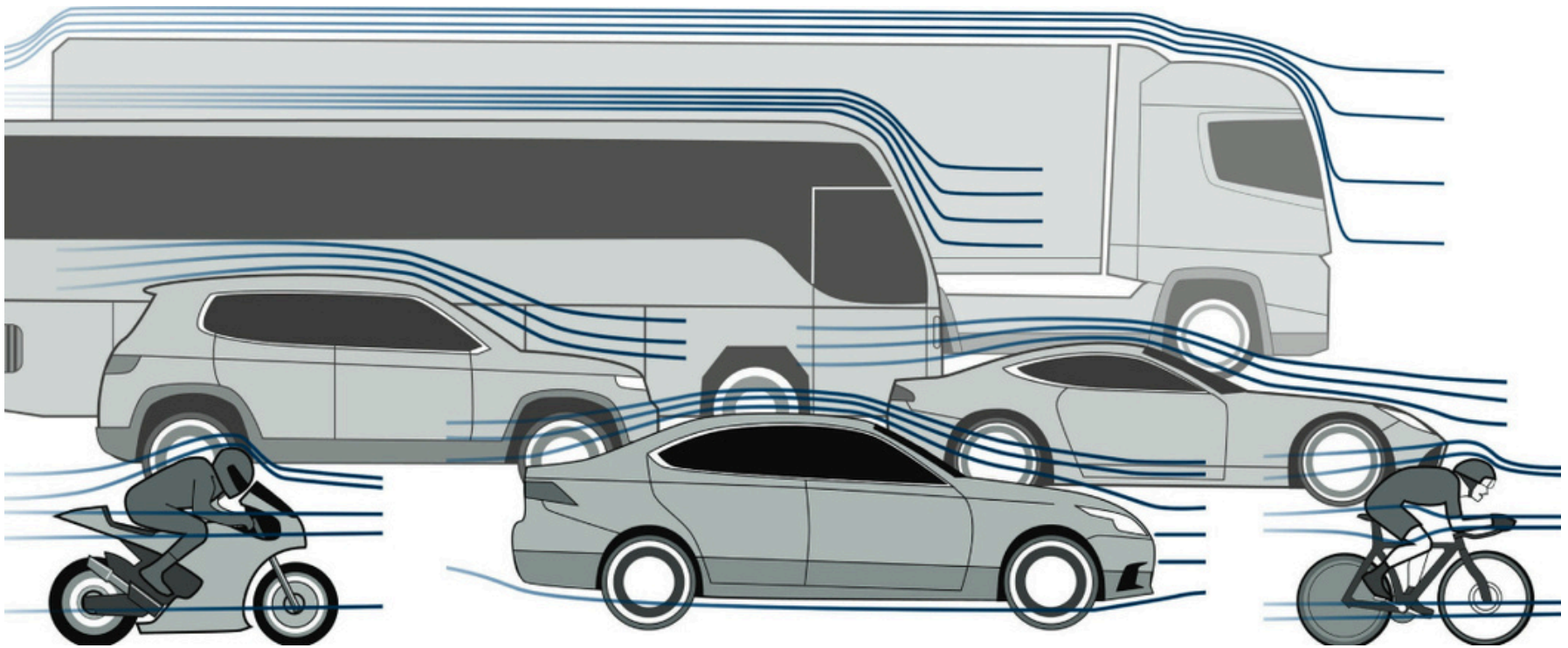
Vehicle-to-Vehicle (V2V) communication enhances emergency braking in truck platooning by enabling **faster and synchronized braking responses** across all vehicles, improving both **safety** and **traffic flow efficiency** compared to traditional systems.



# Vehicle-to-Vehicle (V2V) Communication

What can V2V communication provide ?

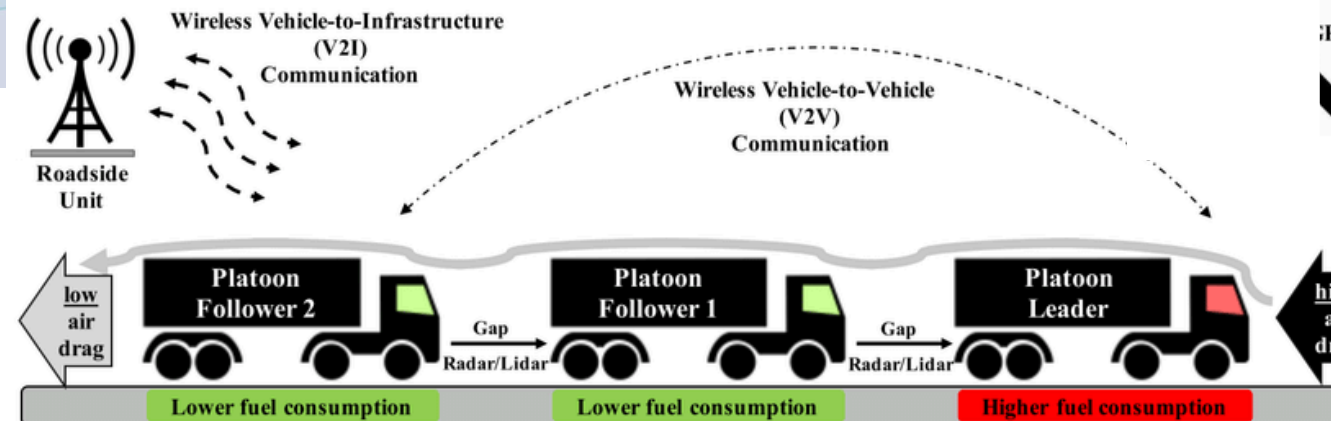
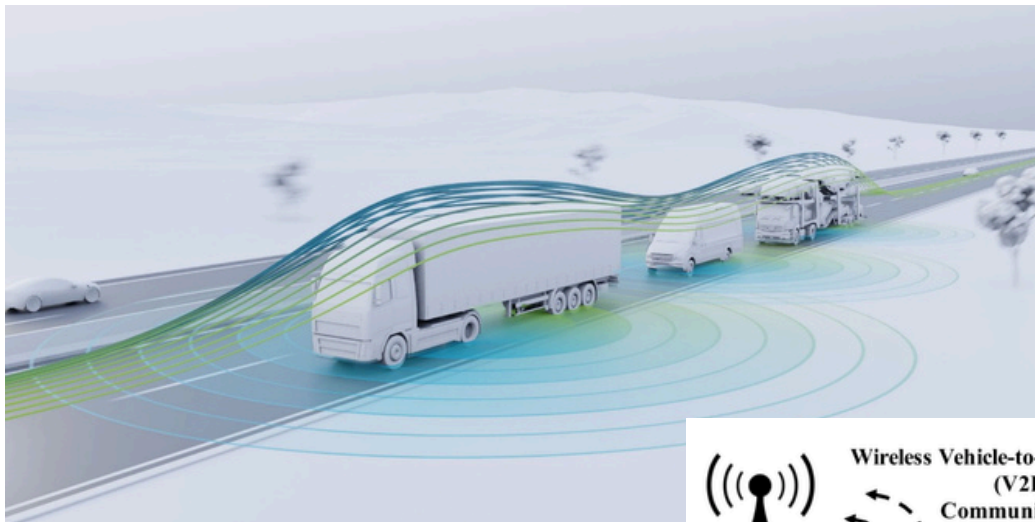
Vehicle Platooning for Improved Fuel Efficiency  
through Reduced Aerodynamic Drag



# Vehicle-to-Vehicle (V2V) Communication

What can V2V communication provide ?

**Vehicle Platooning for Improved Fuel Efficiency**  
through **Reduced Aerodynamic Drag**



# Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

Intelligent Traffic Flow Management

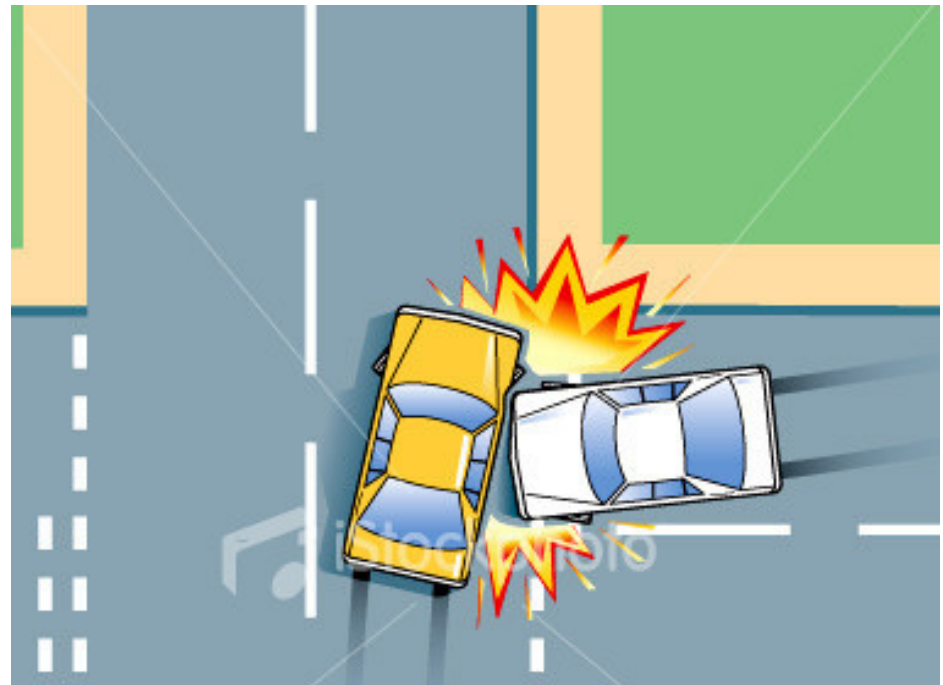
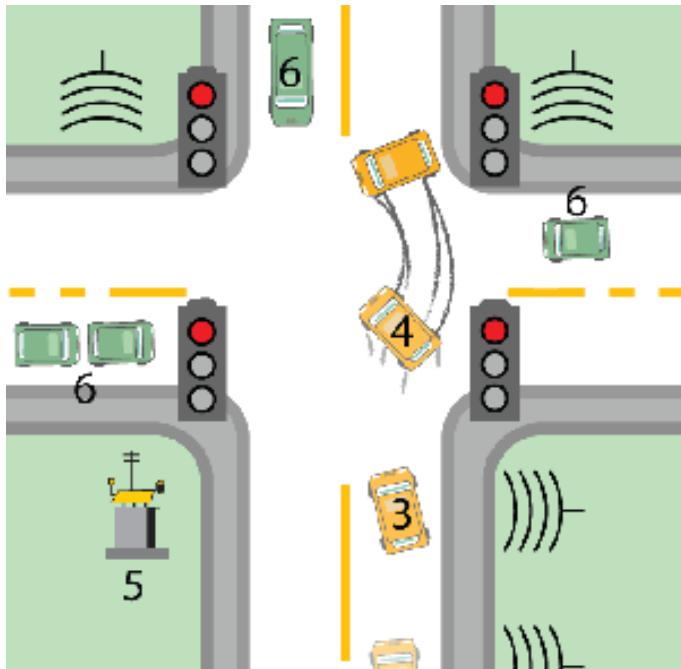




# Car-to-X (C2X) Communication

What can Car-to-X (C2X) communication provide ?

## Road Traffic Accidents Avoidance

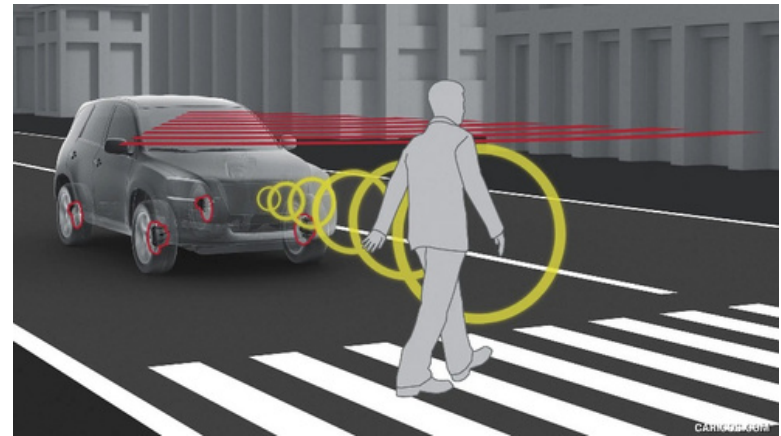


# Car-to-X (C2X) Communication

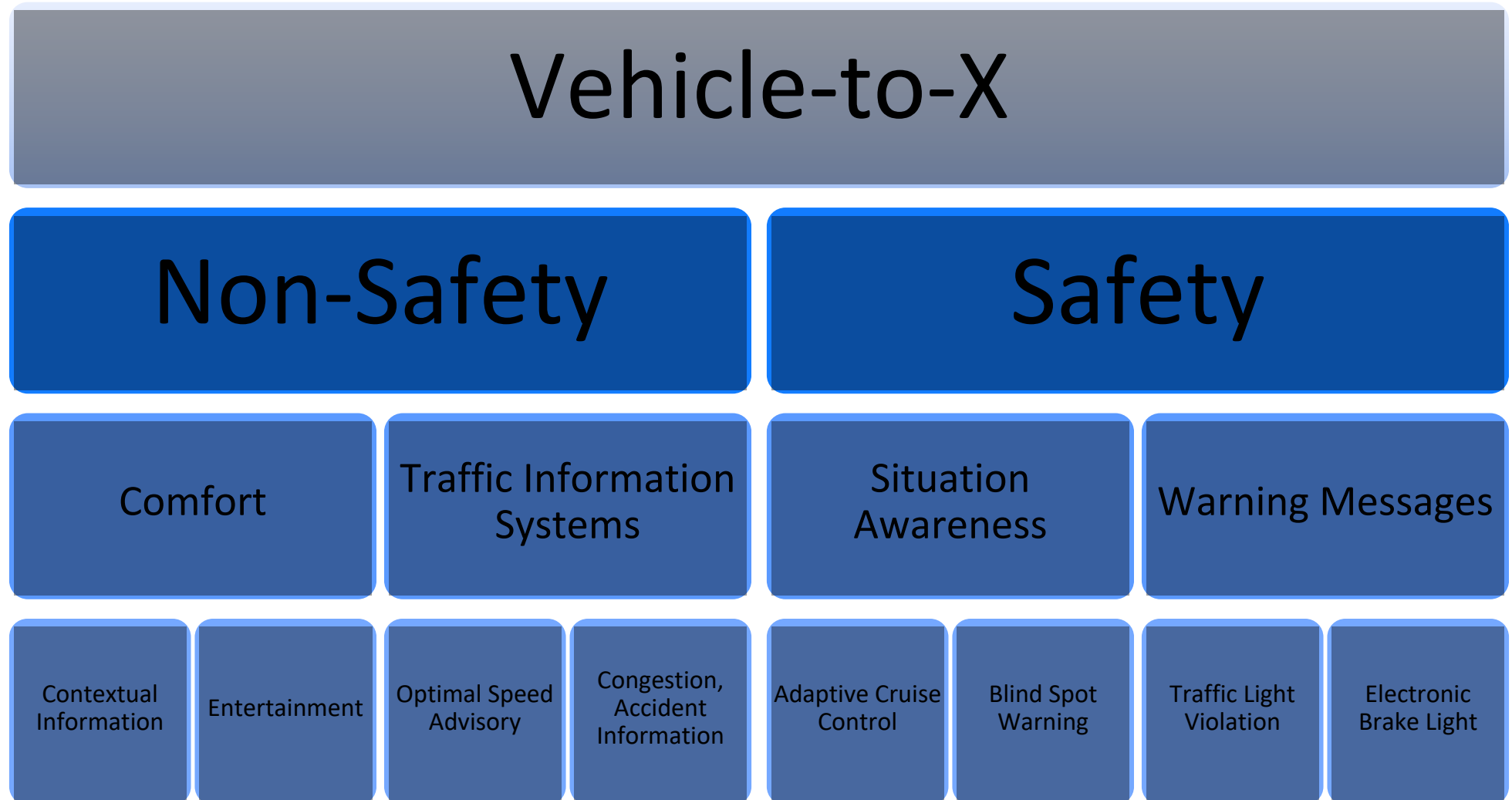
What can Car-to-X (C2X) communication provide ?



Pedestrian and Cyclist Protection



# Car-to-X (C2X) Communication:: Taxonomy of Use Cases



## Vehicle-to-X

### Non-Safety

Many messages  
High data rate

Low latency demands  
Low reliability demands

### Safety

Few messages  
Small packet size

High latency demands  
High reliability demands

# Challenges of C2X Communication

## Communication

- Highly varying channel conditions
- High congestion, contention, interference
- Tightly limited channel capacity

## Networking

- Unidirectional Links
- Multi-Radio / Multi-Network
- Heterogeneous equipment

## Mobility

- Highly dynamic topology
- But: predictable mobility
- Heterogeneous environment

## Security

- No (or no reliable) uplink to central infrastructure
- Ensuring privacy
- Heterogeneous user base

# Distributed Control over Wireless Links

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## Automated Vehicles

- Cars
- **UAVs**(Unmanned Aerial Vehicles)
- Insect flyers



- Packet loss and/or delays impacts controller performance.
- Controller design should be robust to network faults.
- Joint application and communication network design.

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Thank You